Practice Exam 1 MCQ

1. Which of the following best describes the hydrolysis of carbohydrates?

(A) The removal of a water molecule breaks a covalent bond between sugar monomers.

(B) The removal of a water molecule forms a covalent bond between sugar monomers.

(C) The addition of a water molecule breaks a covalent bond between sugar monomers. ✓

(D) The addition of a water molecule forms a covalent bond between sugar monomers.

Answer C

Correct. The hydrolysis of carbohydrates involves the addition of a water molecule, which breaks the covalent bond between monomers.

2. Which of the following best describes a characteristic of DNA that makes it useful as hereditary material?

(A) There are many different types of nucleotide bases that can be incorporated into DNA.

(B) The nucleotide bases can also be used to provide the energy needed for reproduction.

(C) Nucleotide bases can be randomly replaced with different nucleotide bases to increase variation.

(D) Nucleotide bases in one strand can only be paired with specific bases in the other strand. ✓

Answer D

Correct. The pairing of specific bases allows one strand to be used as a template for replicating the other strand, ensuring that DNA can be replicated with a relatively high accuracy.

Practice Exam 1 MCQ

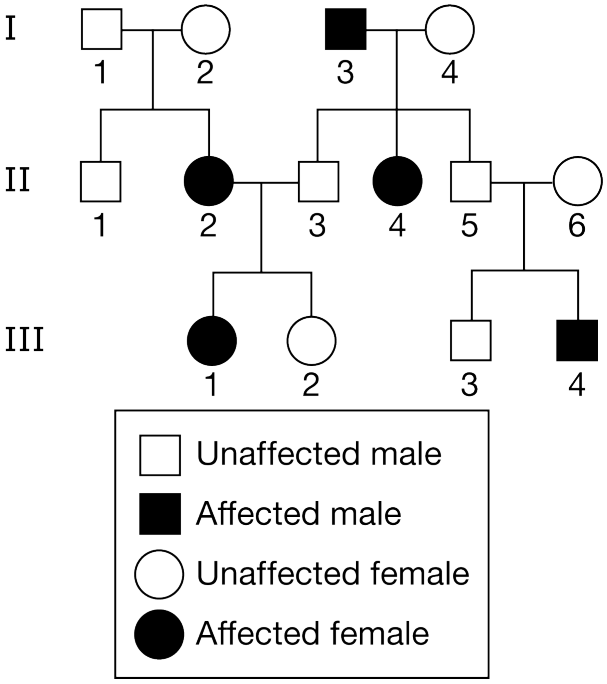


Figure 1. Pedigree of an inherited trait

3. Based on the pedigree in Figure 1, which of the following best explains the observed pattern of inheritance?

(A) The trait is autosomal dominant, because the cross between individuals I-3 and I-4 produced an affected offspring.

(B) The trait is autosomal recessive, because the cross between individuals I-1 and I-2 produced an affected offspring. ✓

(C) The trait is sex-linked dominant, because the cross between individuals II-5 and II-6 produced an affected male.

(D) The trait is sex-linked recessive, because the cross between individuals II-2 and II-3 produced an affected female.

Answer B

Correct. Only an autosomal recessive pattern of inheritance can explain individuals I-1 and I-2, who are unaffected, producing an affected offspring. The autosomal recessive pattern is also consistent with the complete pedigree.

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To investigate bacterial metabolism, a researcher divided a population (culture) of *Staphylococcus capitis* bacteria into two sets of culture tubes containing glucose. The researcher added a chemical to one set of tubes and measured the pH of the cultures at 5-minute intervals as the bacteria metabolized the glucose into lactic acid. The data are shown in Table 1.

TABLE 1. AVERAGE CHANGE IN pH IN CONTROL AND TREATMENT GROUPS OVER A 40-MINUTE PERIOD

Time (min)	Average pH of Control ($\pm 2\ SE_{\bar{x}}$)	Average pH of Treatment ($\pm 2\ SE_{\bar{x}}$)
0	8.04 ± 0.05	8.04 ± 0.06
5	7.96 ± 0.03	7.91 ± 0.04
10	7.88 ± 0.02	7.85 ± 0.04
15	7.82 ± 0.02	7.79 ± 0.06
20	7.76 ± 0.03	7.70 ± 0.04
25	7.71 ± 0.04	7.67 ± 0.02
30	7.63 ± 0.03	7.63 ± 0.02
35	7.65 ± 0.02	7.60 ± 0.02
40	7.65 ± 0.01	7.59 ± 0.02

4. Which of the following best describes the process by which the bacteria are breaking down the glucose to produce lactic acid?

(A) The bacteria are breaking down sugars in the absence of oxygen. ✓

(B) The bacteria are creating a H^+ gradient to synthesize more ATP.

(C) The bacteria are using their mitochondria to break down glucose in the presence of oxygen.

(D) The bacteria are producing CO_2 in the Krebs cycle that is then converted into lactic acid.

Answer A

Correct. Lactic acid is one of the end products of anaerobic metabolism of sucrose.

5. Which of the following was the dependent variable in the researcher's experiment?

Practice Exam 1 MCQ

(A) Time

(B) pH ✓

(C) Glucose concentration

(D) Lactic acid concentration

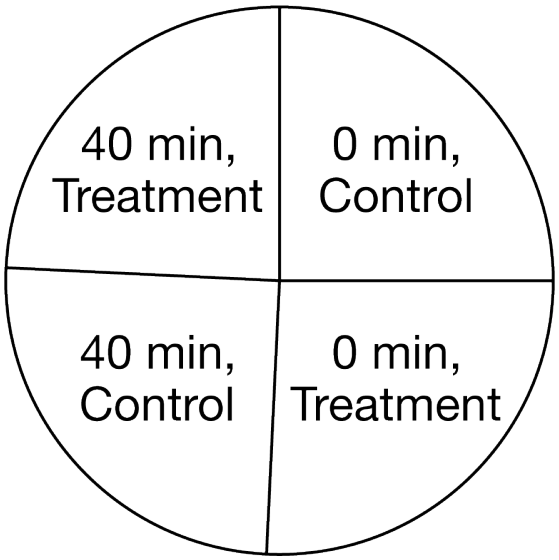
Answer B

Correct. The pH is being measured in this experiment and is therefore the dependent variable.

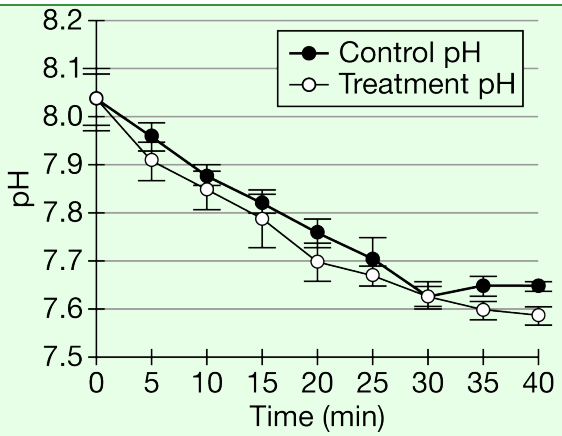
6. Which of the following graphs best represents the data in Table 1 ?

Practice Exam 1 MCQ

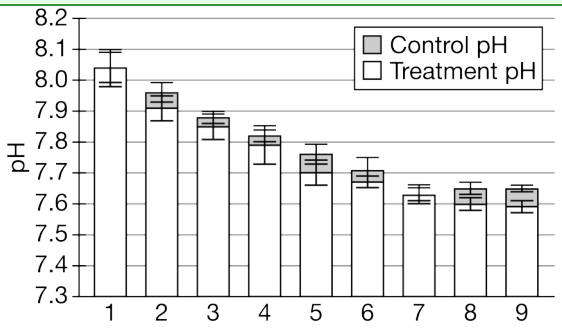
(A)



(B) ✓

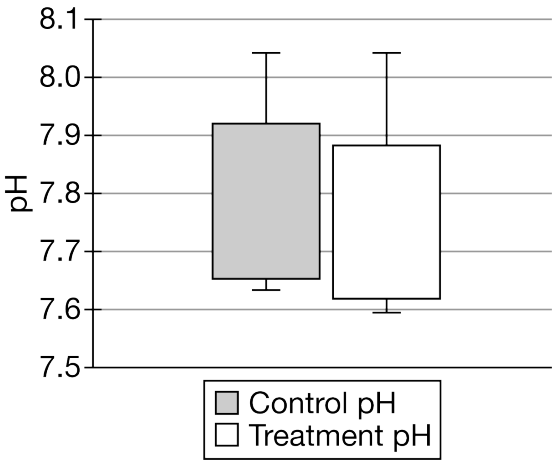


(C)



Practice Exam 1 MCQ

(D)



Answer B

Correct. This graph shows the decreasing pH values and error bars over time for both the control and treatment cultures.

7. Based on the data in Table 1, which of the following is the earliest time point at which there is a statistical difference in average pH between the control and treatment groups?

(A) 5 minutes

(B) 15 minutes

(C) 20 minutes

(D) 35 minutes ✓

Answer D

Correct. The first time the pH values $\pm 2\mathrm{SE}_{\bar{x}}$ of the control and treatment do not overlap is at 35 minutes

8. According to the data, which of the following best explains the results of the experiment?

(A) The pH of the treatment culture was lower than the pH of the control because the chemical increased the bacterial metabolic rate. ✓

(B) The pH of the treatment culture was higher than the pH of the control because the chemical denatured bacterial enzymes and decreased the metabolic rate.

(C) The chemical increased the metabolic rate of the bacteria because it lowered the pH.

(D) The chemical decreased the metabolic rate of the bacteria because it bound all available oxygen.

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Answer A

Correct. An increased bacterial metabolic rate results in the production of more lactic acid, which lowers the pH of the cultures.

A mutation in the gene coding for a single-polypeptide enzyme results in the substitution of the amino acid serine, which has a polar R group, by the amino acid phenylalanine, which has a nonpolar R group. When researchers test the catalysis of the normal enzyme and the mutated enzyme, they find that the mutated enzyme has much lower activity than the normal enzyme does.

9. Which of the following most likely explains how the amino acid substitution has resulted in decreased catalytic activity by the mutated enzyme?

(A) The substitution decreased the mass of the enzyme so that the mutated enzyme binds more weakly to the substrate than the normal enzyme does.

(B) The substitution altered the secondary and tertiary structure of the enzyme so that the mutated enzyme folds into a different shape than the normal enzyme does. ✓

(C) The substitution caused many copies of the mutated enzyme to cluster together and compete for substrate to bind.

(D) The substitution caused the directionality of the enzyme to change such that the amino terminus of the normal enzyme has become the carboxy terminus of the mutated enzyme.

Answer B

Correct. The replacement of a polar amino acid with a nonpolar amino acid will affect the folding of the enzyme because there will be different interactions between the R groups.

Pitcher plants are carnivorous plants that grow in areas where the soil contains low levels of key nutrients such as nitrogen. To obtain these nutrients, most pitcher plants capture prey using traps containing a digestive fluid. The captured prey are then broken down and digested, and the pitcher plant absorbs the nutrients.

The traps of one species of pitcher plant, *Nepenthes hemsleyana*, do not contain digestive fluid. Instead they provide a suitable place for woolly bats (*Kerivoula hardwickii*) to sleep. The feces from the bat are released into the trap where nutrients in the feces are absorbed and provide the plant with the nitrogen it needs.

10. Which of the following best describes the relationship between the pitcher plant and the woolly bat?

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(A) The relationship is an example of parasitism because the bat is harmed while the plant benefits.

(B) The relationship is an example of mutualism because both the plant and the bat benefit. ✓

(C) The relationship is an example of commensalism because the plant benefits but the bat is unaffected.

(D) The relationship is an example of commensalism because the plant is unaffected while the bat benefits.

Answer B

Correct. The pitcher plant benefits from the nitrogen from the bat, and the bat benefits from the safe place to sleep.

A particular genetic disorder is associated with a single gene with two alleles. Individuals with two recessive alleles are affected. The prevalence of the disorder is 1 in $6{,}600$.

11. Assuming the population is in Hardy-Weinberg equilibrium, which of the following is closest to the frequency of carriers in the general population?

(A) 0.00015

(B) 0.01230

(C) 0.02430 ✓

(D) 0.98770

Answer C

Correct. This is the value for $2pq$, which is the frequency of individuals who are heterozygous for the disorder.

Practice Exam 1 MCQ

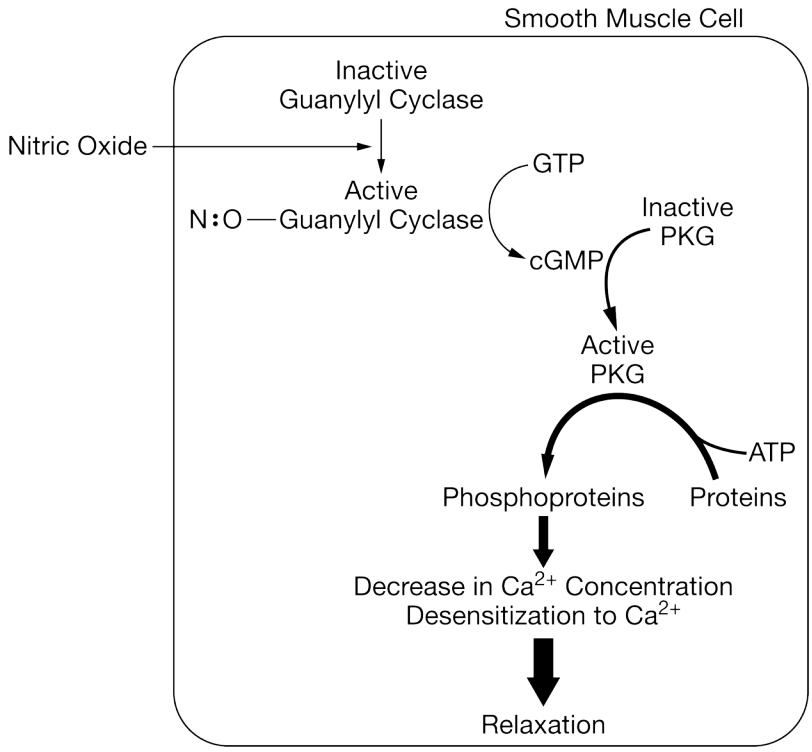


Figure 1. A signaling cascade triggers the relaxation of smooth muscle cells.

Blood vessels are surrounded by cells called smooth muscle cells. Nitric oxide triggers a signaling cascade in smooth muscle cells that causes the cells to relax (Figure 1).

12. Which of the following is represented by the gradual increase in thickness of the arrows from the top to the bottom of Figure 1?

(A) The rate at which nitric oxide triggers signaling gradually increases over time.

(B) The number of signaling molecules that are produced or activated increases with each step in the pathway. ✓

(C) The size of the proteins in the pathway increases as the signaling cascade moves through the cell.

(D) The signaling pathway uses an increase in negative feedback to reduce intracellular Ca^{2+} levels and cell sensitivity to Ca^{2+}.

Answer B

Correct. One of the benefits of a signaling cascade is that a relatively few initial signaling molecules can result in a large change, because at each step, a signaling molecule activates many other signaling molecules.

Practice Exam 1 MCQ

A student peeled the skins from grapes, exposing cells with membranes that are only permeable to water and small diffusible solutes. The student measured the mass of the peeled grapes. The student then placed each peeled grape into one of five solutions. After 24 hours, the student removed the peeled grapes from the solutions, measured their final mass, and calculated the percent change in mass (Table 1).

TABLE 1. PERCENT CHANGE IN MASS OF PEELED GRAPES IN SOLUTIONS

Solution	**Concentration of Solution (weight/volume)**	**Percent Change in Mass**
Distilled water	0%	13.48%
NaCl	20%	-23.39%
Tap water	0.8%	9.46%
Grape juice	2.1%	2.8%
Grape soda	13%	-15.00%

In a second experiment (Table 2), the student placed a peeled grape into a solution containing both small diffusible solutes and solutes to which the membrane is impermeable (nondiffusible solutes).

TABLE 2. CONCENTRATION OF SOLUTES IN SECOND EXPERIMENT

Location	Concentration of Small Diffusible Solutes	Concentration of Nondiffusible Solutes
Inside grape	$0.4\ \mathrm{M}$	$1.2\ \mathrm{M}$
In solution	$1.6\ \mathrm{M}$	$0.8\ \mathrm{M}$

13. Based on Table 1, which of the following best explains the difference in water potential between certain solutions and the grapes?

(A) NaCl and tap water have a lower water potential because these two solutions caused the grape to gain water.

(B) Grape soda and NaCl have a lower water potential because these two solutions caused the grape to lose water. ✓

(C) Tap water and grape juice have a lower water potential because these two solutions caused the grape to lose water.

(D) Grape soda and grape juice have a lower water potential because these two solutions caused the grape to gain water.

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Answer B

Correct. The mass of the grape decreased in both the grape soda and the $NaCl$ solutions, indicating that water left. Since water moves from higher water potential to lower water potential, the grape has a higher water potential than either solution has.

14. Based on Table 1, which of the following percentages is closest to the solute concentration of the grape?

(A) 0.0%
(B) 1.3%
(C) 5.5% ✓
(D) 10.1%

Answer C

Correct. The grape had a small positive change in mass in the 2.1% solution and a large negative change in mass in the 13% solution, suggesting the concentration of the grape is somewhere between those two percentages but closer to 2.1%.

15. A student hypothesizes that the solute concentration of grape juice is higher than the solute concentration of the actual grape because the grape juice has added sugar.

Based on the data in Table 1, which of the following best evaluates the student's hypothesis?

(A) The hypothesis is supported because the mass of the grape decreased in the grape juice.
(B) The hypothesis is supported because the grape juice has a greater solute potential than the grape has.
(C) The hypothesis is not supported because the grape was isotonic to the grape juice.
(D) The hypothesis is not supported because the mass of the grape increased in the grape juice. ✓

Answer D

Correct. The mass of the grape increased in the grape juice, suggesting the grape juice is hypotonic compared with the grape and thus allowed water to move into the grape. Extra sugar would have likely made the grape juice hypertonic.

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16. Assuming a negligible pressure potential, which of the following best predicts the net movement of the small diffusible solutes and water in the second experiment (Table 2) ?

(A) Small diffusible solutes will diffuse into the grape cells, followed by water. ✓

(B) Small diffusible solutes will diffuse out of the grape cells and water will diffuse into the cells.

(C) Small diffusible solutes will diffuse out of the grape cells, followed by water.

(D) Small diffusible solutes will diffuse into the grape cells and water will diffuse out of the cells.

Answer A

Correct. Small diffusible solutes would diffuse into the cell because there is a lower concentration inside the grape. Combined with the concentration of nondiffusible solutes, the interior of the grape will now be hypertonic to the exterior and water would move in.

17. Mercurial sulfhydryl is an inhibitor of aquaporins. Which of the following is the most likely effect of adding mercurial sulfhydryl to the distilled water solution?

(A) The grape cells will burst because of excess water entering by active transport.

(B) The grape cells will gain more water because of the activation of the transport protein.

(C) The grape cells will shrink because active transport has been inhibited.

(D) The grape cells will gain water more slowly because of a lack of facilitated diffusion. ✓

Answer D

Correct. While small amounts of water can pass through the membrane by simple diffusion, aquaporins are channel proteins that allow larger amounts of water to move over a shorter period of time.

18. Which of the following best explains why larger grapes have a different rate of water absorption per gram of mass than smaller grapes do?

(A) The rate is slower because smaller grapes have a larger surface-area-to-volume ratio than the larger grapes do. ✓

(B) The rate is slower because larger grapes have a larger surface-area-to-volume ratio than the smaller grapes do.

(C) The rate is slower because smaller grapes can expand more than larger grapes to hold excess water.

(D) The rate is slower because larger grapes have more volume to hold excess water than smaller grapes do.

Practice Exam 1 MCQ

Answer A

Correct. A grape with a smaller surface-area-to-volume ratio will have a slower rate of water absorption per gram.

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19. Which of the following best explains how some cells of an individual produce and secrete a specific enzyme, but other cells of the same individual do not?

(A) The cells contain different genes and therefore do not make the same proteins.

(B) The cells have evolved under different selective pressures, resulting in some cells making proteins that others cannot.

(C) The cells transcribe and translate different combinations of genes, leading to the production of different sets of proteins. ✓

(D) The cells produce different types of ribosomes that enable the translation of different genes.

Answer C

Correct. The process by which cells specialize for certain functions involves activating different sets of genes.

Practice Exam 1 MCQ

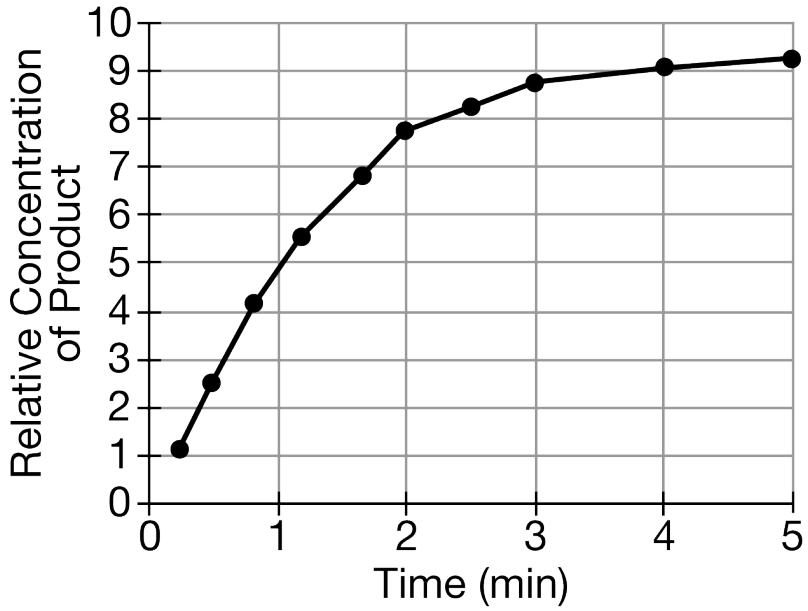


Figure 1. Change in the relative concentration of product over time

Figure 1 shows the amount of product produced in an enzyme-catalyzed reaction over five minutes.

20. Which of the following best explains how the rate of the reaction changes over time?

(A) The rate increases because more products are made over time.

(B) The rate increases because the ratio of product to substrate increases.

(C) The rate decreases because the ratio of product to substrate increases. ✓

(D) The rate decreases because the enzyme is used up as the reaction progresses.

Answer C

Correct. The decreasing slope of the line indicates that the reaction rate is decreasing over time. As the reaction proceeds and the ratio of product to substrate increases, the reaction rate decreases.

Pyruvate kinase, a key enzyme in the glycolysis pathway, is inhibited by the amino acid alanine. The ability of alanine to inhibit the enzyme is not affected by increasing the concentration of substrate.

21. Which of the following best explains the mechanism by which alanine inhibits pyruvate kinase activity?

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(A) Alanine binds to an allosteric site of the enzyme, changing the shape of the enzyme's active site. ✓

(B) Alanine increases the enzyme-substrate binding until the enzyme becomes saturated.

(C) Alanine is a competitive inhibitor that reversibly binds to the active site of the enzyme.

(D) Alanine binds to the substrate, preventing the substrate from being able to bind to the active site of the enzyme.

Answer A

Correct. Changing the shape of the enzyme's active site will prevent it from catalyzing the reaction regardless of the concentration of the substrate.

Different species of bumblebee compete for flower nectar in their ecosystem. While the flowers vary in the length of their petals, bumblebees vary in the length of a mouth part called a proboscis. Bumblebees demonstrate a preference for flowers that correspond to the length of their proboscis. Species with proboscises of similar length tend to occupy different areas. Species that live in the same area tend to have proboscises of different lengths.

22. Which of the following best explains the relationship between the different bumblebee species living in the same area?

(A) The bumblebees have undergone allopatric speciation due to geographic isolation.

(B) The bumblebees have undergone niche partitioning due to competition. ✓

(C) The bumblebees have a mutualistic relationship with other bumblebee species due to natural selection.

(D) The bumblebees have a commensalistic relationship due to the variety of niches that are available.

Answer B

Correct. The bumblebee species rely on different flowers depending on the length of their proboscis. This allows multiple species to occupy the same area while decreasing competition.

Practice Exam 1 MCQ

Researchers investigated whether time of day affects the rate at which certain mRNAs produced by plants are broken down. At either 1 hour after the start of the light period (morning) or 8 hours after the start of the light period (afternoon), the researchers treated identical plant seedlings with a compound that blocks transcription ($\text{time} = 0\ \text{min}$). The researchers measured the percent remaining of two mRNAs, mRNA G and mRNA H, over the course of 120 minutes. The data are shown in Figure 1.

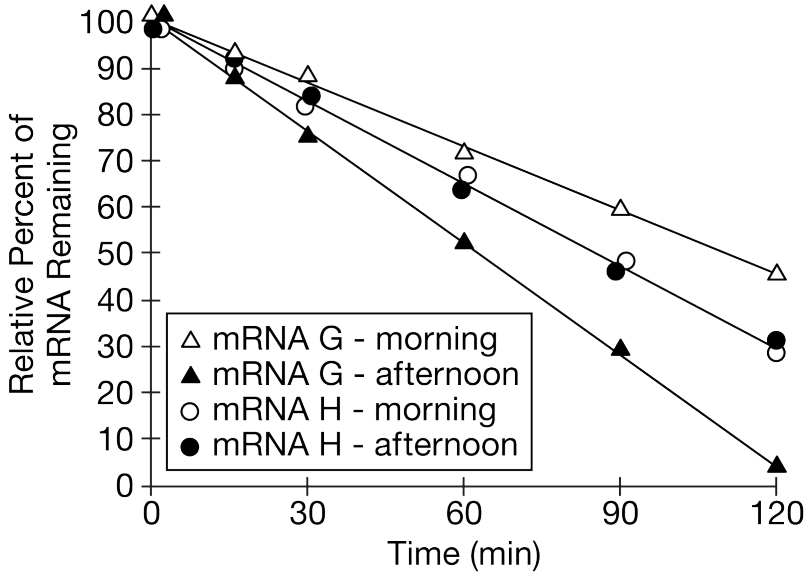


Figure 1. Degradation of mRNA G and mRNA H over time after exposure to light for 1 hour (morning) or 8 hours (afternoon)

23. Based on the data, which of the following best describes the relationship between light and the degradation of mRNA G and mRNA H?

(A) Exposure to light causes the degradation of both mRNA G and mRNA H.

(B) mRNA G and mRNA H degrade at the same rate during morning exposure to light.

(C) A longer exposure to light increases the rate of mRNA G degradation but not of mRNA H degradation. ✓

(D) Exposure to more-intense afternoon light causes both mRNA G and mRNA H to degrade more rapidly in the afternoon than in the morning.

Answer C

Correct. The rate of degradation of mRNA G after the plant has been exposed to light for 8 hours is greater than the rate of degradation of mRNA G after the plant has been exposed to light for only 1 hour. The rate of mRNA H degradation is unaffected by the length of time the plant has been exposed to light.

24. Which of the following observations would provide the strongest evidence that two plants belong to different biological species?

(A) There are variations in their RNA sequences.

(B) The leaves and flowers show morphological differences.

(C) They produce viable but sterile offspring. ✓

(D) They occupy unique habitats in an ecosystem.

Answer C

Correct. Based on the biological species concept, organisms of different species cannot produce hybrid offspring that are capable of reproducing.

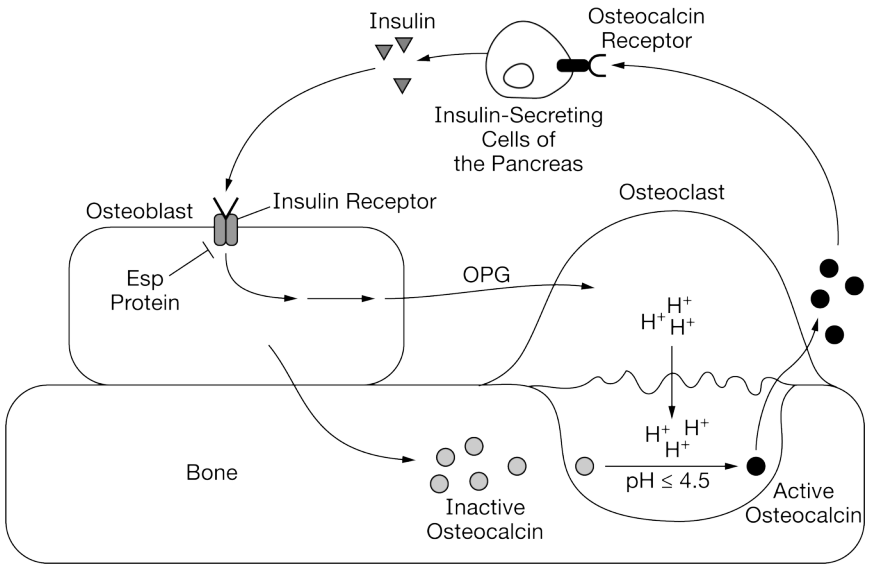


Figure 1. Pathway activated by insulin binding to the insulin receptor

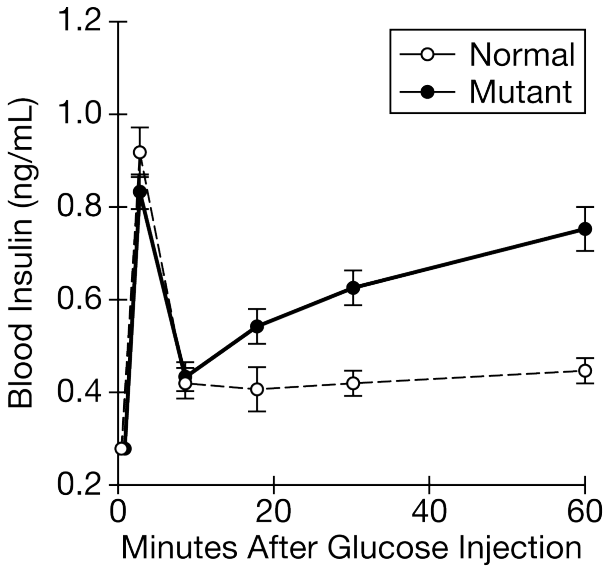


Figure 2: Blood insulin levels in normal mice and *Esp* mutant mice after exposure to glucose

Hormones are chemical signals that are released by cells in one part of the body that travel through the bloodstream to signal cells in another part of the body. Insulin is a hormone that is released by the pancreas that induces the uptake of glucose molecules from the bloodstream into cells. In this way, insulin lowers the overall blood glucose levels of the body. Osteoblasts and osteoclasts are two types of bone cells that play a role in regulating blood glucose levels (Figure 1).

Binding of insulin to the insulin receptor on osteoblasts activates a signaling pathway that results in osteoblasts releasing a molecule, OPG, that binds to neighboring osteoclasts. In response, the osteoclasts release protons (H^+) and create an area of lower pH outside the cell. This low pH activates osteocalcin, a protein secreted in an inactive form by osteoblasts.

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The Esp gene encodes a protein that alters the structure of the insulin receptor on osteoblasts and interferes with the binding of insulin to the receptor. A researcher created a group of osteoblasts with an Esp mutation that prevented the production of a functional Esp product (mutant). The researcher then exposed the mutant strain and a normal strain that expresses Esp to glucose and compared the levels of insulin in the blood near the osteoblasts (Figure 2).

25. Based on the information provided, which of the following best justifies the claim that osteocalcin is a hormone?

(A) The phosphorylation of the insulin receptor causes a response in osteoblast bone cells.

(B) The osteoblasts in the bone secrete osteocalcin, which causes cells in the pancreas to change their activity. ✓

(C) The change in expression of Esp changes the insulin receptor activity of the osteoblast.

(D) The activation of the osteocalcin by a bone cell is pH dependent.

Answer B

Correct. Hormones are signaling molecules that can be secreted from one part of the body and act on another part of the body.

26. Which of the following best describes the effect of insulin binding to the receptor on the osteoblast cells?

(A) Insulin binding ultimately increases pancreatic secretion of additional insulin. ✓

(B) Insulin binding blocks the release of osteocalcin from the osteoblasts.

(C) Insulin binding inhibits the expression of Esp.

(D) Insulin binding increases the pH of the extracellular matrix.

Answer A

Correct. Insulin binding increases the secretion of additional insulin via osteocalcin signaling.

27. Which of the following claims is most consistent with the data shown in Figure 2 ?

(A) Esp expression is necessary to prevent the overproduction of insulin. ✓

(B) Esp protein does not regulate blood-sugar levels in normal mice.

(C) Normal mice require a higher blood concentration of insulin than mutant mice do.

(D) Mutant mice have a cyclical pattern of insulin secretion.

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Answer A

Correct. Mutant mice that do not express Esp have a higher blood insulin level twenty minutes after injection and beyond.

28. Which of the following was a positive control in the experiment?

(A) Minutes after glucose injection

(B) Blood insulin

(C) Mutant strain

(D) Normal strain ✓

Answer D

Correct. Treatment of the normal strain of mouse has a known, expected result to ensure the experiment is functioning and is therefore the positive control.

29. A researcher observes that mice from the mutant strain experience low blood sugar. Which of the following best describes the feedback mechanism in the pathway (Figure 1) causing the low blood sugar in the mutant strain?

(A) The positive feedback of insulin production ✓

(B) The negative feedback of inactive osteocalcin production

(C) The positive feedback of the Esp protein

(D) The negative feedback of insulin-secreting pancreatic cell proliferation

Answer A

Correct. The release of insulin from the pancreas ultimately leads to more insulin being released. The excess insulin production results in too much sugar being removed from the bloodstream and the low blood sugar seen in the mutant mice.

A researcher hypothesizes that, in mice, two autosomal dominant traits, trait Q and trait R, are determined by separate genes found on the same chromosome. The researcher crosses mice that are heterozygous for both traits and counts the number of offspring with each combination of phenotypes. The total number of offspring produced was 64. The researcher plans to do a chi-square analysis of the data and calculates the expected number of mice with each combination of phenotypes.

Practice Exam 1 MCQ

30. Which of the following is the expected number of offspring that will display both trait Q and trait R?

(A) 4

(B) 12

(C) 36 ✓

(D) 48

Answer C

Correct. In a dihybrid cross of mice that are heterozygous for both traits, nine out of sixteen offspring are expected to have the dominant phenotype for both traits. $\frac{9}{16} \times 64 = 36$

Researchers studying new viruses analyzed the genetic material found in four different virus samples to determine the percent nitrogen base composition of each virus. The data are shown in the table.

BASE DISTRIBUTION IN FOUR NUCLEIC ACID SAMPLES

Nitrogen Base	Percent in Sample 1	Percent in Sample 2	Percent in Sample 3	Percent in Sample 4
Adenine	27	33	21	22
Cytosine	23	17	33	28
Guanine	23	17	27	28
Thymine	0	33	0	22
Uracil	27	0	19	0

31. Which of the following samples most likely contains a double-stranded RNA virus?

(A) Sample 1 ✓

(B) Sample 2

(C) Sample 3

(D) Sample 4

Practice Exam 1 MCQ

Answer A

Correct. A double-stranded nucleic acid will contain equal amounts of complementary bases. Sample one contains 23 percent of both cytosine and guanine and 27 percent of both adenine and uracil, consistent with Chargaff's rule. The presence of uracil instead of thymine indicates that the nucleic acid is an RNA molecule, not a DNA molecule.

Goats and sheep belong to the same family but different genera. While they often live together in the same pastures, the hybrid offspring that are occasionally produced between the two species rarely survive. When such a hybrid does survive, it is usually sterile.

32. Which of the following best explains the mechanism that maintains reproductive isolation between goats and sheep?

(A) Gene flow is prevented because the two species belong to different trophic levels and therefore do not share a food source.

(B) Habitat isolation creates a prezygotic barrier between the two species.

(C) The males of one species and the females of the other species are fertile at different times.

(D) The two species have a different number of chromosomes, resulting in a postzygotic barrier. ✓

Answer D

Correct. Closely related species might be able to mate and even produce a viable offspring, but if the chromosome numbers of the two species are not the same, the offspring will not have homologous pairs of chromosomes and will not be able to produce functional gametes.

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33. Ethylene gas is an organic molecule that serves many cell signaling roles in flowering plants. Which of the following best explains how a positive feedback mechanism involving ethylene works?

(A) Cells of ripening fruit produce ethylene, which activates the ripening response in other fruit cells. ✓

(B) Low water stress causes cells to produce ethylene, which binds to root cells and initiates cell division.

(C) Cells damaged by leaf-eating insects produce ethylene, which is released into the air, and repels insects.

(D) Fertilized ovules produce ethylene, which initiates apoptosis in flower petal cells.

Practice Exam 1 MCQ

Answer A

Correct. This is an explanation of positive feedback. The production of ethylene by some cells results in other cells producing ethylene, causing a rapid increase in the ripening process and increasing the overall level of ethylene present.

A student is investigating photosynthesis in plants. The student planted grass seeds in a tray with three sections and grew the grass under artificial lights for 14 days (Figure 1). After 14 days, the student collected all of the grass from section I and recorded its mass (Table 1). The student then placed a clear cover over section II and placed an aluminum foil cover over section III (Figure 2). The student then placed the tray back under the artificial lights for seven additional days. On day 21, the student collected and measured the mass of the grass from sections II and III (Table 1).

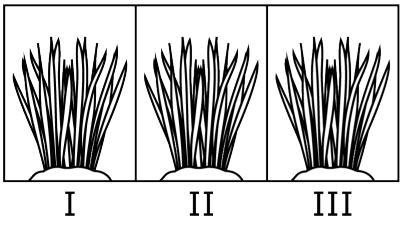


Figure 1. A tray of grass seed, grown under light for 14 days

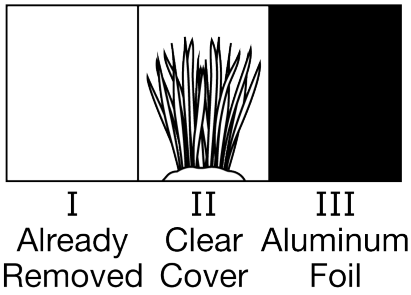


Figure 2. Modifications made to sections II and III on day 14

TABLE 1. MASS OF GRASS GROWN IN EACH SECTION

Section	Day Mass was Measured	Mass (g)
I	14	5.1
II	21	9.6
III	21	4.2

34. Based on the data, which of the following is the best approximation of the rate of growth per week in section II from day 14 to day 21 ?

(A) 3.2 g/week

(B) 4.5 g/week ✓

(C) 5.1 g/week

(D) 9.6 g/week

Practice Exam 1 MCQ

Answer B

Correct. The approximate growth rate of section II is $4.5\ g/week$ for days 14 to 21. Assuming the mass at day 14 of section II is the same as that of section I, then the rate for days 14 through 21 is $9.6 - 5.1\ g/week$.

35. Which of the following best explains the observed difference in mass between the grass in section I after 14 days and the grass in section III after 21 days?

(A) The grass in section III required more energy for metabolic processes than the grass in section I.

(B) The grass in section III performed more photosynthesis than the grass in section I.

(C) Only the grass in section III used energy from the seed for growth and development.

(D) Mass lost by cell respiration in section III was not replaced by photosynthesis. ✓

Answer D

Correct. Grass in all sections used stored mass for cellular respiration, but during days 14 through 21 the grass in section III was not able to produce biomass through photosynthesis.

36. Which of the following best explains the connection between energy, growth, and the maintenance of an ordered system in the experiment?

(A) Energy input from light is required for the grass to grow and maintain an ordered structure. ✓

(B) The grass obtains the energy for growth and maintenance of order from nutrients from the soil.

(C) The grass couples the release of energy from the light reactions with the production of oxygen (O_2) used to produce sugars for growth.

(D) Energy is required for the growth of the grass but not the maintenance of order.

Answer A

Correct. The plant is considered an ordered system, and without the input of new energy by photosynthesis, the plant draws on its stored energy reserves for maintenance at the cost of growth.

37. Based on the data, which of the following is the most likely effect of a period of increased volcanic activity that significantly decreases the amount of sunlight reaching a particular ecosystem for several years?

Practice Exam 1 MCQ

(A) There will be a decrease in the population sizes of most species and a decrease in the number of trophic levels. ✓

(B) The population sizes of plant species will stay the same, but the average biomass of individual plants will increase.

(C) There will be an increase in the population sizes of omnivores because they can eat other animals instead of the plants.

(D) The biodiversity of the ecosystem will increase as new niches arise and become occupied by new species.

Answer A

Correct. The data establish that a lack of adequate light leads to a loss of biomass. This loss would narrow the base of the energy pyramid, with less energy available throughout the ecosystem, leading to a decrease in population sizes of most species and possible extinction of some species and the loss of one or more of the top trophic levels.

38. Which of the following is an appropriate null hypothesis for the student's experiment?

(A) The absence of light negatively affects the mass of the grass.

(B) An increase in light promotes grass growth.

(C) A change in light intensity changes the rate of grass growth.

(D) The presence of light has no effect on the mass of the grass. ✓

Answer D

Correct. A null hypothesis states that a change in the designated independent variable would have no effect on the dependent variable. In this experiment, light is the independent variable and the mass of grass is the dependent variable.

Practice Exam 1 MCQ

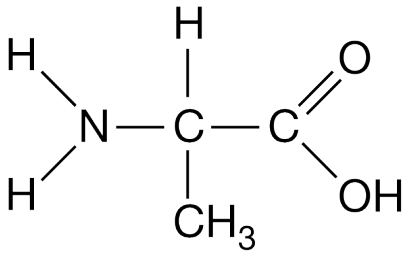


Figure 1. An amino acid

The amino acid in Figure 1 is found in a region of a polypeptide that folds away from water.

39. Which part of the amino acid most likely contributes to the hydrophobic behavior of this region of the polypeptide?

(A) Amine (NH_2) group

(B) Carboxyl ($COOH$) group

(C) Methyl (CH_3) group ✓

(D) Hydrogen (H) atom

Answer C

Correct. The methyl group is one of the functional groups that differentiate the twenty amino acids found in proteins. Since the methyl group is nonpolar, it is likely to make this region of the polypeptide more hydrophobic.

A researcher observes that when two heterozygous plants with red flowers are crossed, the resulting offspring include plants with red, white, or pink flowers. The researcher proposes the null hypothesis that flower color is the result of independent assortment and incomplete dominance. The researcher calculates a chi-square value of 7.3.

40. Assuming two degrees of freedom, which of the following is the correct interpretation of the chi-square analysis, using a p-value of 0.05 ?

(A) The null hypothesis should be rejected because the critical value is less than the calculated value. ✓

(B) The null hypothesis should not be rejected because the critical value is less than the calculated value.

(C) The null hypothesis should not be rejected because the critical value is greater than the calculated value.

(D) The null hypothesis should be rejected because the critical value is greater than the calculated value.

Practice Exam 1 MCQ

Answer A

Correct. The critical value for two degrees of freedom and a p-value of 0.05 is 5.99, which is less than the calculated value of 7.3.

---

41. A researcher is investigating the effects of a chemical that makes thylakoid membranes permeable to hydrogen ions (H^+). Which of the following is the most likely direct effect of adding the chemical to plant cells?

(A) The plant cells will produce less $NADPH$.

(B) The chloroplasts will generate less ATP. ✓

(C) Chlorophyll will require less light energy to excite its electrons.

(D) The plant cells will split fewer water molecules into hydrogen ions and oxygen.

Answer B

Correct. The phosphorylation of ADP to form ATP is dependent on the flow of hydrogen ions through ATP synthase. Making the membrane permeable to hydrogen ions would disrupt the proton gradient and fewer protons would be available to flow through ATP synthase.

Practice Exam 1 MCQ

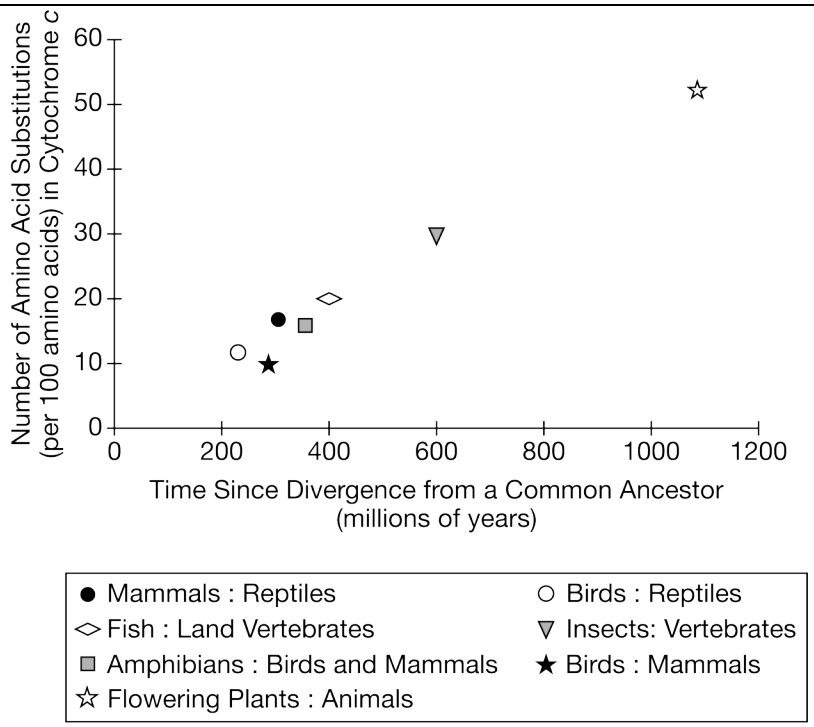


Figure 1. For paired groups of organisms, a comparison of the number of amino acid substitutions in cytochrome *c* with the time since the groups diverged from a common ancestor

Researchers analyzed the amino sequence of the protein cytochrome *c* in various groups of organisms and determined the number of amino substitutions that have occurred in the different groups of organisms. They plotted the data with respect to the time since divergence of the members of paired groups from a common ancestor (Figure 1).

42. Based on the data, which of the following organisms are most distantly related?

(A) Birds and mammals

(B) Mammals and reptiles

(C) Fish and land vertebrates

(D) Insects and vertebrates ✓

Answer D

Correct. The data indicate that insects and vertebrates diverged approximately 600 million years ago.

43. Which of the following best explains how continuity of genetic information in cells is ensured across generations?

Practice Exam 1 MCQ

(A) Replication uses a parental strand of DNA as a template to create a new strand of DNA. ✓

(B) DNA molecules are shaped like a double helix with a constant diameter throughout.

(C) Transcription copies the information in DNA into an RNA transcript.

(D) Cells contain different polymerases for DNA replication and transcription.

Answer A

Correct. The use of a parental DNA template strand ensures the new DNA double helix produced will have the same sequence of bases as the parental strand.

In the spring and summer, the fur of an arctic fox contains a pigment called melanin that gives the fox's fur a dark color. In the fall and winter, the fur of the arctic fox is white.

44. Which of the following most likely explains how the changing seasons result in changing fur color in an arctic fox?

(A) Environmental factors cause changes in gene expression, resulting in seasonal variations in pigment production. ✓

(B) Environmental factors cause different mutations in DNA during different seasons, resulting in seasonal changes in fur phenotype.

(C) Environmental factors cause proteins to be translated using different genetic codes during different seasons, resulting in variations in pigment production.

(D) Environmental factors cause enzymes to react with different substrates, resulting in the accumulation of different pigments.

Answer A

Correct. Environmental factors can influence gene expression, in this case resulting in a reduction of melanin during the fall and winter.

Practice Exam 1 MCQ

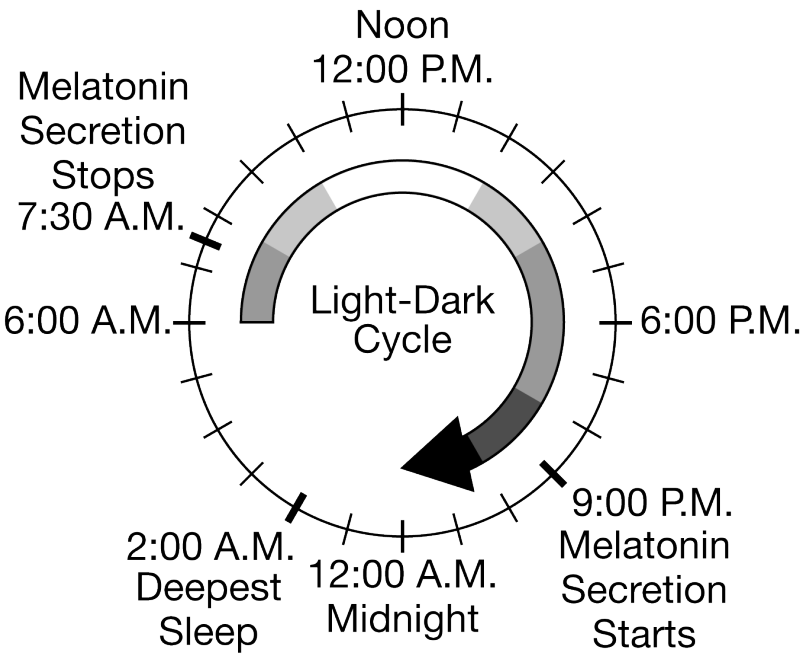


Figure 1. Key physiological points in the human sleep-wake cycle

The human sleep-wake cycle is regulated by melatonin. The synthesis of melatonin is regulated by light exposure (Figure 1). The human body typically develops a sleep-wake rhythm that does not respond quickly to change.

45. Long-term exposure to extended periods of bright light after sunset is most likely to affect a person in which of the following ways?

(A) Melatonin synthesis will be increased, and the entire sleep-wake cycle will be shifted by several hours.

(B) Melatonin synthesis will be increased, and the person will quickly fall asleep.

(C) Melatonin synthesis will be inhibited, and the person will have difficulty sleeping. ✓

(D) Melatonin synthesis will be inhibited, and the person will quickly fall asleep.

Answer C

Correct. Melatonin promotes sleep, and its production is inhibited by light. Exposure to light after sunset will likely prevent melatonin secretion and negatively impact sleep.

---

46. A typical bag of fertilizer contains high levels of nitrogen, phosphorus, and potassium but trace amounts of magnesium and calcium. Which of the following best matches the fertilizer component with the molecule in which it will be incorporated by organisms in the area?

Practice Exam 1 MCQ

(A) Nitrogen will be incorporated into nucleic acids. ✓

(B) Phosphorus will be incorporated into amino acids.

(C) Potassium will be incorporated into lipids.

(D) Magnesium will be incorporated into carbohydrates.

Answer A

Correct. All the bases in the nucleotides that compose nucleic acids contain nitrogen.

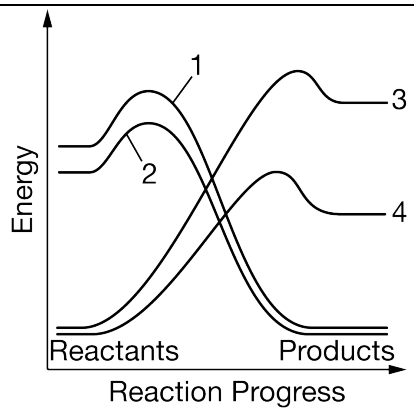


Figure 1. Change in energy over the course of four chemical reactions

47. Based on the data in Figure 1, which of the following most likely represents the change in energy that occurs when ATP hydrolysis is coupled with the phosphorylation of a substrate?

(A) Line 4 represents ATP hydrolysis, and line 1 represents phosphorylation of a substrate.

(B) Line 1 represents ATP hydrolysis, and line 4 represents phosphorylation of a substrate. ✓

(C) Line 2 represents ATP hydrolysis, and line 3 represents phosphorylation of a substrate.

(D) Line 3 represents ATP hydrolysis, and line 2 represents phosphorylation of a substrate.

Answer B

Correct. Line 4, an endergonic (requires energy) reaction that could represent the phosphorylation reaction, is coupled to Line 1, an exergonic reaction (requires energy) that could represent the hydrolysis reaction. Additionally, the total energy of the reactants represented by lines 1 and 4 is less than that of the products, because some energy is converted to heat.

Practice Exam 1 MCQ

The brown anole (*Anolis sagrei*) is a species of lizard native to Cuba that has been introduced into the southeastern United States. The range of brown anoles in the United States has been expanding, and they are now competing with native green anoles (*Anolis carolinensis*). Some of the characteristics of the green and brown anoles are shown in Table 1.

TABLE 1. CHARACTERISTICS OF TWO ANOLE SPECIES

Anole Species	Body Length	Limb Length Relative to Body Length	Toe Pad Size Relative to Body Length	Cold Tolerance
A. sagrei (brown anole)	55–80 mm	Long	Intermediate	30–35°C
A. carolinensis (green anole)	45–85 mm	Short	Very large	25–30°C

A cladogram showing the relationships between some anole species is shown in Figure 1.

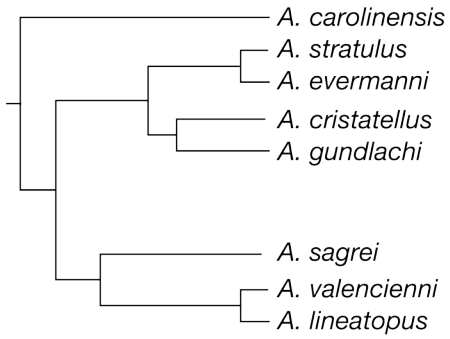


Figure 1. Cladogram of some anole species

48. Based on the data in Table 1, which of the following best explains the observed differences in cold tolerance between brown anoles (*A. sagrei*) and green anoles (*A. carolinensis*) in the United States?

(A) Brown anoles developed a tolerance for cold temperatures and passed on this acquired tolerance.

(B) Brown and green anoles produced a hybrid species that has greater tolerance to cold temperatures.

(C) Green anoles with greater tolerance for cold had greater reproductive success in areas with colder temperatures. ✓

(D) New mutations occurred in brown anoles in response to exposure to colder temperatures.

Practice Exam 1 MCQ

Answer C

Correct. In a population of anoles that showed variation for cold tolerance, individuals with an increased tolerance for colder temperatures would have had an advantage over others in a colder environment. This advantage would have led to increased reproductive success and an increase in the frequency of those alleles permitting increased cold tolerance in future generations.

49. In anoles, small toe pads are better for gripping irregular surfaces such as the forest floor, and large toe pads are better for gripping smooth surfaces such as leaves. Based on the data in Table 1, if brown anoles are introduced into a forest where green anoles currently live, which of the following predictions about the two species is most likely?

(A) Brown anoles will occupy a different area of the forest from green anoles which will minimize competition. ✓

(B) Brown anoles will mostly prey on flying insects, and green anoles will prey on worms and grubs.

(C) Brown anoles and green anoles will evolve to have the same size toe pads because they share the same niche.

(D) Green anoles are better adapted to the forest habitat and will chase away the introduced brown anoles.

Answer A

Correct. Since green anoles have large toe pads, they are likely to occupy a niche on plants. The brown anoles have smaller toe pads than green anoles do, suggesting that they would be less successful on plants.

50. Which of the following is the outgroup in the cladogram shown in Figure 1?

(A) *A. evermanni*

(B) *A. stratulus*

(C) *A. gundlachi*

(D) *A. carolinensis* ✓

Answer D

Correct. Based on the cladogram, *A. carolinensis* is more distantly related to all of the other species in the cladogram than the other species are related to each other.

Practice Exam 1 MCQ

51. Based on Figure 1, which of the following would provide the strongest evidence to support a claim that *A. valencienni* and *A. lineatopus* are the two anole species most closely related to *A. sagrei* ?

(A) Comparing body structures, such as toe pads and limb length

(B) Analyzing homologous DNA sequences among the species ✓

(C) Determining which species occupy a similar niche in Cuba

(D) Observing which species have mating behaviors most similar to those of *A. sagrei*

Answer B

Correct. The rate that mutations appear in the genomes of species is relatively constant, and therefore the number of differences between two species can serve as a molecular clock to measure how long ago species diverged.

52. Based on the cladogram in Figure 1, which of the following conclusions about the evolution of the anole species is most reasonable?

(A) Each species evolved in a different location, and all underwent convergent evolution.

(B) Each species evolved as a result of different selective pressures. ✓

(C) Each species evolved as a result of a genetic bottleneck.

(D) Each species evolved by inheriting random mutations from the common ancestor.

Answer B

Correct. Different populations of anoles likely experienced different selective pressures, which selected for different traits.

Beta thalassemia is a disorder caused by mutations in the HBB gene. Examination of the HBB protein in an individual with beta thalassemia shows that the protein is missing many amino acids at its carboxyl terminus.

53. Which of the following is the most likely explanation for how a mutation in the DNA could result in the loss of the carboxyl terminus of the HBB protein?

Practice Exam 1 MCQ

(A) The mutation keeps the HBB gene wrapped tightly around histones, preventing transcription of the gene.

(B) The mutation changes the promoter sequence of the gene such that different transcription factors initiate transcription of the gene.

(C) The mutation changes a codon in the coding region of the HBB transcript to a stop codon such that translation terminates earlier than it should. ✓

(D) The mutation results in hydrogen bonds between adjacent amino acids instead of covalent bonds, resulting in the production of an unstable protein.

Answer C

Correct. Changing a codon in the coding region to a stop codon would cause the ribosome to stop translating before the polypeptide was fully formed.

A common test for liver function involves sprinkling sulfur powder onto a sample of urine (mostly water with dissolved bodily waste). Sulfur powder sprinkled on a sample from an individual with impaired liver function will sink because the urine contains a high level of bile salts, while the sulfur powder sprinkled on normal urine samples will float.

54. Which of the following best explains why bile salts cause the sulfur powder to sink?

(A) Bile salts decrease the surface tension of the urine sample. ✓

(B) Bile salts increase the water potential of the urine.

(C) Bile salts increase the density of the urine sample.

(D) Bile salts decrease the strength of the covalent bonds within a water molecule.

Answer A

Correct. If bile salts interfere with the hydrogen bonds between surface water molecules, then the surface tension will be reduced and the sulfur power will sink.

Practice Exam 1 MCQ

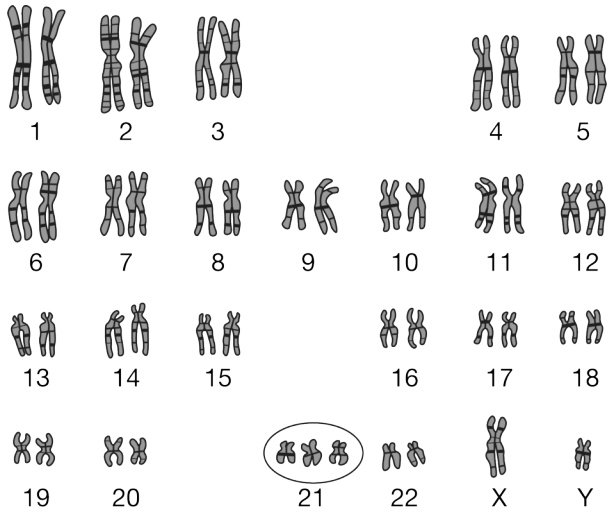


Figure 1. A karyotype obtained from a sample of human cells

A karyotype is a representation of all the metaphase chromosomes in a sample of cells from a particular individual (Figure 1).

55. Which of the following most likely explains how the chromosomes circled in Figure 1 could cause a genetic disorder in the person from whom the cells were obtained?

(A) The extra chromosome causes crowding in the nucleus of the cells and blocks RNA polymerase from binding to and transcribing certain genes.

(B) The extra chromosome will affect the levels of RNA transcribed from certain genes and the amount of protein produced from those genes in each cell. ✓

(C) The cells will not divide and enable growth, because the extra chromosome will interfere with the pairing of homologous chromosomes.

(D) The extra chromosome will cause other chromosomes in the cell to become triploid during future rounds of cell division.

Answer B

Correct. Having an additional copy of all genes on chromosome 21 would result in an increase in the amount of RNA transcribed from those genes. The protein levels for these genes would be higher than is typical.

---

56. Which of the following is the strongest evidence supporting the endosymbiont hypothesis?

Practice Exam 1 MCQ

(A) Mitochondria have their own DNA and divide independently of the cell. ✓

(B) Mitochondria can carry out hydrolytic reactions on organic molecules.

(C) Mitochondria have a highly folded membrane.

(D) Mitochondria are found in both plants and animals.

Answer A

Correct. Existing bacteria all have a DNA genome and the ability to divide. In order for mitochondria to have been free-living bacteria, as suggested by the endosymbiont hypothesis, they would need these traits.

57. Which of the following best explains how the expression of a eukaryotic gene encoding a protein will differ if the gene is expressed in a prokaryotic cell instead of in a eukaryotic cell?

(A) No transcript will be made, because eukaryotic DNA cannot be transcribed by prokaryotic RNA polymerase.

(B) The protein will have a different sequence of amino acids, because prokaryotes use a different genetic code.

(C) The protein will be made but will not function, because prokaryotes cannot remove introns. ✓

(D) The protein will not be made, because prokaryotes lack the ribosomes necessary for translation.

Answer C

Correct. The prokaryotic cell will be able to produce a protein from the transcript, but it will have a different amino acid sequence and therefore a different function, because only eukaryotes remove introns from mRNA transcripts.

High concentrations of cadmium in the water negatively affect steelhead trout. In a river community, steelhead trout are part of the trophic structure shown in Figure 1.

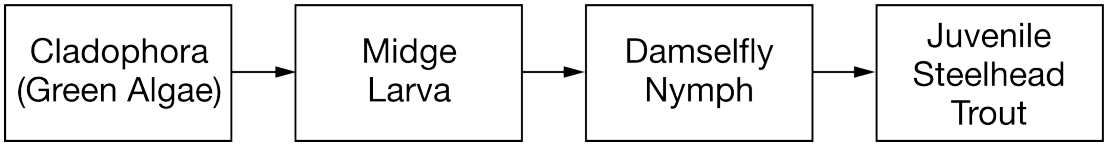


Figure 1. The trophic structure of a river community

Practice Exam 1 MCQ

58. Which of the following is the most likely immediate effect of an increase in runoff containing cadmium on the trophic structure of the river community?

(A) The population of Cladophora will decrease, resulting in an increase in the trout population.

(B) There will be a large decrease in the trout population, resulting in an increase in damselfly nymphs. ✓

(C) Increased stream volume will provide more area for the trout to reproduce, causing a large increase in the population of algae.

(D) The population of trout will decrease because the population of damselfly nymphs will decline.

Answer B

Correct. Increased cadmium in the river will negatively impact the steelhead trout. With fewer trout feeding on the damselfly nymphs, the nymph population will increase.

Poaching is the illegal hunting or capturing of wild animals. Both male and female African elephants have tusks that are a significant source of ivory used for ornaments and jewelry.

59. Assuming that the size of the tusks is heritable, which of the following is the most likely long-term effect of poachers targeting African elephants with large tusks?

(A) An increase in average tusk size because of the need to provide more ivory for the poaching hunters

(B) A decrease in average tusk size because elephants with large tusks are less likely to survive to reproduce ✓

(C) An increase in average tusk size to provide the elephants with a defensive mechanism against poachers

(D) A decrease in average tusk size to reduce the energy investment in the tusks

Answer B

Correct. If poachers targeted the elephants with the largest tusks, then more elephants with small tusks would survive to reproduce and future generations would see an increase in the number of small-tusked elephants.

Practice Exam 1 MCQ

A scientist adds a chemical to a culture of dividing cells in order to disrupt DNA replication. The replicated DNA produced by the cells is double-stranded, but sections of it lack covalent bonds between adjacent nucleotides (Figure 1).

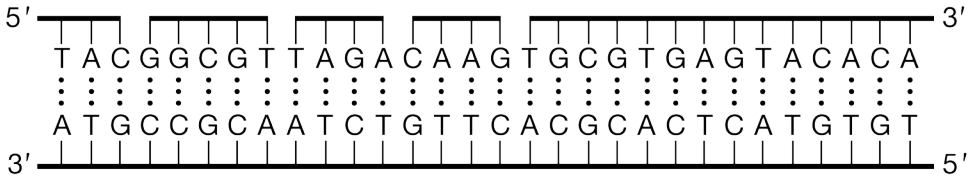


Figure 1. Replicated DNA produced after a chemical is introduced

60. Which of the following claims is best supported by the data?

(A) The chemical prevents the formation of RNA primers.

(B) The chemical inhibits DNA ligase. ✓

(C) The chemical blocks DNA polymerase.

(D) The chemical disrupts hydrogen bonding.

Answer B

Correct. The missing bonds in the backbone of the lagging strand of DNA indicate that DNA ligase was not able to join the fragments.

Practice Exam 1 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph hide_paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Electrons from the oxidation of glucose and other nutrients by glycolysis and the Krebs cycle are added to the electron-carrying compound NAD^+ to form NADH. To meet the continuous requirement for NAD^+ by cells, NADH is oxidized back to NAD^+ through the electron transport chain, in the presence $(+)$ of oxygen, or through fermentation, in the absence $(-)$ of oxygen (Figure 1).

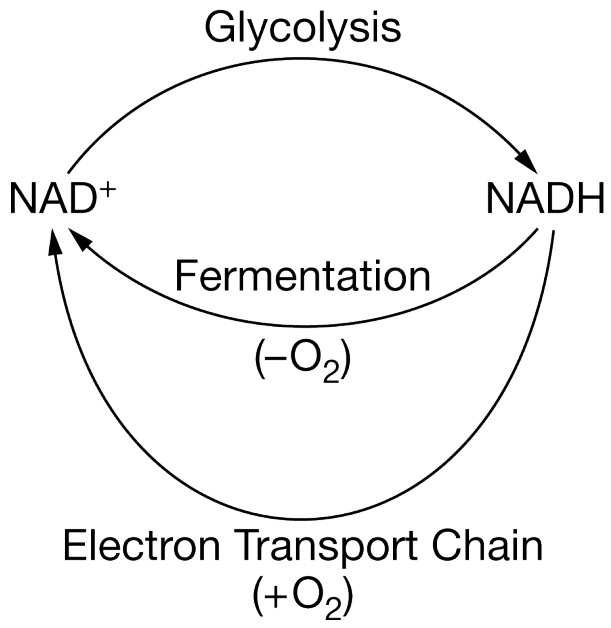


Figure 1. A simplified model of metabolism and the recycling of NAD^+

MT-ND5 is a mitochondrial gene that encodes a subunit of NADH dehydrogenase, the enzyme that catalyzes the initial oxidation of its substrate NADH to NAD^+ and H^+ in the electron transport chain of mitochondria. A mutation in *MT-ND5* is associated with a rare genetic disorder that results in a buildup of lactic acid in the body. A researcher hypothesizes that the mutated NADH dehydrogenase has decreased activity but is not completely nonfunctioning and that, by increasing the pool of NADH in cells, the activity of NADH dehydrogenase will increase.

To test this idea, the researcher treated a group of individuals with this disorder with a vitamin that is similar to NADH and measured the concentration of NAD^+ and lactic acid in the blood over the course of 20 weeks. The results from a representative individual are shown in Figure 2.

Practice Exam 1 FRQ

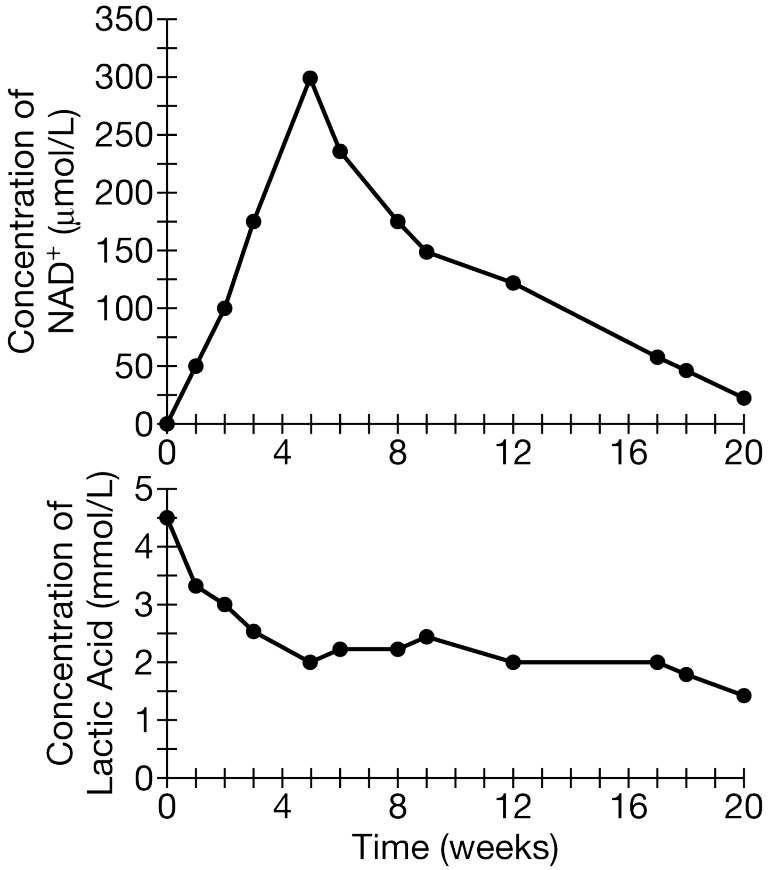


Figure 2. The concentration of NAD^+ (top) and lactic acid (bottom) in the blood of a representative treated individual

1. (a) **Describe** the pattern of inheritance that is most likely associated with a mutation in the *MT-ND5* gene. **Explain** why individuals are not typically heterozygous with respect to mitochondrial genes.

(b) **Identify** a dependent variable measured in the researcher's experiment. **Identify** one control that the researcher could use to improve the validity of the experiment. **Justify** the researcher analyzing blood samples at many intermediate time points instead of at only the beginning and the end of the 20-week period.

(c) **Describe** the relationship between the concentration of NAD^+ in the blood and the concentration of lactic acid in the blood during the first 5 weeks of treatment with the vitamin. Based on Figure 2, **calculate** the average rate of change in blood NAD^+ concentrations from week 5 to week 17.

(d) The researcher performed a follow-up experiment to measure the rate of oxygen consumption by muscle and brain cells. **Predict** the effect of the *MT-ND5* mutation on the rate of oxygen consumption in muscle and brain cells. **Justify** your prediction. The researcher had hypothesized that the addition of the vitamin that is similar in structure to $NADH$ would increase the activity of the mutated $NADH$ dehydrogenase enzyme in individuals with the disorder. **Explain** how the vitamin most likely increased the activity of the enzyme.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Practice Exam 1 FRQ

0	1 ✓

The response includes one of the following.

- [ ] A description that the mutant allele is only inherited from the maternal parent.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following.

- An identification of an acceptable dependent variable. Acceptable dependent variables include the concentration of lactic acid in the blood and the concentration of NAD^+ in the blood.
- An identification of the control as measuring NAD^+ and lactic acid levels in a group of other affected individuals treated with placebo.
- The justification that by collecting more data, the researchers will see a more accurate trend.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all three of the following.

- A description that the concentration of NAD^+ increased and the concentration of lactic acid decreased.
- $125(\mu\mathrm{mol/L})$
- A prediction that the mutation will decrease the rate of oxygen consumption by the electron transport chain.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Practice Exam 1 FRQ

0	1	2 ✓

The response includes all of the following.

- [ ] The justification that the mutation likely inhibits the rate of oxygen consumption because the electron transport chain will be less efficient, so less oxygen will be required as a terminal electron acceptor.
- [ ] The explanation that the enzyme had decreased activity but still functioned. Increasing the amount of substrate should increase the amount of product produced. Because the vitamin was similar to NADH, it could bind to the active site of the enzyme and effectively increase the substrate concentration.

Practice Exam 1 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Atherosclerosis is a disease that results when certain cells and proteins of an individual's body adhere to and damage blood vessels, especially those around the heart. Researchers continue to look for improved ways to treat individuals with the disease. Data obtained from earlier experiments suggested that it might be possible to reduce damage to the blood vessels by increasing the expression of FXR1, an RNA-binding protein in the muscle cells that form the outer surface of the blood vessels. Additional data suggested that IL-19, a protein secreted by certain white blood cells, might regulate expression of the *FXR1* gene.

To investigate the regulation of the *FXR1* gene by IL-19, a researcher added IL-19 to vessel-lining muscle cells growing in the lab. The researcher measured the amount of FXR1 protein produced by the cells over a period of 48 hours in the presence of IL-19. The researcher then calculated the relative amount of FXR1 present at each time point compared with the amount of FXR1 at Time 0, when IL-19 was first added to the cells. Data from three replicate experiments are shown in Table 1.

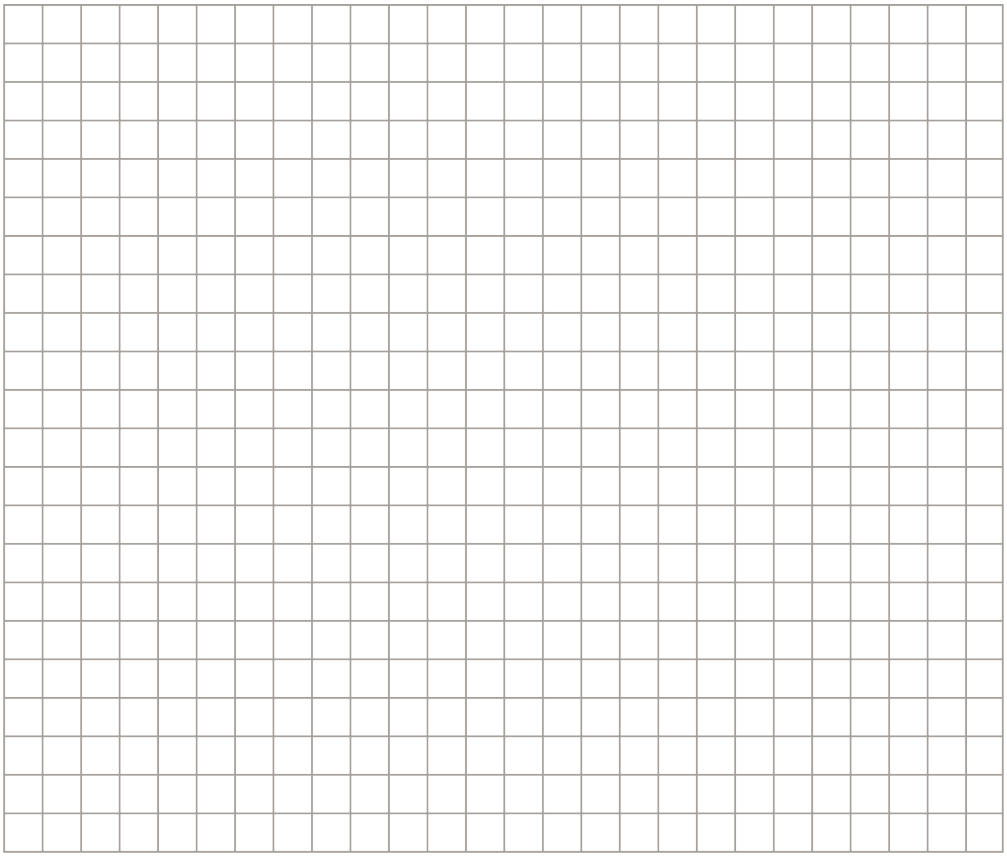
TABLE 1. RELATIVE AMOUNT OF FXR1 PRODUCED BY CELLS IN RESPONSE TO IL-19

Time in the Presence of IL-19 (h)	Relative Amount of FXR1 Protein $\pm 2\mathrm{SE}_{\bar{x}}$
0	0.98 ± 0.070
4	1.70 ± 0.220
8	1.60 ± 0.275
16	3.10 ± 0.800
24	2.15 ± 0.405
48	1.60 ± 0.520

Practice Exam 1 FRQ

2. (a) **Describe** how amino acids are categorized by their chemical properties. **Explain** how a change in the amino acid sequence of the FXR1 protein could decrease the ability of the protein to bind to RNA.

(b) Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data shown in Table 1. **Determine** whether there is a statistical difference in the amount of FXR1 protein produced by the cells after 16 and 24 hours in the presence of IL-19.



(c) Based on the data for the 48-hour period, **describe** the effect of IL-19 on *FXR1* gene expression.

(d) The researcher hypothesizes that the *FXR1* gene codes for a protein that binds to mRNAs that encode some of the proteins that damage arteries. Individuals with a particular mutation of the *FXR1* gene tend to have high levels of these proteins. Based on this information, **predict** how the FXR1 protein most likely interacts with the mRNAs.

Part A

Select a point value to view scoring criteria, solutions, **and/or** examples and to score the response.

0	1 ✓

Practice Exam 1 FRQ

The response includes only one of the following.

- [ ] The description that, based on their R groups, amino acids are categorized as hydrophobic, hydrophilic, or ionic (acidic or basic)

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following.

- Correct axes and labeling on the graph
- Correctly plotted means on the graph
- Correctly plotted error bars on the graph
- The determination that, based on the overlap of error bars, there is no statistical difference

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following.

- A description that IL-19 causes a small increase in expression of the $FXR1$ gene.
- $5.8[(1.7 - 1.6)/1.7]$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following.

- A prediction that the FXR1 protein must inhibit translation of the mRNAs or degrade the mRNAs.

Practice Exam 1 FRQ

- An explanation that the change in amino acid sequence changes the shape of a protein, so that RNA-binding sites formed by the interactions of certain amino acids many change or be lost entirely.

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Researchers grew populations of identical *Escherichia coli* ($E. coli$) bacteria in a growth medium that contained a low concentration of glucose and a high concentration of citrate, a substance that is not typically consumed by $E. coli$. For thousands of generations, the bacteria used only glucose as an energy source and grew relatively slowly and to a low density because of the low concentration of glucose. After about $30{,}000$ generations, one population emerged that began to rapidly grow to a much higher density. The researchers hypothesized that the bacteria evolved the ability to use citrate as an energy source and referred to them as Cit^+. To test the hypothesis, the researchers grew separate populations of the Cit^+ bacteria and bacteria from the original population (Cit^-) in a growth medium that contained only citrate.

3. (a) **Describe** one outcome that would demonstrate that a given population has evolved.

(b) **Identify** the dependent variable measured in the experiments.

(c) **Predict** the results obtained by the researchers when they grew the Cit^+ and Cit^- bacteria in the medium that contained only citrate.

(d) The researchers claim that the Cit^+ mutation increases the fitness of the bacteria. **Provide reasoning** to support the claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response describes that there is a change in allele frequency.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response identifies one of the following as the independent variable.

Practice Exam 1 FRQ

- [ ] Growth rate of the bacteria
- [ ] Density of the bacteria

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response predicts that only the Cit^+ bacteria grew.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the reasoning that the Cit^+ bacteria have an energy source unavailable to other strains and could grow and reproduce at a more rapid rate due to reduced competition for resources.

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The Tasmanian devil (*Sarcophilus harrisii*) is a species of mammal whose only wild population is native to the isolated island of Tasmania, off the coast of Australia. Tasmanian devils have a very low level of genetic diversity across the entire species. Researchers hypothesize this lack of diversity is the result of several genetic bottlenecks in the species' history.

In 1996 a new disease called devil facial tumor disease (DFTD) appeared in the wild Tasmanian devil population. DFTD can spread from one animal to another and causes the death of most affected animals. The impact of DFTD on the Tasmanian devil population has resulted in a decrease in the total population of more than 85% over the last two decades.

Practice Exam 1 FRQ

4. (a) **Describe** the process that maintained a stable Tasmanian devil population size before the appearance of DFTD in 1996.

(b) **Explain** how the huge reduction of the Tasmanian devil population since 1996 affects the susceptibility of the current population to new diseases in comparison with the susceptibility of the population before 1996.

(c) Tasmanian devils are top predators and are considered a keystone species in their community. **Predict** the effect of the rapid reduction of the Tasmanian devil population on the rest of the community.

(d) **Justify** the prediction of part (c).

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response describes that the Tasmanian devil population would have been maintained at a stable population size by density-dependent factors.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response explains that the population underwent a genetic bottleneck and thus has reduced genetic diversity, which makes the current population more likely to be susceptible to a new disease than it would have been in the past.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes one of the following predictions.

- ☐ The community will collapse or be destabilized.
- ☐ The population sizes of some species will increase and others will decrease.

Practice Exam 1 FRQ

- [ ] Species diversity of the community will decrease.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the justification that the prey species of the Tasmanian devil would have increased in number as the Tasmanian devil population decreased, likely leading to population decreases of the food sources for these prey species.

Practice Exam 1 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph hide_paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The p53 protein is a transcription factor that regulates a cell's progression through the cell cycle after DNA is damaged by ultraviolet (UV) radiation. The p53 pathway is shown in Figure 1.

Figure 1. The role of p53 in Regulating Cell Cycle Progression in Response to DNA Damage

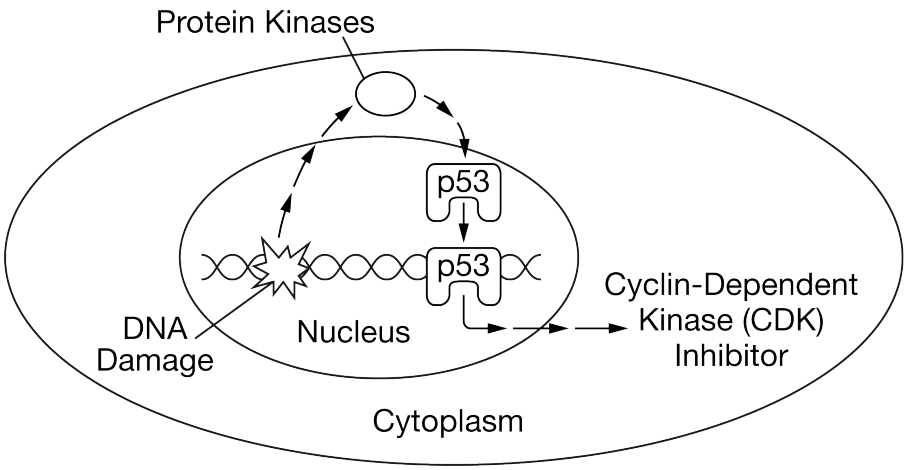
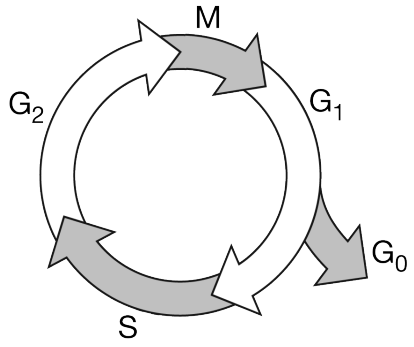


Figure 2. The Cell Cycle



Practice Exam 1 FRQ

5. (a) A skin cell completes one round of the cell cycle. **Describe** the products.

(b) Based on Figure 1, **explain** how p53 regulates the cell cycle in the presence of damaged DNA.

(c) Based on Figure 2, **identify** the phase during which the replication of damaged DNA would occur (G_1, S, or G_2).

(d) Based on Figure 1, **explain** how a mutation to p53 may lead to an increased risk of cancer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response describes the products as two diploid cells with identical sets of DNA.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response explains that p53 stimulates expression of the CDK inhibitor, preventing progression of the cell cycle.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates the phase during which the replication of damaged DNA would occur with an X placed on any portion of S phase in Figure 2.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Practice Exam 1 FRQ

0	✓ 1

The response explains that a mutation may prevent activation of p53, which will then not stimulate the CDK inhibitor, allowing the cell cycle to progress through the cell cycle with damaged or mutated DNA.

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The human kidney filters the blood to retain useful substances and remove waste products and excess water as urine. *SLC3A1* and *SLC7A9* are two different genes that each encode one part of a transporter protein complex typically found in the plasma membrane of human kidney cells. This transporter complex moves certain amino acids across the plasma membrane of the cells. Mutations in both the *SLC3A1* and *SLC7A9* genes are associated with increased levels of the polar amino acid cysteine in the urine, a condition called cystinuria. A researcher identified seven individuals with a family history of cystinuria. The researcher measured the concentration of cysteine in the urine of the seven individuals and sequenced their *SLC3A1* and *SLC7A9* genes to determine whether they carry functional (+) or mutant (−) alleles of genes. The results are shown in Table 1.

TABLE 1. CHARACTERISTICS OF SEVEN INDIVIDUALS WHO CARRY AT LEAST ONE MUTANT ALLELE

+ indicates the presence of a functional allele and − indicates the presence of a mutant allele.

Individual	Concentration of Cysteine in Urine Relative to Normal Values	*SLC3A1* Alleles	*SLC7A9* Alleles
1	32.2	−/−	+/−
2	29.8	+/−	−/−
3	11.0	−/−	+/+
4	8.8	+/+	−/−
5	7.7	−/−	+/+
6	2.3	+/+	+/−
7	0.5	+/−	+/+

Practice Exam 1 FRQ

6. (a) **Identify** the individual who most likely exhibits symptoms of cystinuria.

(b) **Describe** the relationship between the total number of mutant alleles in an individual and the concentration of cysteine in the urine.

(c) **Evaluate** the hypothesis that mutations in $SLC7A9$ have a greater effect on the transport of cysteine across the plasma membrane of kidney cells than do mutations in $SLC3A1$.

(d) **Explain** how the data support the claim that cysteine is a large polar molecule.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response identifies individual 1 as the individual who most likely exhibits symptoms of cystinuria.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response describes that the more mutant alleles an individual has, the higher the concentration of cysteine in the urine.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates both of the following.

· The hypothesis is not supported by the data.

· While the individual with a single mutant allele for $SLC7A9$ has a higher concentration than the individual with a single mutant allele for $SLC3A1$, the individual with two mutated $SLC7A9$ alleles does not have a higher concentration than one of the individuals with two mutant $SLC3A1$ alleles.

Practice Exam 1 FRQ

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response explains that cysteine builds up in the urine when the transporter proteins are mutated, which suggests that cysteine requires specific channel proteins through which to move, as do most large polar molecules.

Practice Exam 2 MCQ

1. Humans produce sweat as a cooling mechanism to maintain a stable internal temperature. Which of the following best explains how the properties of water contribute to this physiological process?

(A) The high specific heat capacity of water allows the body to absorb a large amount of excess heat energy.

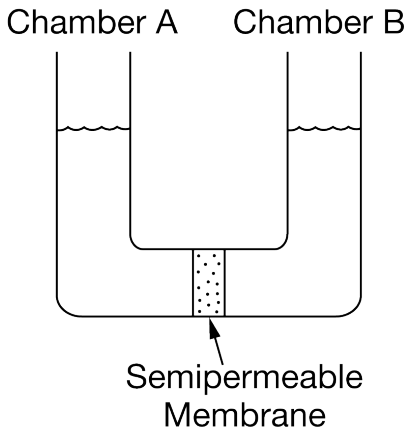
(B) The high heat of vaporization of water allows the body to remove excess heat through a phase change of water from liquid to gas. ✓

(C) The high surface tension of water contributes to the physical process by which water leaves the body.

(D) The high melting temperature of water allows the body to remove excess heat through a phase change of water from solid to liquid.

Answer B

Correct. Water has a high heat of vaporization. Sweat produced by the body decreases the temperature of the body by absorbing a large quantity of heat energy during the liquid to gas phase change.



A student placed a semipermeable membrane inside a U-shaped channel with two chambers, as shown. The membrane permits the movement of water but not salt. The student wants to vary the rate of osmosis that occurs across the membrane.

2. Which of the following experimental designs will result in the fastest net rate of water movement into chamber A?

(A) Placing salt water in chamber A and distilled water in chamber B ✓

(B) Placing distilled water in both chambers

(C) Placing distilled water in chamber A and salt water in chamber B

(D) Placing salt water in both chambers

Practice Exam 2 MCQ

Answer A

Correct. Distilled water from chamber B will move to chamber A because chamber A has a higher solute concentration.

---

3. Which of the following best describes the role of water in photosynthesis?

(A) Water is the only source of protons for the formation of a proton gradient.

(B) Water molecules donate electrons to the electron transport chain. ✓

(C) Water molecules combine with stored carbon molecules to produce glucose.

(D) Water is the terminal electron acceptor for electrons that pass through the electron transport chain.

Answer B

Correct. Electrons are released from water molecules and donated to photosystem II.

4. What evolutionary advantage does compartmentalization of core metabolic processes offer eukaryotes?

(A) Evolution of the mitochondria allowed eukaryotes to perform respiration.

(B) With the evolution of mitochondria in eukaryotes, the Krebs cycle and electron transport chain also evolved.

(C) Evolution of a nucleus in eukaryotes separates the processes of transcription and translation and they can be regulated separately. ✓

(D) A nucleus in bacteria provides separation of respiration from transcription.

Answer C

Correct. The evolution of a nuclear membrane that contains genetic material allowed for cells to separate the copying of genetic material and the production of proteins.

Practice Exam 2 MCQ

Pyruvate dehydrogenase is an enzyme that converts pyruvate to acetyl-CoA. Acetyl-CoA is further metabolized in the Krebs cycle. A researcher measured the accumulation of acetyl-CoA in a reaction containing pyruvate and pyruvate dehydrogenase under several different conditions (Figure 1).

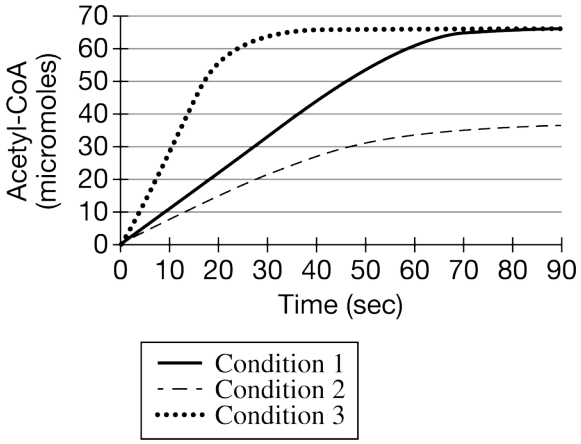


Figure 1. Accumulation of acetyl-CoA under different conditions

5. Which of the following best describes the cellular location where pyruvate dehydrogenase is most likely active?

(A) The cytosol

(B) The lysosomes

(C) The nucleus

(D) The mitochondrial matrix ✓

Answer D

Correct. Pyruvate dehydrogenase plays a role in the Krebs cycle, which occurs in the mitochondrial matrix.

6. The maximum production rate of acetyl-CoA under condition 1 is closest to which of the following?

(A) 1 micromole/sec ✓

(B) 24 micromoles/sec

(C) 35 micromoles/sec

(D) 65 micromoles/sec

Practice Exam 2 MCQ

Answer A

Correct. This value is closest to the steepest slope of the curve under condition 1.

7. Which of the following observations provides the best evidence that acetyl-CoA negatively regulates pyruvate dehydrogenase activity?

(A) The rate of the pyruvate dehydrogenase–catalyzed reaction is slower in the presence of a higher concentration of acetyl-CoA. ✓

(B) The gene that encodes pyruvate dehydrogenase is transcribed when excess acetyl-CoA is detected.

(C) The accumulation of acetyl-CoA stops after 70 seconds, regardless of the reaction mixture.

(D) Acetyl-CoA is continuously broken down in the Krebs cycle.

Answer A

Correct. This observation is an example of a negative feedback mechanism.

8. Pyruvate dehydrogenase deficiency is a genetic disease most commonly linked to a mutation in the α-subunit of the mitochondrial enzyme that causes the enzyme to cease functioning. As a result of the mutation, affected individuals build up dangerous amounts of lactic acid. Which of the following best explains the buildup of lactic acid in individuals with the mutation?

(A) Cells use lactic acid to shunt electrons from pyruvate to the electron transport chain in the mitochondria.

(B) Cells undergo glycolysis because there is a buildup of pyruvate in affected individuals.

(C) Cells cannot transport pyruvate to the mitochondria in the absence of pyruvate dehydrogenase activity, so the pyruvate is broken down to lactic acid and ethanol.

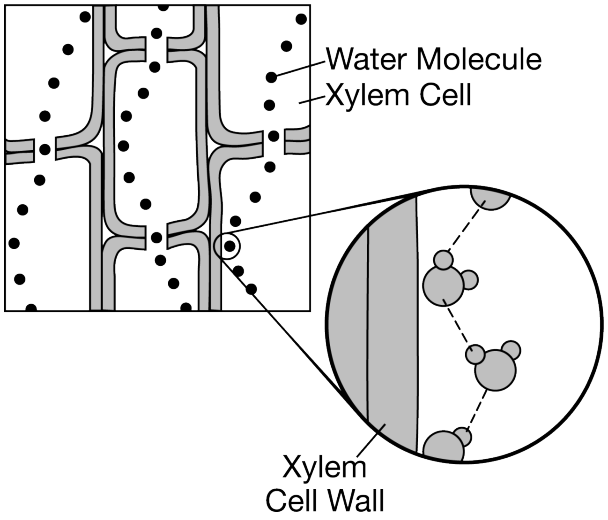
(D) Cells undergo fermentation because pyruvate cannot be metabolized to proceed into the Krebs cycle. ✓

Answer D

Correct. Pyruvate that cannot be broken down by the Krebs cycle will be further metabolized by fermentation.

Practice Exam 2 MCQ

The diagram shows how water can adhere to the xylem in the stems of plants, which contributes to water movement in the plant.



9. Which of the following best explains how water is able to move upward from the roots of a plant, through its xylem in the stem, and out to the leaves?

(A) Water is polar, and the walls of the xylem are nonpolar. Water molecules have the ability to form hydrogen bonds with one another but not with the xylem walls.

(B) Water is nonpolar, and the walls of the xylem are polar. Water molecules are able to form hydrogen bonds with the xylem walls, and they are pulled up the xylem.

(C) Water and the xylem are both nonpolar. Water molecules have the ability to form hydrogen bonds with one another but not with the xylem walls.

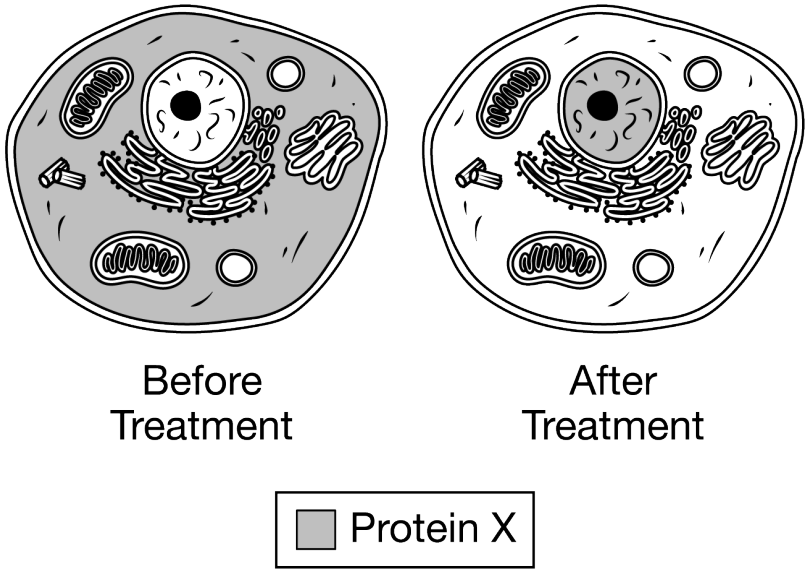
(D) Water and the xylem are both polar. Water molecules have the ability to form hydrogen bonds with each other and with the walls of the xylem. ✓

Answer D

Correct. Water has a high surface tension as a result of the hydrogen bonds that water molecules form between each other because both the xylem and the water are polar.

Practice Exam 2 MCQ

Protein X activates gene expression only in cells exposed to a specific signaling molecule. In a study, researchers determined the intracellular location of Protein X in cultured cells both before and after exposing the cells to the signaling molecule. The results of the study are shown in the diagram.



10. Based on the results, which of the following best describes what Protein X is?

(A) Protein X is an RNA splicing enzyme.

(B) Protein X is a cell membrane receptor protein.

(C) Protein X is a transcription factor. ✓

(D) Protein X is a hormone.

Answer C

Correct. The movement of Protein X from the cytoplasm to the nucleus indicates that it is a transcription factor.

The hormone prolactin has varying effects in many different animal species. All vertebrates produce prolactin, which is involved in signal transduction pathways. In mammals, prolactin stimulates the production of milk in mammary glands. In fish, prolactin plays an important role in osmoregulation. In birds, prolactin is involved in lipid metabolism.

11. Which of the following best explains the presence of prolactin in various vertebrate species?

Practice Exam 2 MCQ

(A) Though all vertebrates produce prolactin, its varied uses indicate it arose as a result of convergent evolution and not as a result of common ancestry.

(B) Prolactin is a homologous hormone because it has a common origin but different functions in various species. ✓

(C) Prolactin will bind only to intracellular receptors in animal species with phospholipid bilayers, so its effects are varied in different species.

(D) Because of different receptors activating different signal transduction pathways within the same species, it is likely that prolactin production is a trait with highly selective pressure.

Answer B

Correct. Prolactin is not highly conserved, because it has multiple functions in different organisms. Therefore, its presence in these organisms indicates a common origin.

Practice Exam 2 MCQ

Certain reef-building corals contain photosynthetic, symbiotic algae that have the ability to make dimethylsulphoniopropionate (DMSP), a chemical involved in the marine sulfur cycle. DMSP is released into the surrounding water, where it is converted to the gas dimethyl sulfide (DMS) by microorganisms and enters the atmosphere. Once in the atmosphere, it triggers the formation of sulfate aerosols, which induce cloud formation and block sunlight from heating up the water.

The symbiotic algae produce DMSP when they are stressed by a high water temperature. If water temperature is too high, corals will expel the symbiotic algae that produce DMSP. Researchers measured the amount of DMSP produced by juvenile and adult coral and their symbionts under normal and thermally stressed conditions. The data are shown in the graphs in Figure 1.

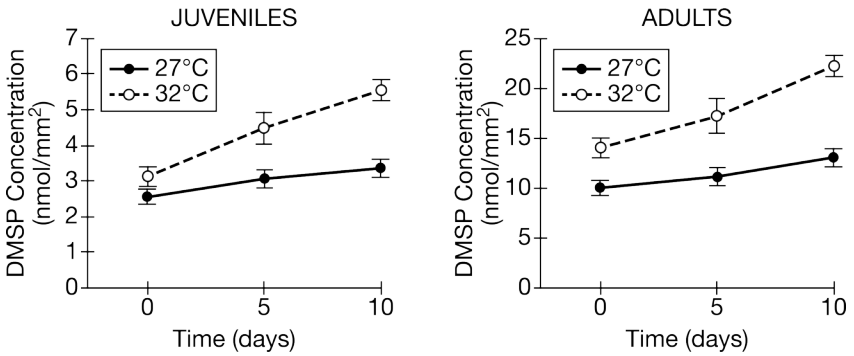


Figure 1: DMSP concentration in juvenile and adult corals and their symbionts in normal and thermally-stressed conditions. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

The researchers also measured the density of the symbiont as well as the photosynthetic yield in adult corals at the two temperatures. Photosynthetic yield is an index measure of energy output compared to sunlight energy input in which larger photosynthetic yield values represent photosynthetic organisms producing more energy.

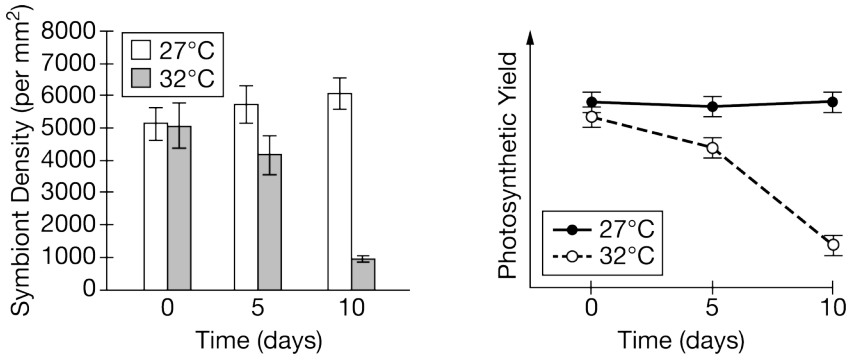


Figure 2: Variation in symbiont density and photosynthetic yield in adult corals grown in normal and thermally-stressed conditions. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

12. Which of the following best describes the production of DMSP by coral and coral symbionts?

Practice Exam 2 MCQ

(A) A negative feedback mechanism that increases the environmental change

(B) A negative feedback mechanism that reverses the environmental change ✓

(C) A positive feedback mechanism that increases the environmental change

(D) A positive feedback mechanism that reverses the environmental change

Answer B

Correct. This is a negative feedback mechanism in which DMSP is produced to prevent, not encourage, further environmental change as a result of increasing water temperatures.

13. Which of the following best describes the effect of temperature on corals' ability to produce DMSP as shown in Figure 1?

(A) Both juvenile and adult corals produce less DMSP at $27^\circ C$ than at $32^\circ C$. ✓

(B) Both juvenile and adult corals produce less DMSP at $32^\circ C$ than at $27^\circ C$.

(C) The amount of DMSP produced over time increases at $32^\circ C$ in juveniles only.

(D) The amount of DMSP produced over time increases at $32^\circ C$ in adults only.

Answer A

Correct. The graphs show that both age groups of corals produce less DMSP at $27^\circ C$ than they do at $32^\circ C$.

14. Which of the following best describes the difference between the total amount of DMSP produced by adults compared to juveniles at the start of the $32^\circ C$ trial?

(A) Adult corals produced 3 times more DMSP than juveniles produced.

(B) Adult corals produced 3 times less DMSP than juveniles produced.

(C) Adult corals produced 5 times more DMSP than juveniles produced. ✓

(D) Adult corals produced 5 times less DMSP than juveniles produced.

Answer C

Correct. The DMSP produced by adult corals at $32^\circ C$ was approximately $14\ \text{nmol}/\text{mm}^2$, and the DMSP produced by juveniles was approximately $3\ \text{nmol}/\text{mm}^2$.

Practice Exam 2 MCQ

15. In addition to the effect of temperature on DMSP produced by corals and their symbionts, which of the following relationships is also being considered in this experiment?

(A) Effect of varying light levels and coral species

(B) Effect of additional DMSP produced by symbionts and the corals' age ✓

(C) Effect of age and varying light levels

(D) Effect of coral species and additional DMSP produced by symbionts

Answer B

Correct. This experiment is also measuring the difference between the amount of DMSP produced by corals versus algae and corals together. It is also looking at the difference in DMSP produced by different-aged corals.

16. Which of the following best describes the scientists' findings concerning the density of symbionts presented in Figure 2 ?

(A) The symbiont density at $32^{\circ}C$ on day 5 was less than the density on day 0 of the experiment.

(B) The symbiont density at $27^{\circ}C$ on day 0 was less than the density on day 5 of the experiment.

(C) The symbiont density at $32^{\circ}C$ was different from the density at $27^{\circ}C$ on days 5 and 10 of the experiment. ✓

(D) The symbiont density at $27^{\circ}C$ was higher than the density at $32^{\circ}C$ for the entire length of the experiment.

Answer C

Correct. The error bars for days 5 and 10 do not overlap between temperature treatments, indicating that they are statistically different.

17. Which of the following best explains the result of adult corals being exposed to elevated temperatures for extended periods?

(A) They are able to obtain more energy from their symbionts because the algae are receiving more light.

(B) They are able to obtain more energy from their symbionts because the efficiency increases slightly over time.

(C) They are able to obtain less energy from their symbionts because the algae have been expelled. ✓

(D) They are able to obtain less energy from their symbionts because more DMSP is being produced at lower temperatures.

Practice Exam 2 MCQ

Answer C

Correct. The corals and algae have less photosynthetic ability at higher temperatures because the algae density has been reduced.

A particular gene has two alleles, a dominant allele A and a recessive allele a. The frequency of allele A is 0.55.

18. If the population is in Hardy-Weinberg equilibrium with respect to the gene, then what is the expected frequency of genotype Aa?

(A) 0.203

(B) 0.303

(C) 0.405

(D) 0.495 ✓

Answer D

Correct. This option correctly calculates the heterozygote frequency from the dominant allele frequency.

Practice Exam 2 MCQ

A particular genetic disorder results from a single change in the amino acid sequence coded for in a gene. Parts of the sequence in normal and mutated genes are shown below.

Normal: TAC CTC GTG GAC TGA GGT CTC

Mutated: TAC CTC GTG GAC TGA GGT CAC

First Base	Second Base: U	Second Base: C	Second Base: A	Second Base: G	Third Base
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	Stop	Stop	A
	Leu	Ser	Stop	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

19. Based on the codon chart, which of the following amino acid changes is most likely found in the mutated protein?

(A) Glu → Val ✓

(B) Val → Glu

(C) Glu → Pro

(D) Pro → Val

Answer A

Correct. The change in the gene code from a T to an A results in an amino acid change from Glu to Val.

Practice Exam 2 MCQ

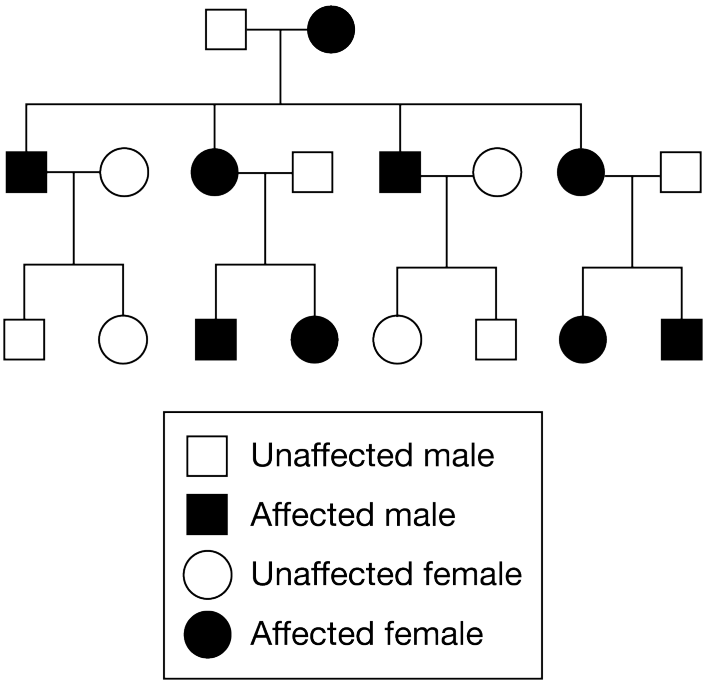


Figure 1. A pedigree of three generations of a family that have a high frequency of a particular genetic condition

20. Figure 1 shows the inheritance of a particular genetic condition in three generations of one family. Which of the following best explains the observed pattern of inheritance?

(A) The condition is passed randomly because of the independent assortment of chromosomes.

(B) The condition is passed from fathers to sons via a Y-linked gene.

(C) The condition is passed from mothers to sons via an X-linked gene.

(D) The condition is passed from mothers to offspring via a mitochondrial gene. ✓

Answer D

Correct. This condition is inherited by all children of affected mothers, regardless of sex, which indicates a mitochondrial inheritance pattern.

---

21. Oncogenes are genes that can cause tumor formation as a result of a particular mutation. Which of the following potential therapies would be most effective at preventing the expression of an oncogene?

Practice Exam 2 MCQ

(A) Reducing the number of ribosomes in the cell to prevent the creation of the oncogene's proteins

(B) Blocking membrane-bound receptors of transcription factors

(C) Introducing a chemical that binds to transcription factors associated with the oncogene's promoter ✓

(D) Producing additional transcription factors for tumor suppressor genes in the cell

Answer C

Correct. If a transcription factor is bound to a chemical, it cannot bind to the oncogene promoter, thus preventing gene expression that would result in tumor formation.

22. Ultraviolet (UV) radiation can damage DNA by breaking weak bonds. Which of the following best explains how this occurs?

(A) UV radiation disrupts the double helix structure by breaking the covalent bonds between the nitrogenous base pairs.

(B) UV radiation disrupts the double helix structure by breaking the hydrogen bonds between the nitrogenous base pairs. ✓

(C) UV radiation is able to break DNA strands in two by breaking covalent bonds between the sugar-phosphate backbone molecules.

(D) UV radiation is able to break DNA strands in two by breaking hydrogen bonds between the sugar-phosphate backbone molecules.

Answer B

Correct. Hydrogen bonds, like those found between nitrogenous base pairs, are weaker than covalent bonds and are easily broken by UV radiation.

Practice Exam 2 MCQ

Scientists are interested in determining the evolution of seven lizard species found on different islands of the Canary Island group. They isolated DNA from individuals of each species and sequenced the mitochondrial gene that encodes cytochrome *b*. The numbers of genetic differences between species are shown in the table below.

	Species A	Species B	Species C	Species D	Species E	Species F	Species G
Species A							
Species B	34						
Species C	39	23					
Species D	38	21	6				
Species E	38	17	8	4			
Species F	43	22	17	17	13		
Species G	47	26	17	19	15	2	

23. Based on the data in the table, which of the following lizard species are most closely related?

(A) Species C and species B

(B) Species E and species D ✓

(C) Species F and species B

(D) Species G and species A

Answer B

Correct. These species have the fewest mitochondrial genetic differences.

Practice Exam 2 MCQ

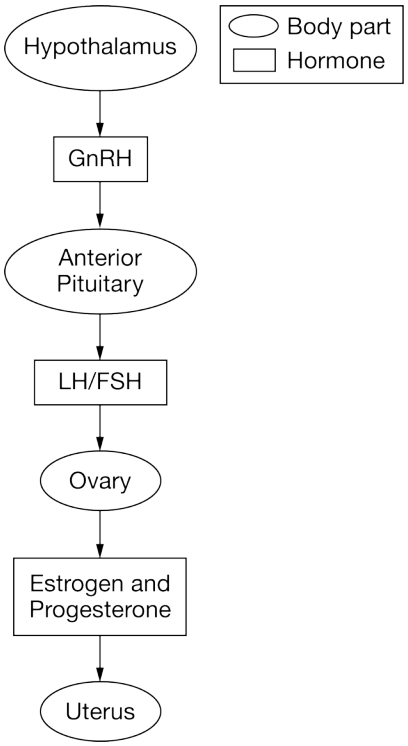


Figure 1. A model of an endocrine signaling pathway showing involved body parts and hormones. GnRH = gonadotropin-releasing hormone, LH = luteinizing hormone, and FSH = follicle-stimulating hormone.

Figure 1 shows a model of the endocrine signaling pathway that regulates ovulation.

24. Which of the following observations would provide evidence of a positive feedback mechanism in this system?

(A) Estrogen from the ovaries inhibits the release of GnRH from the hypothalamus.

(B) Progesterone from the ovaries stimulates the thickening of the uterine lining.

(C) Progesterone from the ovaries inhibits the release of LH and FSH from the anterior pituitary.

(D) Estrogen from the ovaries stimulates the hypothalamus and anterior pituitary to secrete more GnRH, LH, and FSH. ✓

Answer D

Correct. This choice describes an observation in which the production of estrogen stimulates systems at earlier stages in the signaling pathway to produce hormones that further stimulate the ovaries.

Practice Exam 2 MCQ

Dichlorophenolindophenol (DCPIP) is a chemical dye. When DCPIP is chemically reduced, it changes color from blue to clear. DCPIP can be used as an electron acceptor in experiments that measure the rate of electron transport through the electron transport chain. A student performed an experiment to study the effects of a chemical, DCMU, on photosynthesis.

The student prepared four tubes with a liquid buffer and chloroplasts that had been extracted from spinach leaves. The student then added DCPIP to three of the tubes and added DCMU to one of them. Additionally, tube 3 was wrapped in tin foil. The contents of each tube are shown in the table. The student then incubated each tube for 60 minutes and measured the absorbance (A_{600}) of each solution at five-minute intervals. The absorbance readings of each solution are shown in Figure 1.

Tube 1:	Buffer and water
Tube 2:	DCPIP, buffer, and water
Tube 3:	DCPIP, buffer, and water (wrapped in foil)
Tube 4:	DCPIP, buffer, and 1.0 mM DCMU

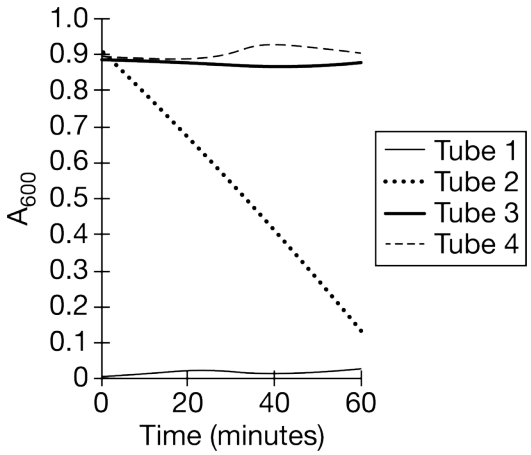


Figure 1. Absorbance readings of four prepared tubes with various solutions over a 60-minute period.

25. In which of the following tubes did the greatest reduction of DCPIP occur after 60 minutes?

(A) Tube 1

(B) Tube 2 ✓

(C) Tube 3

(D) Tube 4

Answer B

Correct. Over the 60-minute experiment, the absorbance of tube 2 decreased, indicating that DCPIP

was reduced.

26. Which of the following claims is best supported by the experimental results?

(A) Light is required for the electron transport chain to transfer electrons. ✓

(B) Water, not carbon dioxide, is the source of electrons used in the light-dependent reaction of photosynthesis.

(C) Carbon dioxide is the source of carbon used by green plants to build carbohydrates.

(D) DCPIP provides a significant source of electrons to the electron transport chain of the light reaction in the absence of light.

Answer A

Correct. Tube 2 showed a reduction in absorbance while tube 3 did not, which suggests that light was required for the reaction.

27. Which of the following best explains how DCMU affected the reaction?

(A) DCMU acts as a second buffer in the reaction.

(B) DCMU acts as an additional carrier of electrons from photosystem II to DCPIP.

(C) DCMU acts as an additional source of electrons to the light reaction of photosynthesis.

(D) DCMU acts as an inhibitor to the movement of electrons within the light reaction of photosynthesis. ✓

Answer D

Correct. The lack of change in absorbance over the length of the tubes containing DCMU indicates that it prevents the flow of electrons. These results are similar to those observed in tube 3, when the lack of light prevent electrons from flowing.

28. Which of the following best justifies the use of tube 2 as a control treatment?

(A) It was a negative control for the accuracy of the spectrophotometer, ensuring that an accurate reading for treatment cuvette 2 would be made.

(B) It was a negative control for the chemical stability of the chlorophyll suspension, ensuring that changes in absorbance could only be attributed to changes in chlorophyll content.

(C) It was a positive control for the change in DCPIP color associated with changes in light intensity.

(D) It was a positive control for measuring the effect of DCMU on the reaction. ✓

Practice Exam 2 MCQ

Answer D

Correct. Tubes 4 and 2 differed only by the presence of DCMU, so any changes in the outcome of the experiment could be attributed only to DCMU.

A scientist designed an experiment to test an artificial membrane that mimics the phospholipid bilayer of a cell.

The scientist built a tube that was divided by an artificial membrane and filled with distilled water. The scientist put a known amount of a protein into the water on one side of the membrane. After some time, the scientist measured the concentration of the protein on either side of the membrane but found that there had been no change.

29. Which of the following experimental changes would allow the scientist to observe transport of a solute across the artificial membrane?

(A) Increase the solute concentration in the solution

(B) Use a small, nonpolar solute instead of a protein ✓

(C) Increase the temperature of the solution

(D) Add artificial aquaporins to the membrane

Answer B

Correct. Small, nonpolar molecules are able to diffuse through cell membranes, while large proteins are not.

---

30. Which of the following best illustrates the flow of information required for the synthesis of proteins encoded in the genome of a retrovirus?

(A) DNA → RNA → Protein

(B) RNA → DNA → Protein

(C) RNA → DNA → RNA → Protein ✓

(D) DNA → RNA → DNA → Protein

Practice Exam 2 MCQ

Answer C

Correct. The genome of a retrovirus is composed of RNA. It must first be reverse-transcribed into DNA and integrated into a host cell's DNA before it is transcribed and the RNA is translated.

31. Which of the following best describes the process by which gas from the atmosphere is obtained by plants and used to build lipids?

(A) Gas is fixed by plants as part of the sulfur cycle.

(B) Gas is fixed by plants as part of the nitrogen cycle.

(C) Gas is directly obtained by plants as part of the carbon cycle. ✓

(D) Gas is directly obtained by plants as part of the magnesium cycle.

Answer C

Correct. Lipids are made of hydrocarbon chains whose main component is carbon. Plants obtain this carbon directly from the atmosphere in the form of carbon dioxide.

32. Which of the following best explains how the phospholipid bilayer of a transport vesicle contributes to cellular functions?

(A) The phospholipid bilayer allows the vesicle to fuse with the Golgi apparatus and the plasma membrane, allowing the exocytosis of proteins. ✓

(B) The phospholipid bilayer physically connects the nuclear envelope to the rough endoplasmic reticulum, thus increasing the rate of transcription and translation.

(C) The phospholipid bilayer of a transport vesicle contains chemicals that digest the proteins made in the rough endoplasmic reticulum.

(D) The phospholipid bilayer contains enzymes that catalyze the conversion of hydrogen peroxide to water and oxygen.

Answer A

Correct. The phospholipid bilayer allows for the transport of proteins from inside the cell to outside through the use of vesicles.

33. The diploid number of chromosomes in the cell of a domesticated dog is 78. Which of the following options includes the correct number of chromosomes in a cell after each cellular process (G_2 checkpoint, meiosis, and fertilization, respectively)?

Practice Exam 2 MCQ

	After G_2 Checkpoint	After Meiosis	After Fertilization	
(A)	156	78	39	
(B)	78	39	78	✓
(C)	156	39	78	
(D)	78	78	39	

Answer B

Correct. This choice correctly identifies that although chromatids are duplicated after the G_2 checkpoint, they still form 78 chromosome pairs. After meiosis, the chromosome number is divided in half. After fertilization, the chromosomes from two gametes are combined to form 78 chromosomes.

Practice Exam 2 MCQ

Scientists were interested in testing the effects of rotenone, a broad-spectrum pesticide, on a cell culture. Cell culture A was used as a control, while culture B was treated with rotenone. After a period of time, the scientists measured the concentration of several metabolites in the mitochondria of cells in both cultures. Their results are shown in the table below.

Metabolites	Concentration in Culture A (µM)	Concentration in Culture B (µM)
Pyruvate	25	25
NADH	55	550
NAD^+	55	5
ATP	85	5
$ADP+P_i$	55	100
$FADH_2$	25	26
FAD	25	25

34. Based on the data in the table, which of the following best explains the effects of rotenone on cellular respiration?

(A) Rotenone acts as an inhibitor of the enzymes in the Krebs cycle.

(B) NADH, produced during glycolysis, is not able to enter the mitochondria because transport proteins are blocked from entering.

(C) Treated cells are not able to break down NADH because certain enzymes of the electron transport chain are inhibited. ✓

(D) Rotenone acts as an allosteric inhibitor of glycolytic enzymes, thus inhibiting cellular respiration.

Answer C

Correct. The data indicate that NADH concentrations are significantly increased in cell culture B.

35. Which of the following best explains why triploid bananas do not produce seeds?

(A) The cells of the banana plant are unable to replicate DNA, thus preventing cell division and limiting growth.

(B) The banana plants lack enough genetic diversity to properly hybridize.

(C) The production of gametes is disrupted because of unequal pairing of homologous chromosomes during meiosis. ✓

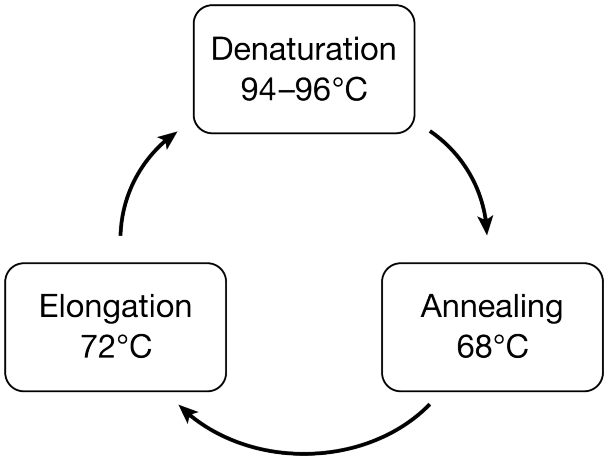
(D) The production of seeds is not required because triploid plants produce gametes without fertilization.

Practice Exam 2 MCQ

Answer C

Correct. If an organism has three copies of its genes, chromosomes cannot properly pair and separate during meiosis.

PCR is a technique used to copy a target DNA sequence. Researchers often use DNA polymerase from *Thermus aquaticus*, a bacteria species that lives around hydrothermal vents.



36. Which of the following best explains why the polymerase from the species *T. aquaticus* is often used for PCR?

(A) *T. aquaticus* polymerase has an optimal temperature of $68^{\circ}C$.

(B) *T. aquaticus* polymerase does not denature at high temperatures. ✓

(C) *T. aquaticus* polymerase can be used more than once without degrading.

(D) *T. aquaticus* polymerase adds nucleotides to both the $3\prime$ and $5\prime$ ends of DNA.

Answer B

Correct. The polymerase enzyme of *T. aquaticus* is used because it evolved to function at temperatures close to boiling without denaturing.

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37. A researcher is studying two different species. Which of the following sets of observations would best support the claim that the two species have structural similarities as a result of convergent evolution?

Practice Exam 2 MCQ

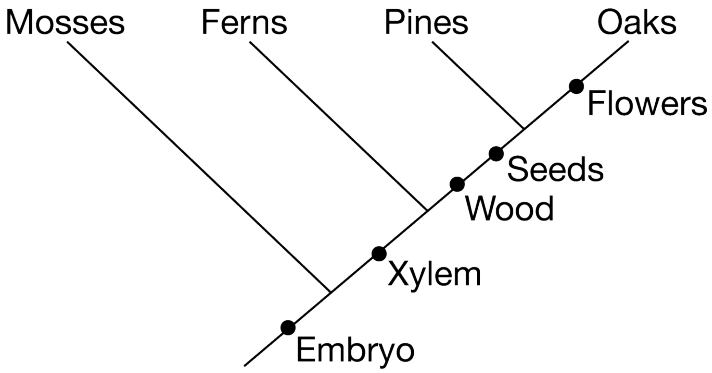
(A)	Genetic Similarity Low	Function of Structure Same	Form of Structure Different	✓
(B)	Genetic Similarity Low	Function of Structure Different	Form of Structure Same	
(C)	Genetic Similarity High	Function of Structure Same	Form of Structure Different	
(D)	Genetic Similarity High	Function of Structure Different	Form of Structure Same	

Answer A

Correct. Convergent evolution results in similarly functioning traits whose structure differs as a result of genetic differences between distantly related species.

Practice Exam 2 MCQ

The cladogram shows shared traits among four plant taxa.



38. Based on the cladogram, which of the following are the traits shared between ferns and pines?

(A) Xylem only

(B) Seeds and wood only

(C) Embryo and xylem only ✓

(D) Embryo, xylem, wood, and seeds only

Answer C

Correct. The cladogram shows that ferns and pines have two shared characteristics: embryos and xylem tissue.

Practice Exam 2 MCQ

Snapdragons are a type of flowering plant. The flower color of the plants is determined by a single gene with two alleles: an incompletely dominant R allele and a recessive r allele. Researchers established a new population of snapdragon plants that they hypothesize will be in Hardy-Weinberg equilibrium. At the start of the study, the researchers randomly picked 100 individuals from a large population and recorded their flower color. They planted the snapdragons in an isolated area and returned after seven generations had passed. They randomly sampled 100 plants from the population and recorded their flower color. The results are shown in the table.

DISTRIBUTION OF SNAPDRAGON FLOWER COLOR PHENOTYPES			
Time	Number of Red (RR) Snapdragons	Number of Pink (Rr) Snapdragons	Number of White (rr) Snapdragons
At the start of study	34	43	23
After seven generations	39	34	27

The researchers performed a chi-square analysis to test their hypothesis.

39. Assuming the researchers use a significance level of 0.05, which of the following is closest to the critical value the researchers should use in the chi-square analysis?

(A) 3.84 ✓

(B) 5.99

(C) 7.82

(D) 9.49

Answer A

Correct. This is the critical value with 1 degree of freedom and a significance level of $.05$. Since this Hardy-Weinberg equilibrium question involves only 2 alleles and the frequency of 1 allele determines the frequency of the other allele, there is a single independent variable.

Practice Exam 2 MCQ

Sea otters are native to the western coast of North America. Between 1750 and 1850, hunting had reduced the population from hundreds of thousands to only one thousand individuals. In the early 1900s, a small population of sea otters was discovered in Elkhorn Slough, an estuary in central California near a large human population center. The otters were then protected by the international fur seal treaty, which banned sea otter hunting. The sea otter population has rebounded to nearly three thousand individuals today.

Otters live in kelp forests and eelgrass beds and feed on crabs and shellfish (Figure 1). Most herbivores in the habitat eat algae that grows on the eelgrass and not the eelgrass itself. If there is too much algae, the eelgrass does not receive enough light for photosynthesis. As the otter population has increased, the eelgrass habitat has increased.

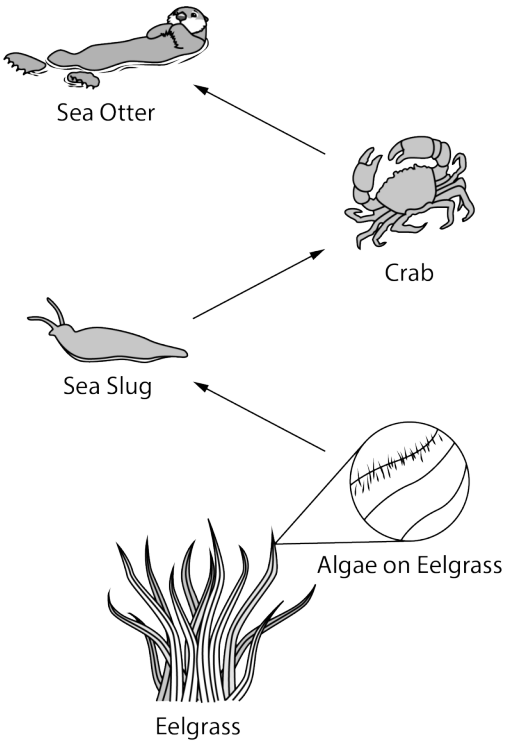


Figure 1. Partial food chain in eelgrass habitats

Recently, however, scientists have noticed the presence of two nonnative, predatory invertebrate species that may be colonizing the Elkhorn Slough, which would have been too cold for them three decades ago. Scientists have also observed that otters in the area are experiencing increased mortality because of an increase in harmful algal blooms, which occur as a result of nutrient pollution. The harmful algae are ingested by shellfish, which the otters eat.

40. Which of the following best describes what happened to the otter population between 1750 and 1850 ?

(A) The population decreased in size as a result of a loss of genetic diversity.

(B) The population decreased in size as a result of habitat loss.

(C) The population lost genetic diversity as a result of a bottleneck effect. ✓

(D) The population lost genetic diversity as a result of the founder effect.

Practice Exam 2 MCQ

Answer C

Correct. The rapid decrease in population size as a result of hunting likely caused a decrease in genetic diversity. This is an example of a genetic bottleneck.

41. Based on the information provided in the passage, which of the following best describes the effect of harmful algal blooms on otter populations?

(A) They are a density-dependent factor that increases otter mortality in larger populations.

(B) They are a density-dependent factor that reduces otter numbers at lower population sizes.

(C) They are a density-independent factor that negatively affects the otter population regardless of its size. ✓

(D) They are a density-independent factor that increases otter mortality in only larger populations.

Answer C

Correct. The effect of harmful algal blooms is a factor negatively affecting otter populations that is independent of the otter population size.

42. As otters were removed during the hunting years, there was a large decrease in the catches of fish species from the eelgrass habitats. Which of the following best explains why this decrease happened?

(A) Otters are a keystone species, so their disappearance from the area affected the population size of one other species.

(B) Otters are a keystone species, so their disappearance from the area resulted in the collapse of an entire community. ✓

(C) Otters have mutualistic relationships with many other species, so their disappearance from the area affected the population size of another species.

(D) Otters have mutualistic relationships with many other species, so their disappearance from the area resulted in the collapse of an entire ecosystem.

Answer B

Correct. Otters are a keystone species because their disappearance negatively affected the entire community.

43. Climate change could affect the ecosystem of the Elkhorn Slough in many ways. From the information provided, which of the following predictions about the direct, local effects of climate change is most likely?

Practice Exam 2 MCQ

(A) Ocean warming will favor population growth of nonnative species as their habitats shift northward. ✓

(B) Ocean warming will decrease eelgrass habitat area as a result of increased herbivory by nonnative species.

(C) Harmful algal blooms will decrease otter populations as a result of increased mortality of otter prey species.

(D) Harmful algal blooms will decrease the availability of nutrients for eelgrass and other algae species.

Answer A

Correct. The passage indicates that ocean warming has allowed for the introduction of nonnative species that could not previously live in Elkhorn Slough.

44. Based on the information, an increase in the sea slug population would most likely be directly related to which of the following?

(A) An increase in the eelgrass population

(B) The introduction of nonnative invertebrates

(C) A decrease in algae availability

(D) A decrease in the crab population ✓

Answer D

Correct. An increase in the sea slug population is likely the result of a decrease in their predator population, the crabs, as a result of increased sea otter populations.

Many fish species, such as fathead minnows, release a pheromone when their skin cells are damaged. Researchers placed pike, a predator of fathead minnows, in a choice chamber and released the minnow pheromone at one end of the chamber. The researchers observed that the pike oriented themselves toward the end of the chamber where the pheromone was released.

45. Which of the following questions will best guide a follow-up investigation about the role of pheromones in locating prey?

(A) How do pike determine that the fathead minnow pheromone is present in the water? ✓

(B) Why do pike prey on fathead minnows?

(C) Do pike have natural predators in the environment?

(D) Do pike release pheromones that are detected by fathead minnows?

Practice Exam 2 MCQ

Answer A

Correct. This question focuses on the role of the pheromone in prey detection.

---

46. Which of the following best explains why a cell's plasma membrane is composed of two layers of phospholipids rather than just a single layer?

(A) Having two oppositely oriented layers of phospholipids allows only the hydrophilic heads to interact with water inside and outside of the cell. ✓

(B) Having two oppositely oriented layers of phospholipids allows the hydrophilic heads to repel water both inside and outside of the cells.

(C) Having two identically oriented layers of phospholipids gives cells more protection from the exterior environment than just a single layer would.

(D) Having two identically oriented layers of phospholipids allows for the production of vacuoles while still maintaining a protective barrier.

Answer A

Correct. Phospholipids are oriented so the hydrophilic heads are on the membrane exterior and can interact with water inside and outside of the cell. The hydrophobic tails in the middle of the membrane create a protective barrier for the cell.

A student calculated the average surface area-to-volume ratio of four different types of human epithelial cells. The results are shown in the table below.

Cell Type	Surface Area-to-Volume Ratio
Simple squamous	9
Simple cuboidal	6
Simple columnar	4
Simple spherical	3

47. Based on the data, which type of cell would be best suited for the lining the alveoli of the lungs, where diffusion of carbon dioxide and oxygen must occur very rapidly?

(A) Simple squamous cells ✓

(B) Simple cuboidal cells

(C) Simple columnar cells

(D) Simple spherical cells

Answer A

Correct. Diffusion happens very quickly in cells that have a high surface area-to-volume ratio. The data in the table show that the simple squamous cell has the highest ratio.

---

48. Which of the following best explains how the extensive folding of the inner mitochondrial membrane benefits a eukaryotic cell?

(A) It enlarges the volume of the matrix, which allows for more enzymatic reactions.

(B) It increases the area available for proteins involved in energy transfer. ✓

(C) It allows for greater area for the diffusion of water into and out of the mitochondria.

(D) It provides better insulation for reactions in the matrix from conditions outside the mitochondria.

Answer B

Correct. The extensive folding provides more surface area for proteins involved in energy transfer.

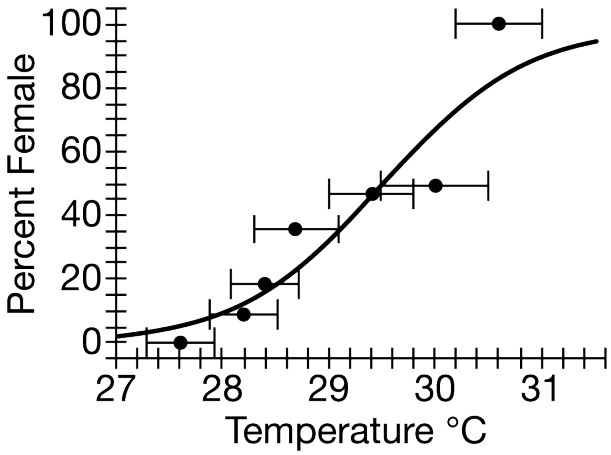


Figure 1. Percentage of sea turtles that develop as females at different average nest temperatures. The bars represent the temperature range observed within the nests.

Some green sea turtle females deposit their eggs in nests that are dug on warm, tropical beaches. Researchers have studied the effects of temperature on the hatching rate and the proportion of female to male hatchlings produced at different average nest temperatures in Suriname (Figure 1). At temperatures below $23^{\circ}\mathrm{C}$ or above $33^{\circ}\mathrm{C}$, the eggs do not develop.

49. Which of the following best describes the results shown in Figure 1?

(A) The percentage of hatchlings that survive to adulthood is directly proportional to average nest temperature.

(B) Female sea turtles search for cooler beaches in order to have more male offspring.

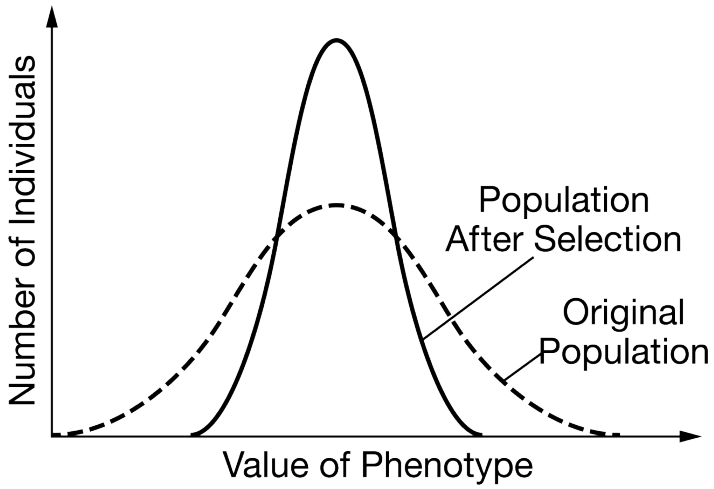
(C) Warmer nests produce more female sea turtles than do cooler nests. ✓

(D) The sex ratio of sea turtles is genetically determined.

Answer C

Correct. The graph indicates that more female sea turtles are produced in nests with a higher average nest temperature.

Practice Exam 2 MCQ



50. Which of the following describes a scenario that would result in the phenotypic change shown in the graph?

(A) A fish that has mouth parts adapted to eat moderately sized prey is introduced into a lake in which there are only large and small prey.

(B) A population of mice live on the ground and in short trees. An invasive, tree-dwelling snake that preys on the mice is introduced into the area.

(C) Climate change-induced warming of arctic tundra reduces average snow cover that lighter-colored arctic foxes rely on for camouflage. Darker-colored arctic foxes are better suited to the exposed moss and grass habitat.

(D) House sparrows that lay smaller-than-average clutches of eggs produce fewer viable offspring, while larger-than-average clutches of eggs result in malnourished chicks that have a higher mortality rate. ✓

Answer D

Correct. This scenario would result in stabilizing selection, as shown in the graph.

Practice Exam 2 MCQ

Researchers have identified a molecule produced by *Ecteinascidia turbinata*, a marine invertebrate, from which the drug trabectedin is produced. Soft-tissue tumors treated with trabectedin rapidly decrease in size. In a preliminary study, healthy cells and tumor cells sampled from skin cancer patients treated with trabectedin were collected, and several characteristics of the cells were observed. The observed results of the study are shown in the table.

Characteristic	Healthy Cell	Tumor Cell
DNA (in picograms)	12 pg	6 pg
Nuclear envelope	Thinning	Present
Nucleolus	Absent	Present
Cyclin level	High	Low
General appearance	Round, firm	Shrunken

51. Which of the following best explains the most likely method by which this antitumor drug works?

(A) Trabectedin increases the production of cyclin proteins that signal the cancer cells to enter prophase.

(B) Trabectedin interferes with the plasma membrane, causing it to break down and expose the DNA to damage.

(C) Trabectedin interferes with the duplication of DNA during interphase and thus prevents cancer cells from passing the G_2 checkpoint. ✓

(D) Trabectedin interferes with the regulations of cyclin proteins, causing their levels to increase and creating errors in DNA.

Answer C

Correct. The data indicate that tumor cells treated with the drug are not replicating their DNA and thus prevents the cells from passing the G_2 checkpoint.

Practice Exam 2 MCQ

Friedreich's ataxia is an inherited disorder. Friedreich's ataxia is caused by an insertion mutation in a noncoding portion of the FXN gene where a GAA triplet is repeated hundreds of times. The FXN gene encodes the protein frataxin. A pedigree of a family with members affected by this disorder is shown in Figure 1.

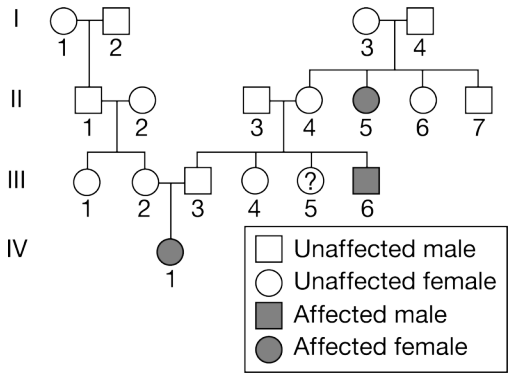


Figure 1. A pedigree of a family affected by Friedreich's ataxia

A researcher collected DNA from several members of the family and used PCR to amplify the FXN genes from each individual's DNA. The researcher then used DNA gel electrophoresis to separate the DNA. The results are shown in Figure 2.

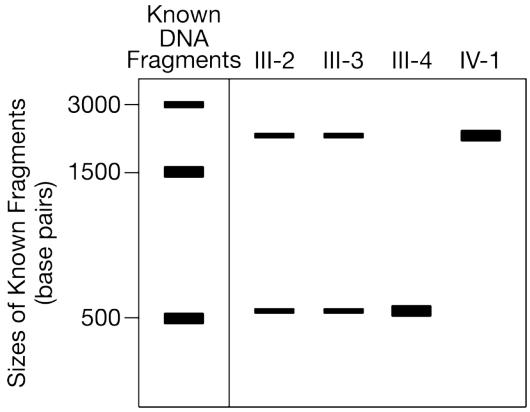


Figure 2. FXN gene fragment sizes for several family members. A sample of DNA with fragments of known lengths was used for comparison.

The researcher also used a computer to model the structure of the mutant FXN allele. The model suggests that the repeated GAA triplets in the mutant FXN gene may lead to the formation of an unusual triple-stranded configuration of DNA (Figure 3).

Practice Exam 2 MCQ

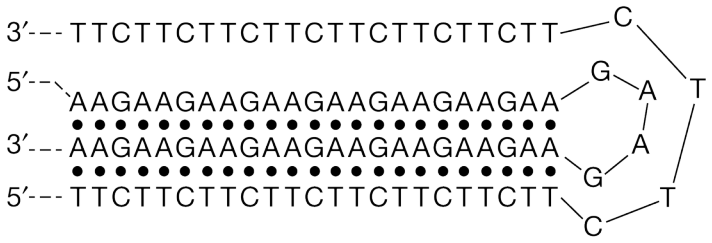


Figure 3. The modeled DNA triple-helix structure that can form in areas with multiple GAA triplets

52. Based on the data in Figure 1, which of the following best describes the inheritance pattern of Friedreich's ataxia?

(A) Autosomal recessive ✓

(B) Autosomal dominant

(C) Sex-linked recessive

(D) Sex-linked dominant

Answer A

Correct. Affected offspring were born to unaffected parents, and unaffected males had affected offspring.

53. The probability that individual III-5 will develop Friedreich's ataxia is closest to which of the following?

(A) 0%

(B) 25% ✓

(C) 50%

(D) 75%

Answer B

Correct. Both parents must be carriers because individual III-6 is affected. There is a 25% chance that III-5 inherited two recessive alleles.

54. Which of the following statements best describes the results seen in Figure 2 ?

Practice Exam 2 MCQ

(A) Individuals III-2 and III-3 carry two different alleles of the FXN gene, a mutant allele and a wild-type allele. Individual IV-1 inherited two copies of the mutant allele. ✓

(B) Individuals III-2 and III-3 carry two different alleles of the FXN gene, a mutant allele and a wild-type allele. Individual IV-1 inherited two copies of the wild-type allele.

(C) Individuals III-2 and III-3 both carry two wild-type alleles. Individual IV-1 inherited two copies of the wild-type allele.

(D) Individuals III-2 and III-3 both carry two wild-type alleles, but individual IV-1 inherited one copy of the wild-type allele and one copy of the mutant allele.

Answer A

Correct. Individuals with two different bands of DNA have two different alleles. Individuals with one band of approximately 2500 base pairs have two copies of the mutant allele, which is larger because of the insertion mutation.

55. Which of the following best describes the most likely effect of the formation of a triplex DNA structure (Figure 3) on the synthesis of the frataxin protein?

(A) The binding of the ribosome to the mRNA is prevented, resulting in a decrease in frataxin translation.

(B) The DNA will not degrade in the cytoplasm, leading to an increase in frataxin translation.

(C) RNA polymerase is prevented from binding to the DNA, resulting in a decrease in frataxin mRNA transcription. ✓

(D) The protein will include extra amino acids, resulting in a protein with an altered secondary structure.

Answer C

Correct. The formation of a triple-strand structure prevents the binding of RNA polymerase, which requires a single strand to bind to.

56. Which of the following types of bonds is most likely responsible for the unusual base pairing shown in Figure 3 that results in the formation of a triplex DNA structure?

(A) Hydrogen ✓

(B) Polar covalent

(C) Ionic

(D) Nonpolar covalent

Practice Exam 2 MCQ

Answer A

Correct. Hydrogen bonds form between the bases on complementary strands of DNA.

A scientist is studying a population of lizards with three different color phenotypes. The color phenotype is controlled by a single gene with two alleles: an incompletely dominant allele A and a recessive allele a. The scientist collected data on the color of 100 lizards. The data are shown in the table.

Phenotype	Genotype	Number Observed
Red	AA	75
Purple	Aa	15
Blue	aa	10

To test the hypothesis that the population is at Hardy-Weinberg equilibrium, the scientist performed a chi-square test.

57. Which of the following values is closest to the chi-square value the scientist calculated?

(A) 0.1

(B) 1.9

(C) 18.3

(D) 23.1 ✓

Answer D

Correct. This choice correctly approximates the chi-square value from the data.

Practice Exam 2 MCQ

Scientists compared the chemical structure of several molecules that various bacterial species use for quorum sensing. Quorum sensing is an ability some bacteria have to detect the number of related cells nearby. The chemical structure of some of these molecules found in certain species of bacteria are shown in Figure 1.

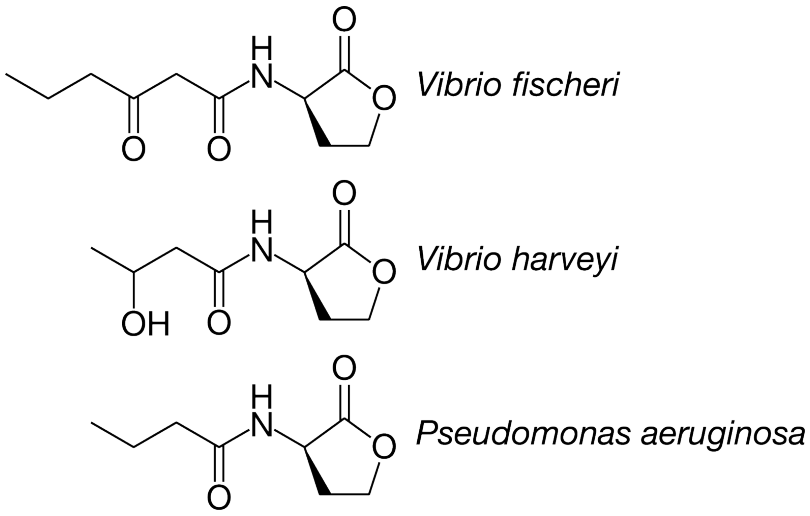


Figure 1. The chemical structure of several molecules used for quorum sensing in three species of bacteria

58. Which of the following research questions would best guide an investigation of the link between the structure of the signaling molecules and the evolution of quorum sensing?

(A) Do these molecules require the same receptors in each bacteria species to generate a response?

(B) Did these species evolve from a common ancestor that used a similar signaling molecule? ✓

(C) Do these species all perform the same action when the concentration of the signaling molecules is high enough?

(D) Did these species evolve from the same common ancestor that is still living today and uses the same receptors?

Answer B

Correct. Bacteria species likely have similarly structured signaling molecules as a result of common ancestry.

---

59. Carbon dioxide most likely enters a cell through which of the following processes?

Practice Exam 2 MCQ

(A) Simple diffusion through the membrane ✓

(B) Facilitated diffusion through membrane proteins

(C) Active transport through membrane proteins

(D) Active transport through aquaporins

Answer A

Correct. Carbon dioxide is a small molecule without a charge and so can pass through a cell membrane by simple diffusion.

In pea seeds, yellow color (Y) is dominant to green color (y), and a round shape (R) is dominant to a wrinkled shape (r). A dihybrid cross between a true-breeding plant with yellow, round seeds ($YYRR$) and a true-breeding plant with green, wrinkled seeds ($yyrr$) results in an F_1 generation of plants with yellow, round seeds. Crossing two F_1 plants produces an F_2 generation with approximately nine times as many plants with yellow, round seeds as plants with green, wrinkled seeds.

60. Which of the following best explains these results?

(A) The allele pairs of each parent stay together, resulting in gametes that are identical to the parents.

(B) Gene segments on sister chromatids cross over.

(C) Alleles that are on nonhomologous chromosomes recombine.

(D) The genes for seed color and seed shape assort independently. ✓

Answer D

Correct. A ratio of nine to one of offspring with yellow, round seeds to those with green, wrinkled seeds is expected as a result of genes that are independently assorted.

Practice Exam 2 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph hide_paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Scientists studying transcription in yeast (*Saccharomyces cerevisiae*) created an experimental strain that produced a modified RNA polymerase containing a single amino acid substitution. The scientists determined the maximum elongation rate during transcription with and without the modified RNA polymerase enzyme (Figure 1).

The compound amanitin, which is commonly found in toxic mushrooms, is a specific RNA polymerase inhibitor. Amanitin binds to the RNA polymerase active site and inhibits transcription. In a second experiment, the scientists treated the wild-type and experimental strains of *S. cerevisiae* with a $40\,\mu g/mL$ solution of amanitin and recorded the maximum elongation rate of the mRNA (Figure 2). Error bars represent $\pm 2SE_{\bar{x}}$.

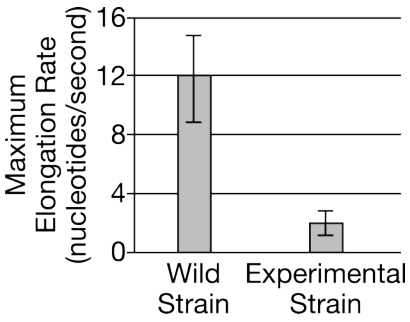


Figure 1. Maximum elongation rate under natural conditions

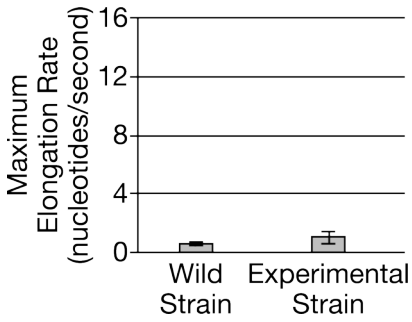


Figure 2. Maximum elongation rate in a $40\,\mu g/mL$ amanitin solution

1. (a) **Describe** the three structural components of an RNA nucleotide monomer. **Explain** the role of RNA polymerase during transcription.

(b) **Identify** the dependent variable in the experiments. **Identify** a control group missing from the second experiment. **Justify** the need for this control group in the second experiment.

(c) **Describe** the effect of amanitin on the maximum elongation rate for the wild-type and modified RNA polymerases. **Determine** the ratio of the average maximum elongation rate for the modified RNA polymerase compared to the wild strain RNA polymerase in Figure 1.

(d) **State the null hypothesis** for the experiment in Figure 1. **Provide reasoning to justify** the claim that the change in the amino acid sequence in the modified RNA polymerase affected the shape of the active site on the enzyme.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Practice Exam 2 FRQ

0	1	2 ✓

The response includes both of the following criteria.

- ☐ A description that an RNA nucleotide has the following three structural components: a five-carbon sugar (ribose), a phosphate group, and a nitrogen base (adenine, cytosine, guanine, or uracil)
- ☐ An acceptable explanation of the role of RNA polymerase.
  Acceptable explanations include the following.
  - RNA polymerase synthesizes a new RNA molecule based on a DNA template by matching the current DNA base with the proper RNA complement.
  - RNA polymerase joins/bonds the newly paired RNA nucleotide and the growing RNA strand with a covalent bond.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- ☐ The dependent variable is identified as the maximum elongation rate of the mRNA.
- ☐ An acceptable control group missing from the second experiment is identified. Acceptable control groups include the following.
  - Wild strain from the first experiment without amanitin
  - Experimental strain from the first experiment without amanitin
- ☐ The justification that the missing control is needed because the effect of amanitin on the maximum elongation rate cannot be determined without comparison to the maximum elongation rate under the same conditions without amanitin

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes all of the following criteria.

Practice Exam 2 FRQ

- [ ] A description that amanitin decreases the maximum elongation rate for the wild strain and does not affect the rate in the experimental strain
- [ ] The ratio of the average maximum elongation rate for the modified RNA polymerase compared to the wild strain RNA polymerase is determined to be 1 : 6 / 1/6 / 1 to 6.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes all of the following criteria.

- [ ] An acceptable null hypothesis is provided. Acceptable null hypotheses include the following.
  - The modified RNA polymerase would not affect the maximum elongation rate.
  - Any difference between the two elongation rates is due to chance.
- [ ] A description that a change to the active site would explain the decrease in the elongation rate is provided as reasoning to justify the claim.

Practice Exam 2 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Scientists studied soybean plants grown in sunlit, controlled-environment chambers with a carbon dioxide (CO_2) concentration of $700\ \mu mol\ CO_2/mol$ of air. The CO_2 concentration in the chambers was approximately double that of normal concentrations of environmental CO_2.

Photosynthetic rate is measured by the amount of CO_2 exchanged in an area of leaf surface over time. The rate is directly influenced by the activity of the plant enzyme rubisco. During photosynthesis, rubisco fixes atmospheric carbon into an organic molecule as one of the first steps in the Calvin cycle.

Scientists measured the amount of carbon fixation that occurred in the leaves of soybean plants grown at each temperature shown in Table 1.

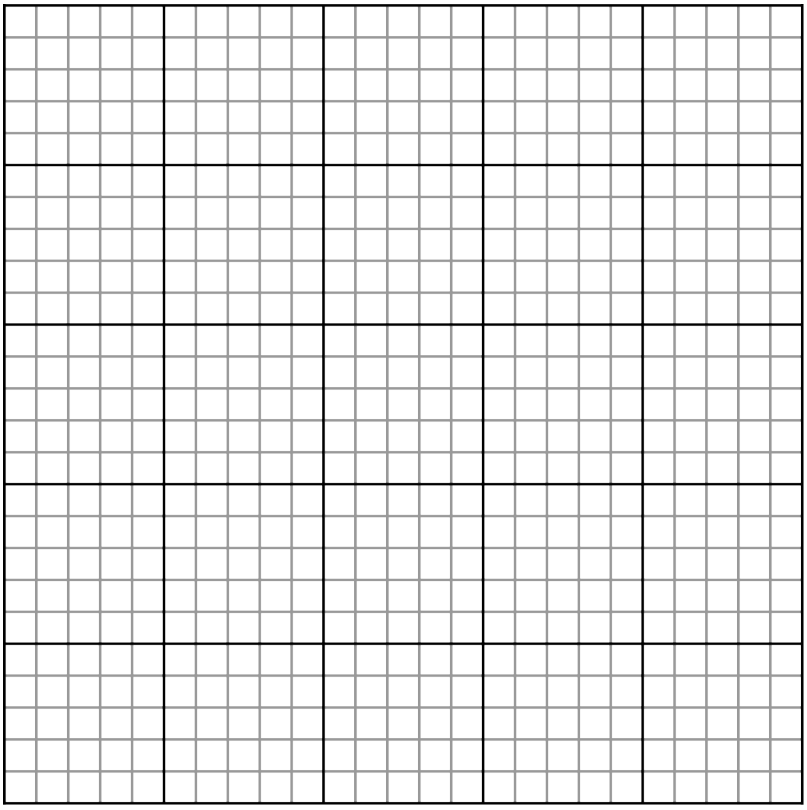
Table 1. Carbon Fixation by Rubisco Under Different Temperatures

Temperature ($^\circ C$)	**Carbon Fixation** $\pm 2SE_{\bar{x}}\ (\mu mol/m^2/s)$
25	91 ± 9
30	96 ± 5
35	93 ± 8
40	104 ± 7
45	71 ± 7
50	59 ± 4

Practice Exam 2 FRQ

2. (a) **Describe** the effect on an enzyme of an increase in environmental temperature above the optimum temperature range of the enzyme.

(b) Using the template in the space provided for your response, **construct** an appropriately labeled graph to represent the data in Table 1. **Determine** whether there is a statistical difference in the amount of CO_2 fixed by rubisco at $30^\circ C$ and $40^\circ C$.



(c) Based on the data, **identify** the temperature range at which the soybean leaves are producing the lowest amount of carbohydrate.

(d) Increased amounts of carbon dioxide in the atmosphere are correlated with increased global surface temperatures. Based on the data, **predict** how a surface temperature that continues to rise above $40^\circ C$ most likely affects the amount of energy available to primary consumers in an ecosystem. Provide reasoning to **justify** your prediction. One model predicts that an increase in greenhouse gases will lead to a $2^\circ C$ increase in average surface temperature on Earth. Based on the data from the experiment and the prediction of an increase in average surface temperature by the model, **predict** how the locations of plant species are expected to change over time.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

Practice Exam 2 FRQ

The response includes one of the following descriptions.

- [ ] An increase in temperature (above the optimum range) will denature the enzyme **and** reduce the reaction rate.
- [ ] An increase in temperature (above the optimum range) can alter the shape of the enzyme/active site **and** reduce the reaction rate.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following criteria.

- [ ] Properly labeled and scaled axes
- [ ] A line graph with accurately plotted points
- [ ] Error bars of appropriate magnitude
- [ ] A determination that there is no significant difference in the amount of CO_2 fixed by rubisco at $30^\circ C$ and $40^\circ C$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The temperature at which the soybean leaves are producing the lowest amount of carbohydrate is identified as any value above $45^\circ C$ and equal to or below $50^\circ C$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following criteria.

- [ ] The prediction that less energy would be available if surface temperature continues to rise above $40^\circ C$

Practice Exam 2 FRQ

- ☐ The reasoning that carbon fixation decreases at temperatures above $40^{\circ}C$
- ☐ An acceptable prediction of how the locations of plant species are expected to change over time. Acceptable predictions include the following.
  - As temperatures warm, the ranges where individual plant species are found will move toward the poles.
  - As temperatures warm, the ranges where individual plant species are found will move toward higher altitudes.
  - As temperatures warm, the ranges where individual plant species are found will move toward cooler areas.

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Pigeon lice (*Columbicola columbae*) are small parasites that feed on the feathers of the rock pigeon (*Columba livia*). The lice are host-specific, and transmission of lice between pigeons requires direct contact. Pigeons remove the lice with their beaks as they clean their feathers. Figure 1 shows the black and white phenotypes for pigeon feather color and the dark and light phenotypes for louse body color.

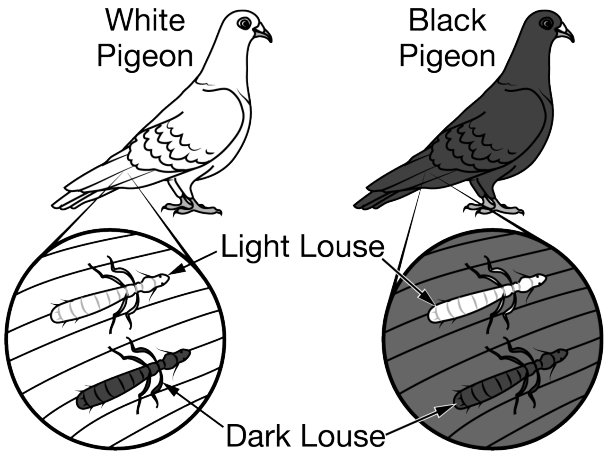


Figure 1. Pigeon and louse phenotypes, showing lice against pigeon feathers

Researchers designed an experiment to test the hypothesis that the evolution of coloration in lice is driven by birds. Researchers transferred 25 lice from a random sample of lice obtained from wild-caught rock pigeons to each individual in groups of black pigeons and white pigeons. Half of the pigeons from each color group were fitted with a harmless mouthpiece that stopped the beaks from entirely closing. The mouthpiece prevented pigeons from removing lice but did not affect the pigeons' ability to feed. Each pigeon was isolated and kept in identical environmental conditions for temperature, humidity, and amount of light. The lice have a generation time of 24 days. The researchers observed the phenotypes of the lice over a period of four years.

Practice Exam 2 FRQ

3. (a) **Describe** the importance of phenotypic variation in louse body color among individuals in a population of lice.

(b) **Justify** the decision of the researchers to hold lighting conditions constant in their experiment.

(c) **Predict** the most likely effect of the pigeons cleaning their feathers on the phenotypes of the lice after four years.

(d) **Provide reasoning** to support your prediction in part (c).

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response describes that increased variation increases the probability that some members of population will survive (and reproduce) in a changing environment.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The decision is justified by describing that, since pigeons clean their feathers by sight, variable light conditions would affect the effectiveness of this behavior.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes one of the following predictions.

- ☐ Louse phenotypes that blended with their host's feather color would increase in frequency.
- ☐ Louse phenotypes that contrasted with their host's feather color would decrease in frequency.

Part D

Practice Exam 2 FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the reasoning that lice with a contrasting coloration to the host feathers were removed before they could reproduce while lice with a similar coloration to the host feathers survived to reproduce.

Practice Exam 2 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph hide_paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Daphnia longicephala is a freshwater crustacean that can detect chemical cues produced by the predator *Notonecta glauca*. Researchers raised *Daphnia* clones and observed that those exposed to *Notonecta* chemical cues during their development produced larger protective crests, as described in Figures 1 and 2. Crest height and width are measured as ratios to body length. The increase in the size of the crest width and height is not associated with a corresponding change to the *Daphnia*'s genome. Researchers hypothesize that growing a crest is energetically costly, so *Daphnia* do not develop the large crests in the absence of *Notonecta*.

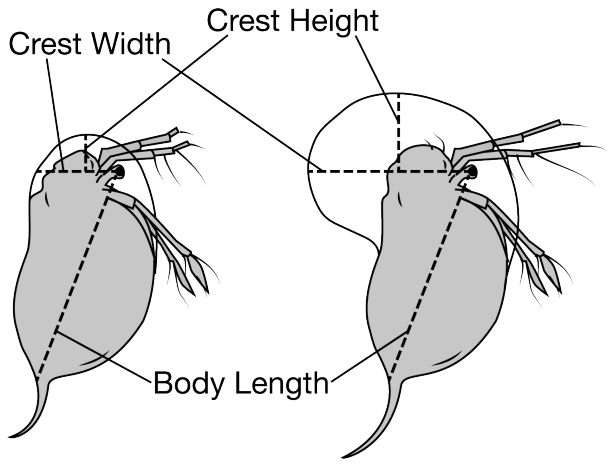


Figure 1. Comparison of *Daphnia* that developed in the absence (left) and the presence (right) of *Notonecta* chemical cues

Practice Exam 2 FRQ

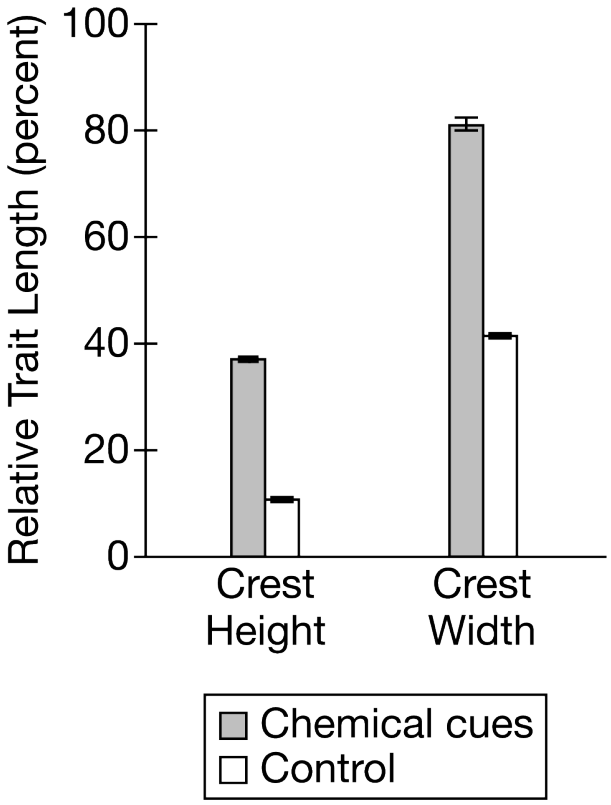


Figure 2. Comparison of mean relative crest height and width between control *Daphnia* and *Daphnia* exposed to *Notonecta* chemical cues during development. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

4. (a) **Describe** the factors that influence an individual's phenotype.

(b) **Explain** how the presence of *Notonecta* chemical cues affects gene expression in the *Daphnia*.

(c) As a follow-up experiment, researchers placed the *Daphnia* that were exposed to the *Notonecta* chemical cues into a tank without chemical cues. The *Daphnia* reproduced asexually, and the offspring developed in the tank without chemical cues. **Predict** the relative size of the crest height and width of offspring raised in the tank without chemical cues as compared to the parent *Daphnia*.

(d) Provide reasoning to **justify** your prediction in part (c).

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The factors that influence an individual's phenotype include both of the following.

Practice Exam 2 FRQ

· Genetic information

· Environmental influences

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the explanation that the presence of *Notonecta* chemical cues enhanced expression of genes associated with crest development in the *Daphnia*.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes a prediction that the relative size of the crest height and width of the offspring will resemble that of the original population.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes both of the following criteria.

- ☐ The reasoning that the *Notonecta* chemical cues only affected the phenotype of the *Daphnia* parents during their development, leaving the genotype unchanged
- ☐ The reasoning that because there was no change to the heritable genetic sequence of the parents, the offspring are not expected to exhibit the larger crest heights and widths in the predator-free environment

Practice Exam 2 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph hide_paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

A student collected samples of cells from a woody plant during the spring and produced the drawings below. Sample 1 was collected from the woody stem of the plant. Sample 2 was collected from the root tip of the plant. Cells in different steps of mitosis are labeled with letters A, B, C, D, and E in Sample 2.

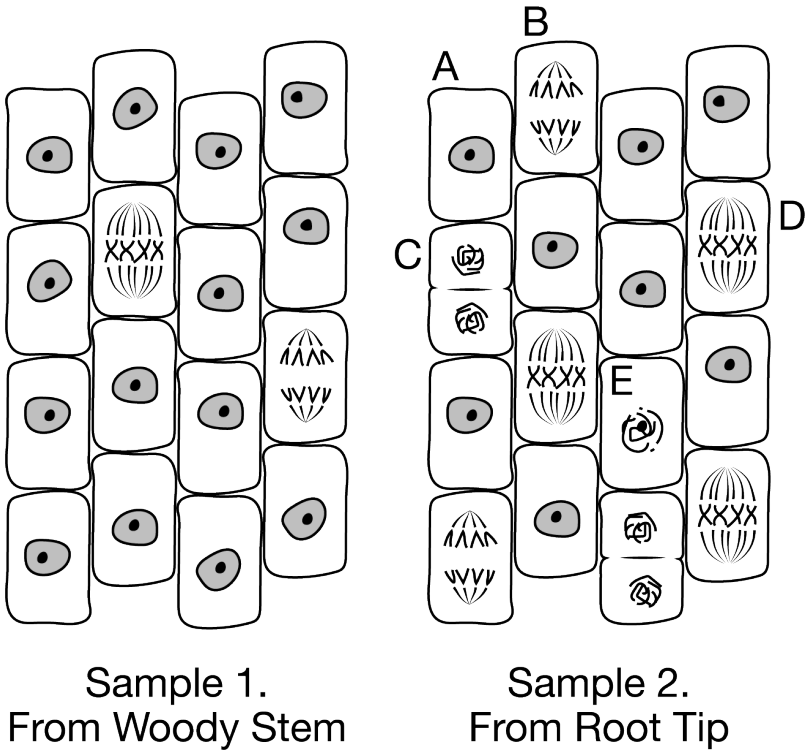


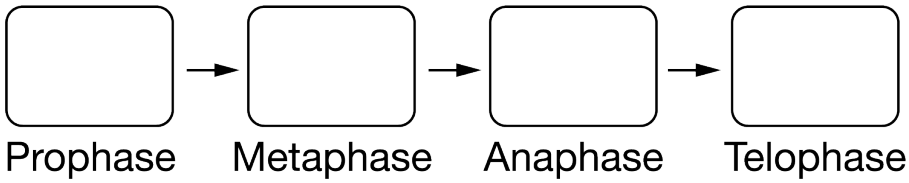
Figure 1. Drawings of samples of plant cells collected from two locations in a woody plant during the spring

Practice Exam 2 FRQ

5. (a) **Describe** the role of mitosis in the growth of a plant.

(b) **Explain** why there are more cells observed in mitosis in the root tip than in the woody stem.

(c) **Refine** the model in the space provided for your response by using the letters from four of the cell labels of Sample 2 to represent the sequential steps of mitosis.



(d) **Explain** how the arrangement of cellular components during the step of mitosis portrayed by cell D in Sample 2 facilitates the proper distribution of chromosomes to the two daughter cells.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes both of the following criteria.

- ☐ A description that plant growth occurs when new cells are produced
- ☐ A description that mitosis is a process that produces two genetically identical daughter cells, which can develop into different tissues or further divide

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that more growth occurs in the root tips and that very little new cell development occurs in woody stems after they are formed.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Practice Exam 2 FRQ

0	✓ 1

From left to right, the boxes in the model contain the letters E, D, B, and C in that order.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	✓ 1

The response indicates that during stage D all chromosomes line up on the equator, and that the spindle fibers enable the two chromatids of each chromosome to move to opposite ends of the cell before the cell divides.

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Students were using the formulas for the surface area and volume of a rectangular solid to model cells of different sizes and shapes. They produced the data shown in the table.

Rectangular Solid	Length (cm)	Height (cm)	Width (cm)	Surface Area (cm^2)	Volume (cm^3)	Surface Area-to-Volume Ratio
1	2.00	1.00	1.00	10.00	2.00	5.00
2	2.67	0.50	1.50	12.18	2.00	6.08
3	4.00	0.25	2.00	19.00	2.00	9.50
4	6.05	0.13	2.65	34.24	2.00	17.09
5	10.68	0.06	3.00	65.79	2.00	32.85

Practice Exam 2 FRQ

6. (a) **Identify** the rectangular solid that represents the cell with the smallest amount of plasma membrane per volume of cytoplasm.

(b) **Describe** the relationship between the dimensions of the rectangular solids and their surface area-to-volume ratios.

(c) Using the data from the model, **evaluate** the hypothesis that cells shaped like rectangular solids 1 and 2 are better suited for slow metabolism and the long-term storage of energy-containing molecules than are cells shaped like rectangular solids 4 and 5.

(d) Based on the trends shown in the data, **explain** why some rectangular cells include folds or projections in their plasma membranes.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response identifies rectangular solid 1, with the smallest surface area-to-volume ratio, as the cell with the smallest amount of plasma membrane per amount of cytoplasm.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response describes that as the rectangles become longer, wider, **and** thinner, the surface area-to-volume ratio increases.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the following criteria.

- ☐ Rectangular solid 1/rectangular solid 2 is identified as better for storage.

Practice Exam 2 FRQ

- [ ] The explanation that the rectangular solid identified is better for storage because of a lower surface area-to-volume ratios.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that folds and projections produce cells with increased surface area-to-volume ratios without reducing cell size or changing the basic cell shape.

Practice Exam 3 MCQ

Many species of corals are threatened by the increasing temperatures and decreasing pH of ocean waters. One species, *Stylophora pistillata*, has been found to thrive in water that is warmer and has a lower pH than the water that corals typically thrive in. Additionally, researchers have found that the tolerance for the new water conditions is heritable.

1. Which of the following statements best explains the changes seen in *S. pistillata* in response to the changing water conditions?

(A) The corals' adaptation is an example of natural selection because the tolerance is in response to a changing environment and has a genetic basis. ✓

(B) The corals' adaptation is an example of the founder effect because the majority of corals do not have a tolerance for warmer water.

(C) The corals' adaptation is an example of genetic drift because the change was a chance event and not the result of selection in response to environmental change.

(D) The corals' adaptation is an example of adaptive radiation because it has resulted in a wide range of species adapting to the new ocean conditions.

Answer A

Correct. The changes in the ocean water have put selective pressure on the corals. Those with tolerance for warm water had an advantage.

A researcher measured the temperature at which two different samples of double-stranded DNA denature (separate into single strands). Sample 1 denatured at a significantly lower temperature than sample 2 did. Based on the data, the researcher claims that the DNA in sample 2 is composed of a higher percentage of guanine and cytosine than the DNA in sample 1 is.

2. Which of the following best supports the researcher's claim?

(A) The bonds between guanine and cytosine are covalent bonds, which require more energy to disrupt than those between adenine and thymine.

(B) Guanine-cytosine pairs denature at a higher temperature because they have more hydrogen bonds between them than adenine-thymine pairs do. ✓

(C) Adenine-thymine pairs require less energy to separate because adenine and thymine are both single-ring bases.

(D) Guanine-cytosine pairs require more energy to separate because one is a purine and one is a pyrimidine.

Answer B

Correct. Three hydrogen bonds form between guanine and cytosine, while only two hydrogen bonds

form between adenine and thymine.

Researchers studying the bacterium *Escherichia coli* split a population of the bacteria into two samples. Sample 1 was transformed with a plasmid containing a gene that makes the bacteria resistant to the antibiotic kanamycin. Sample 2 was transformed with a plasmid lacking the antibiotic resistance gene. A portion of each sample was then added to growth plates containing just nutrients or growth plates containing nutrients and kanamycin.

After being allowed to grow for 24 hours at 37°C, the number of colonies on each growth plate was counted (Table 1).

TABLE 1. BACTERIAL GROWTH FOLLOWING TRANSFORMATION WITH OR WITHOUT THE KANAMYCIN RESISTANCE GENE

Plate #	Transforming Plasmid Contains Kanamycin Resistance Gene	Growth Media	Results
1	No	Nutrient only	Too many colonies to count (lawn)
2	Yes	Nutrient only	Too many colonies to count (lawn)
3	No	Nutrient with antibiotic	No colonies observed
4	Yes	Nutrient with antibiotic	7 colonies observed

3. Which of the following claims is best supported by the data in Table 1 ?

(A) The transformation procedure killed all the bacteria that were added to plate 3.

(B) More bacteria on plates 1 and 2 were successfully transformed than on any other plate.

(C) None of the bacteria on plate 2 were successfully transformed with the kanamycin resistance gene.

(D) Only the bacteria that were successfully transformed with the kanamycin resistance gene grew on plate 4. ✓

Answer D

Correct. Plate 4 contains kanamycin, which will kill any bacteria that do not contain the kanamycin resistance gene. The bacteria that grew are expected to contain the gene.

Practice Exam 3 MCQ

4. The fertilization of ovules from plant Q by pollen from plant R results in the production of seeds. What percent of the genes in each offspring's chloroplasts will have been inherited from plant R?

(A) 0% ✓

(B) 25%

(C) 50%

(D) 100%

Answer A

Correct. Chloroplasts are transmitted in the ovule, not the pollen. Therefore chloroplast-determined traits would be 100% inherited from plant Q.

High levels of certain plant nutrients in runoff can lead to rapid growth of algae (an algal bloom) in aquatic ecosystems. These algal blooms are generally followed by algal death and decomposition, which consumes large amounts of dissolved oxygen in the water and results in oxygen levels insufficient to support aerobic respiration. This process is known as eutrophication. The amount of algae present in a body of water can be estimated from the amount of chlorophyll *a* in a sample of the water. A researcher studying eutrophication collected samples at different times of the year in a freshwater ecosystem. The samples were analyzed for total nitrogen and chlorophyll *a* concentration (Figure 1) as well as total phosphorus and chlorophyll *a* concentration (Figure 2).

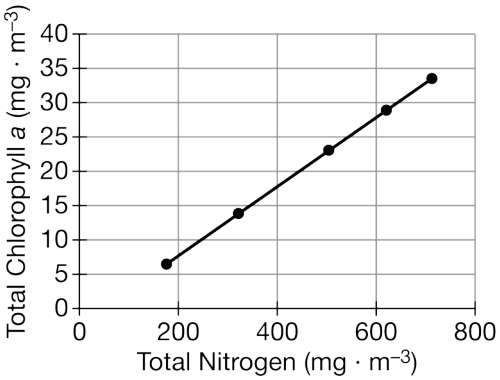


Figure 1. Amount of chlorophyll *a* in relation to the amount of total nitrogen

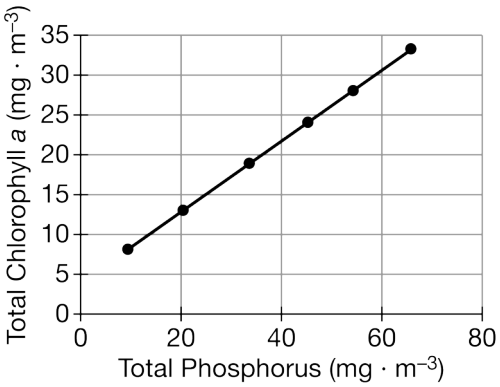


Figure 2. Amount of chlorophyll *a* in relation to the amount of total phosphorous

5. Which of the following best explains how higher concentrations of nitrogen and phosphorus contribute to eutrophication?

Practice Exam 3 MCQ

(A) An increase in the population of algae results in more nitrogen and phosphorus in the water, causing severe eutrophication.

(B) Both bacteria and algae require nitrogen and phosphorus, so the algae must grow faster to compete with bacteria.

(C) Nitrogen and phosphorus stimulate oxidative phosphorylation, which consumes the available oxygen in the water.

(D) Algae require nitrogen and phosphorus to build macromolecules, so higher concentrations of these nutrients can result in algal blooms. ✓

Answer D

Correct. Algae use nitrogen and phosphorus in the synthesis of macromolecules, so the addition of these nutrients increases the growth of the algae.

6. Which of the following was the dependent variable in the researcher's study?

(A) The concentration of chlorophyll *a* ✓

(B) The concentration of total nitrogen and phosphorus

(C) The slope of the trend line showing the rate of change

(D) The variance of the data points from the trend line

Answer A

Correct. The concentration of chlorophyll *a* is dependent on the concentrations of nitrogen and phosphorus.

7. Which of the following describes the relationship between the amount of chlorophyll *a* in a water sample and the concentration of nitrogen in that sample?

(A) As the concentration of chlorophyll *a* increases, the concentration of nitrogen decreases.

(B) The concentrations of nitrogen and chlorophyll *a* are directly correlated. ✓

(C) The concentrations of chlorophyll *a* and nitrogen increase throughout the year.

(D) There is no relationship between the concentrations of chlorophyll *a* and nitrogen.

Practice Exam 3 MCQ

Answer B

Correct. The data indicate that as the amount of nitrogen increases, the amount of chlorophyll in the water sample increases.

8. Which of the following investigations would enable researchers to test the claim that an increased concentration of algae has a negative effect on the number of aquatic invertebrates in the ecosystem?

(A) Examining the contents of the digestive tracts of aquatic invertebrates and looking for the presence of algae

(B) Examining the growth rate of algae in the absence of aquatic invertebrates

(C) Counting the number of aquatic invertebrates at different concentrations of chlorophyll *a* in the water ✓

(D) Counting the number of aquatic invertebrates at different concentrations of nitrogen and phosphorus in the water

Answer C

Correct. If the number of aquatic invertebrates decreases as chlorophyll *a* increases, it suggests the increased concentration of algae has a negative effect on the invertebrates.

Amylase is an enzyme that converts carbohydrate polymers into monomers. Glycogen synthase is one of the enzymes involved in converting carbohydrate monomers into polymers.

9. Which of the following best explains the reactions of these enzymes?

(A) Amylase aids in the removal of a water molecule to break covalent bonds whereas glycogen synthase aids in the addition of a water molecule to form covalent bonds.

(B) Amylase aids in the addition of a water molecule to break covalent bonds whereas glycogen synthase aids in the removal of a water molecule to form covalent bonds. ✓

(C) Amylase aids in the addition of a water molecule to form covalent bonds whereas glycogen synthase aids in the removal of a water molecule to break covalent bonds.

(D) Amylase aids in the removal of a water molecule to form covalent bonds whereas glycogen synthase aids in the addition of a water molecule to break covalent bonds.

Answer B

Correct. Amylase breaks bonds in polymers (hydrolysis) and glycogen synthase forms bonds between

monomers (dehydration synthesis). Water is added to break bonds and removed to form them.

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10. Which of the following describes the most likely location of cholesterol in an animal cell?

(A) Embedded in the plasma membrane ✓

(B) Dissolved in the cytosol

(C) Suspended in the stroma of the chloroplast

(D) Bound to free ribosomes

Answer A

Correct. Cholesterol is a nonpolar steroid that can be found embedded in the membrane between phospholipids.

Lactase is the enzyme needed to digest lactose, the sugar found in milk. Most mammals produce lactase when they are young but stop once nursing ends. In humans however, many people continue to produce lactase into adulthood and are referred to as lactase-persistent.

11. Which of the following mutations is most likely to cause lactase persistence in humans?

(A) A nucleotide substitution in the coding region of the lactase gene that interferes with the interaction between lactase and lactose

(B) A mutation that turns off the expression of transcription factors that activate the expression of lactase

(C) A mutation that increases the binding of transcription factors to the promoter of the lactase gene ✓

(D) The insertion of a single nucleotide into the lactase gene that results in the formation of a stop codon

Answer C

Correct. A mutation that results in an increase in transcription-factor binding will result in a higher expression of the lactase gene.

Practice Exam 3 MCQ

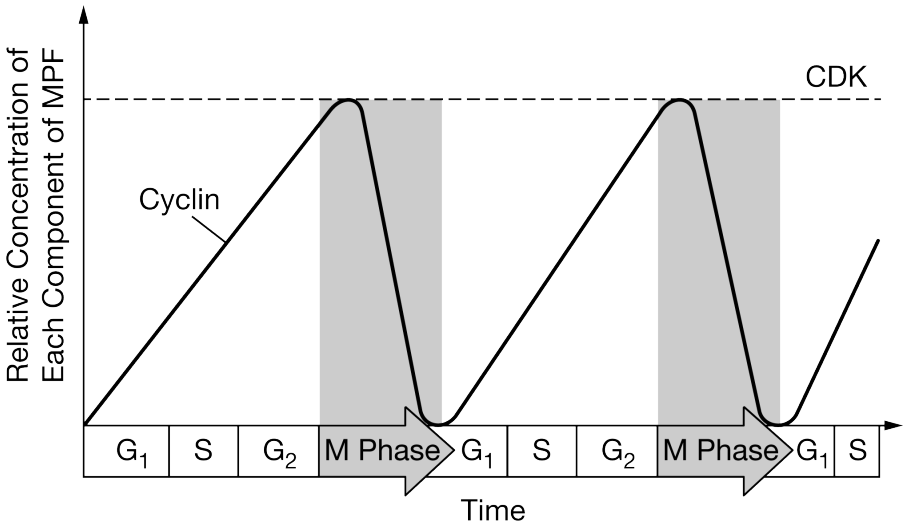


Figure 1. The relative concentrations of both the cyclin and CDK components of MPF

Maturation promoting factor, MPF, is a cyclin-CDK complex that catalyzes the phosphorylation of other proteins to start mitosis. The activity level of MPF is dependent on the relative concentrations of the cyclin and CDK components of MPF (Figure 1).

12. Based on Figure 1, which of the following describes the role of cyclin in the regulation of the cell cycle?

(A) During G_1 phase, the cyclin level decreases to signal the start of the resting phase of the cell cycle.

(B) During M phase, the cyclin level peaks, resulting in an increased binding frequency with CDK. ✓

(C) During S phase, the cyclin level remains the same because DNA replication is occurring.

(D) During G_2 phase, the cyclin level remains low, causing MPF activity to decrease, which leads cells to initiate mitosis.

Answer B

Correct. The cyclin and the CDK must bind for the MPF to be active and lead to the initiation of mitosis. The higher the concentration of cyclin, the more likely this binding will occur.

Modern bananas originated from a cross between a tetraploid banana species and a diploid banana species. The product of this cross was the triploid Cavendish banana strain, a sterile hybrid that is only grown asexually. Recently, the Panama fungus has been observed more frequently parasitizing the Cavendish banana, and scientists claim that this type of banana is on the brink of extinction.

Practice Exam 3 MCQ

13. Which of the following provides the best explanation of the scientists' claim regarding the threat to the Cavendish banana?

(A) The triploid genome of the Cavendish banana makes it susceptible to the negative impact of recessive alleles.

(B) Having three of each chromosome interferes with normal function of the cells and increases the likelihood of contracting a disease.

(C) The lack of genetic diversity of the Cavendish banana decreases the chance that a variation exists in the population that is immune to the fungus. ✓

(D) Asexual reproduction increases the mutation rate during replication, resulting in a greater chance that the offspring have a dysfunctional immune system.

Answer C

Correct. Species and populations with little genetic diversity, like the asexually grown Cavendish hybrid, are at a greater risk when environmental changes occur.

Belding's ground squirrels (*Spermophilus beldingi*) live in closely related groups. When they feed in the open, certain individuals (guard squirrels) watch for predators instead of feeding. The guard squirrels give an alarm call when a predator is sighted, allowing the rest of the group to run to safety. Researchers have noted that, because the alarm call draws attention to the guard, the guard is more likely to be caught by the predator and therefore has a lower survival rate in comparison with the other squirrels.

14. Which of the following best explains the behavior of the guard squirrels?

(A) The behavior of the guard squirrels increases the survival of close relatives that share the genes of the guard squirrels. ✓

(B) The guard squirrels confuse the predator, lowering the predator's success rate because the predator cannot tell which squirrel is producing the sound.

(C) Guard squirrels typically have recessive alleles, and by sacrificing themselves, they lessen the chance that recessive alleles will get passed on.

(D) Guard squirrels are typically females who have already reproduced, so they are no longer needed by the group.

Answer A

Correct. Protecting those that are genetically similar contributes to the presence of many of the same genes in the next generation.

Practice Exam 3 MCQ

15. If 30% of the nucleotides in a single-stranded RNA molecule are adenine, then what percent are expected to be thymine?

(A) 0% ✓

(B) 20%

(C) 30%

(D) 70%

Answer A

Correct. RNA does not contain the nitrogenous base thymine.

Practice Exam 3 MCQ

Melanocytes are skin cells that can become cancerous and develop into a cancer known as melanoma. Some cancerous melanocytes have developed resistance to the drugs currently used to treat melanoma. As a result, researchers are investigating the effects of a new compound (drug X) on four different melanoma cell lines. Researchers analyzed cell survival in two cell lines (Figure 1) and oxygen consumption in the presence of drug X in all four cell lines (Figure 2). Figure 3 shows the proposed mechanism by which drug X affects cells.

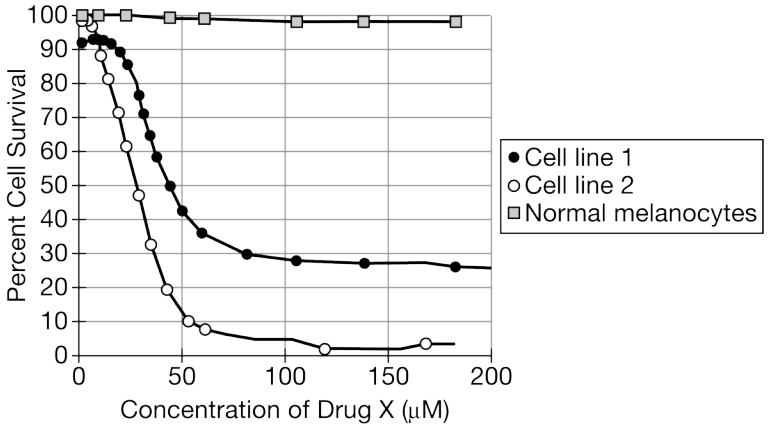


Figure 1. Percent survival of normal melanocytes and cancerous melanocyte (melanoma) lines 1 and 2 after treatment with different concentrations of drug X

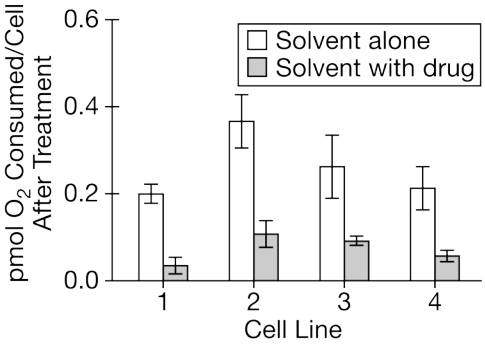


Figure 2. Oxygen consumption per cell in four melanoma lines after treatment with either solvent alone or solvent containing drug X. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

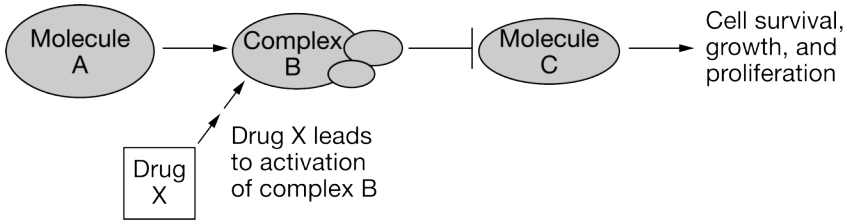


Figure 3. Pathway leading to cell survival, growth, and proliferation and the likely effect of drug X

Practice Exam 3 MCQ

16. Which of the following best describes the data in Figure 1 ?

(A) As the concentration of drug X increases, there is an increase in melanoma cell survival.

(B) At a concentration above 10 µM, drug X reduces melanoma cell survival. ✓

(C) At a concentration below 25 µM, drug X increases survival in all melanoma cell lines.

(D) At a concentration of 25 µM, drug X has a greater effect on melanoma line 1 than on melanoma line 2.

Answer B

Correct. Fewer melanoma cells survive in the presence of 10 µM drug X or more than when the drug is not present.

17. Based on Figure 2, which of the following best supports the claim that drug X inhibits oxygen consumption?

(A) In the absence of drug X, melanoma lines 1 and 4 consume similar amounts of oxygen.

(B) In the presence of drug X, melanoma line 2 consumes statistically more oxygen per cell than does melanoma line 3.

(C) Melanoma line 3 consumes statistically less oxygen per cell in the presence of drug X than it does in the presence of the solvent alone. ✓

(D) Melanoma line 2 in the presence of drug X consumes statistically less oxygen than does melanoma line 4 in the absence of drug X.

Answer C

Correct. To determine the effect of the drug on oxygen consumption, the oxygen consumption of cells in the presence of the drug would be compared to the oxygen consumption of cells from the same cell line in the absence of the drug to see if the data are statistically different (error bars do not overlap).

18. Based on the information presented, which of the following best explains why the researchers measured oxygen consumption as an indicator of the effectiveness of drug X?

(A) Oxygen provides the source of electrons for cellular respiration and is necessary for energy production.

(B) Oxygen consumption increases the mutation rate and causes cells to become cancerous.

(C) Oxygen activates apoptosis, which results in the death of melanoma cells.

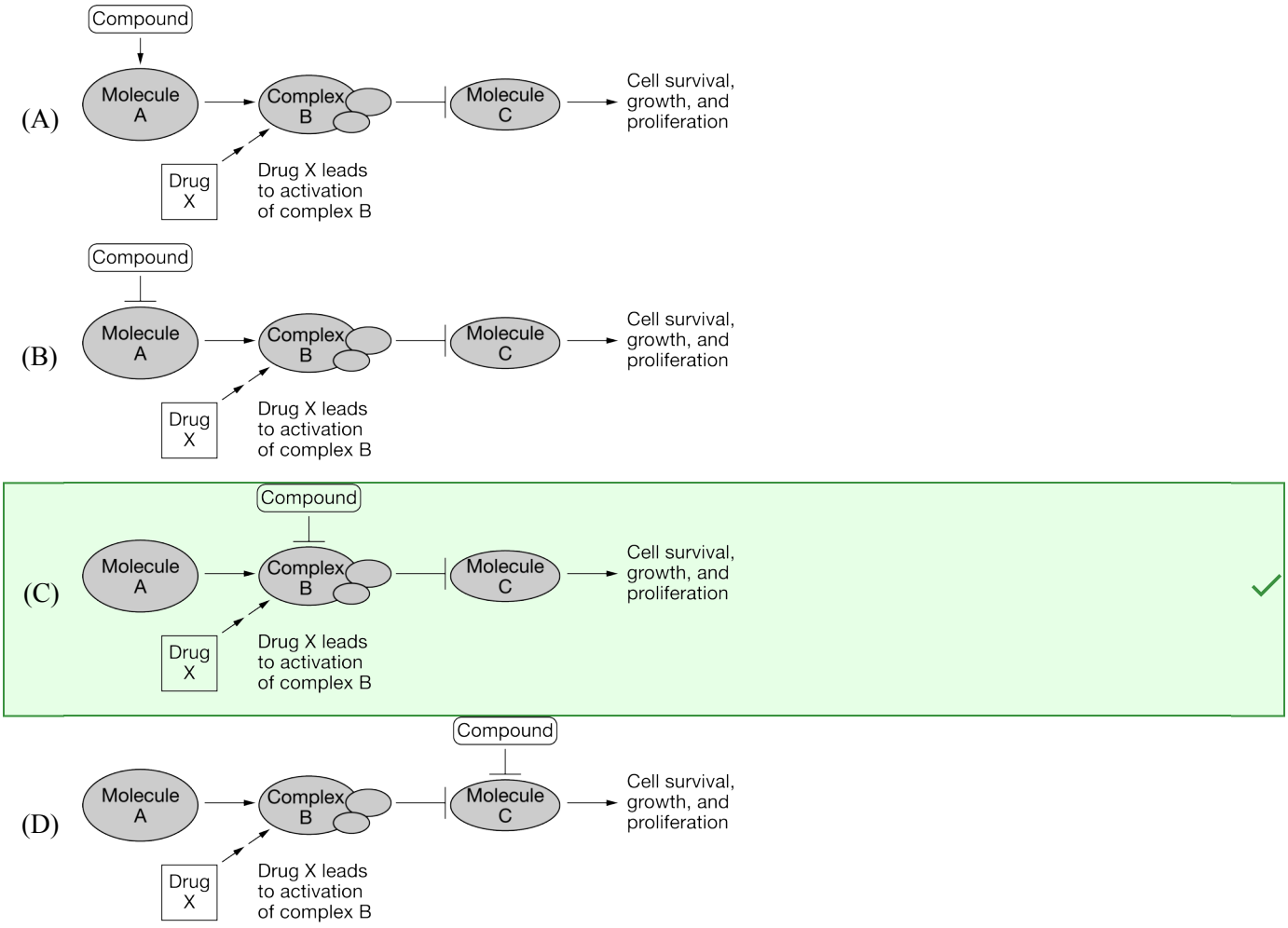
(D) Oxygen accepts electrons in oxidative phosphorylation, a process necessary for melanoma cell survival. ✓

Practice Exam 3 MCQ

Answer D

Correct. Melanoma cells require oxygen for normal metabolism, so oxygen consumption is a good indicator of metabolism.

19. A researcher has identified a compound that reverses the effect of drug X. Based on Figure 3, which of the following best explains how the compound acts in the pathway to reverse the effects of drug X?



Answer C

Correct. Drug X activates complex B, which inhibits molecule C and thus cellular division. Reversing the effect of drug X requires the inhibition of complex B.

Practice Exam 3 MCQ

20. Which of the following best explains how molecules such as O_2 and CO_2 can move across the membrane of a cell?

(A) The majority of the cell membrane contains protein channels that allow this type of molecule into the cell.

(B) The majority of the cell membrane is nonpolar, which allows small, nonpolar molecules to freely cross. ✓

(C) The phospholipids of the membrane are tightly packed, so only small molecules and ions can fit between phospholipids.

(D) ATP is hydrolyzed to provide energy to help O_2 and CO_2 move against their concentration gradient and across the membrane.

Answer B

Correct. Small, nonpolar molecules can pass through the cell membrane by simple diffusion because the membrane is composed of phospholipids with nonpolar tails.

Scientists investigated the effect of oxygen levels on the net rate of carbon fixation in two types of plants. The plants were grown in either well-watered soil (control) or dry soil and then exposed to either 21% or 1% O_2. The net rate of CO_2 fixation for both types of plants was measured. Data are shown in Figure 1 and Figure 2.

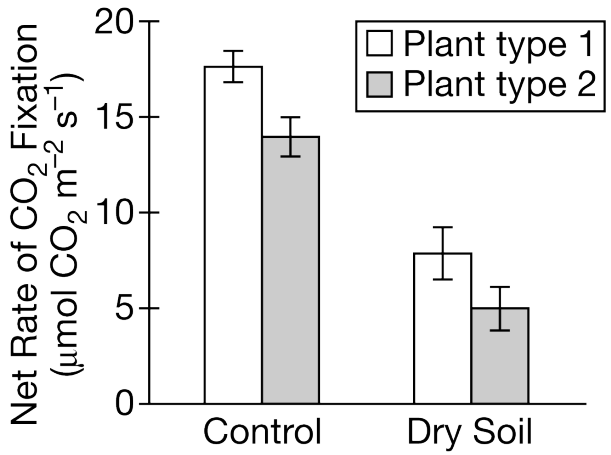


Figure 1. Net rate of CO_2 fixation in two types of plants grown in wet (control) or dry soil at 21% O_2

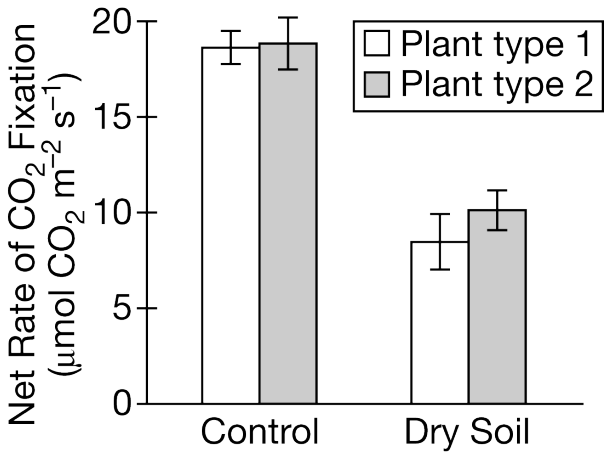


Figure 2. Net rate of CO_2 fixation in two types of plants grown in wet (control) or dry soil at 1% O_2

21. Which of the following statements about the rate of CO_2 fixation in the two types of plants is supported by the data shown in the figures?

Practice Exam 3 MCQ

(A) At 21% O_2, plant type 2 has a lower rate of CO_2 fixation than plant type 1 does in both types of soil. ✓

(B) At 1% O_2, plant type 2 has a higher rate of CO_2 fixation than plant type 1 does in the dry soil but not in the control soil.

(C) Plant types 1 and 2 have a statistically different rate of CO_2 fixation in both soil types at both oxygen levels.

(D) The rate of CO_2 fixation is the same in both types of plants in the control soil at both oxygen levels.

Answer A

Correct. The standard error bars do not overlap for the two types of plants at 21% O_2 in either soil type, indicating that there is a statistical difference between the rates of CO_2 fixation.

A researcher is conducting an experiment in which cells in different phases of the cell cycle are fused together. The researcher then records what happens to the nuclei of the resulting cell (Table 1).

TABLE 1. COMBINATIONS OF CELLS THAT WERE FUSED AND THE PHASE OF NUCLEI IN THE RESULTING CELL

Phase of Cell 1	Phase of Cell 2	Phase of Nuclei in Resulting Cell
S phase	G_1 phase	Two S-phase nuclei
S phase	G_2 phase	One S-phase nucleus and one G_2-phase nucleus
G_1 phase	G_2 phase	One G_1-phase nucleus and one G_2-phase nucleus
Interphase	M phase	Two M-phase nuclei

22. Which of the following research questions is best addressed by the experiment?

(A) How do chemical messengers affect a cell's transition between the phases of the cell cycle? ✓

(B) How does the number of chromosomes affect when a cell transitions to the next phase of the cell cycle?

(C) How does the amount of genetic information change throughout the cell cycle?

(D) How does the checkpoint at G_2 serve to prevent the transmission of mutations?

Practice Exam 3 MCQ

Answer A

Correct. By fusing these cells, the researcher was able to determine which phases had chemical messengers and how those messengers could influence cells in other phases.

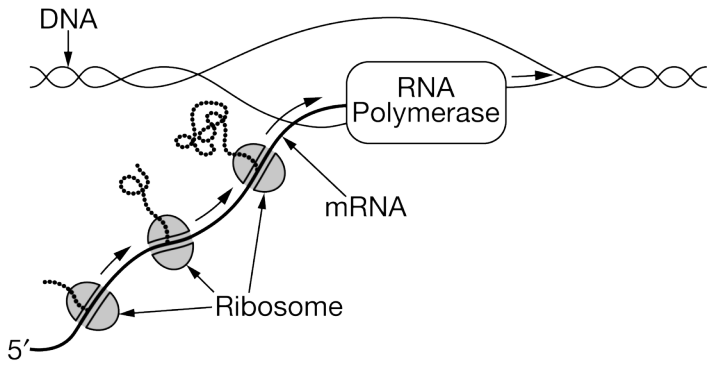


Figure 1. Protein synthesis in a prokaryotic cell

23. Which of the following best describes a characteristic of the process shown in Figure 1 that is unique to prokaryotes?

(A) The mRNA is synthesized in a 5′ to 3′ direction.

(B) A single strand of the DNA is being used as a template for the transcription of the mRNA.

(C) The translation of the mRNA is occurring while the mRNA is still being transcribed. ✓

(D) The enzyme that is transcribing the mRNA is RNA polymerase.

Answer C

Correct. In eukaryotes, RNA transcripts need to be modified before they can be translated, but prokaryotes can translate the beginning of an mRNA molecule while the end is still being transcribed.

Practice Exam 3 MCQ

A scientist is investigating the possibility that two traits in a particular plant are determined by genes that are on the same chromosome. The scientist crossed a plant that is homozygous dominant for both traits with a plant that is homozygous recessive for both traits. The heterozygous offspring in the F_1 generation were then crossed with a plant that is homozygous recessive for both traits. The results expected if the genes independently assort and the observed results are presented in the table.

Phenotype	Expected Number in F_2	Observed Number in F_2
Long stems, white flowers	25	17
Short stems, red flowers	25	19
Long stems, red flowers	25	31
Short stems, white flowers	25	33
Total number of plants	100	100

24. Which of the following critical values should the scientist use for the chi-square analysis of the data?

(A) 3.00

(B) 3.84

(C) 7.81 ✓

(D) 8.00

Answer C

Correct. This is the critical value for an analysis with three degrees of freedom.

Sugar gliders and Northern flying squirrels are mammals that have many morphological similarities despite being genetically different. Both are nocturnal and have large eyes to help them see in the dark. Both live in trees and have flaps of skin connecting their front and back legs, and when their front and back legs are extended, the flaps allow the animals to glide from one tree branch to another. Sugar gliders are native to Australia, and Northern flying squirrels are native to North America.

25. Which of the following best explains the morphological similarities between these two species?

Practice Exam 3 MCQ

(A) The similarities in traits indicate the species are the result of divergent evolution from a common ancestor that had the same traits that these two species share.

(B) The traits evolved through sympatric speciation, which often results in species being highly similar because they evolve in the same area.

(C) The two species evolved as the result of two separate bottleneck events that reduced the existing populations to a few individuals who happened to have the same traits.

(D) The similarities between the species evolved independently as a result of similar selective pressures in each species' environment. ✓

Answer D

Correct. The lack of genetic similarity between the two species suggests that the similarities arose independently in the two species and that the similarities are the result of convergent evolution.

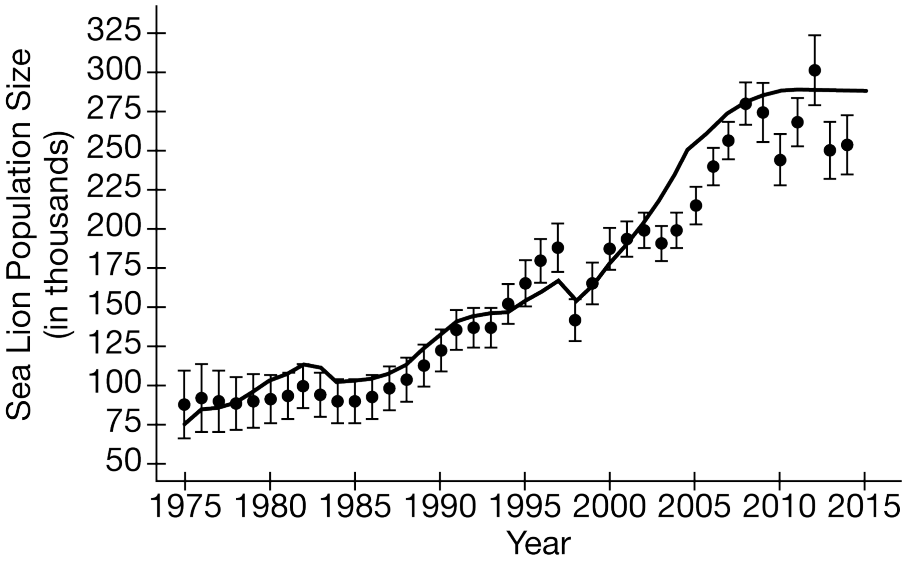


Figure 1. Change in the population size of sea lions over time. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

26. Which of the following best estimates the population size of the sea lions in 2000 based on the data shown in Figure 1?

(A) 100,000

(B) 125,000

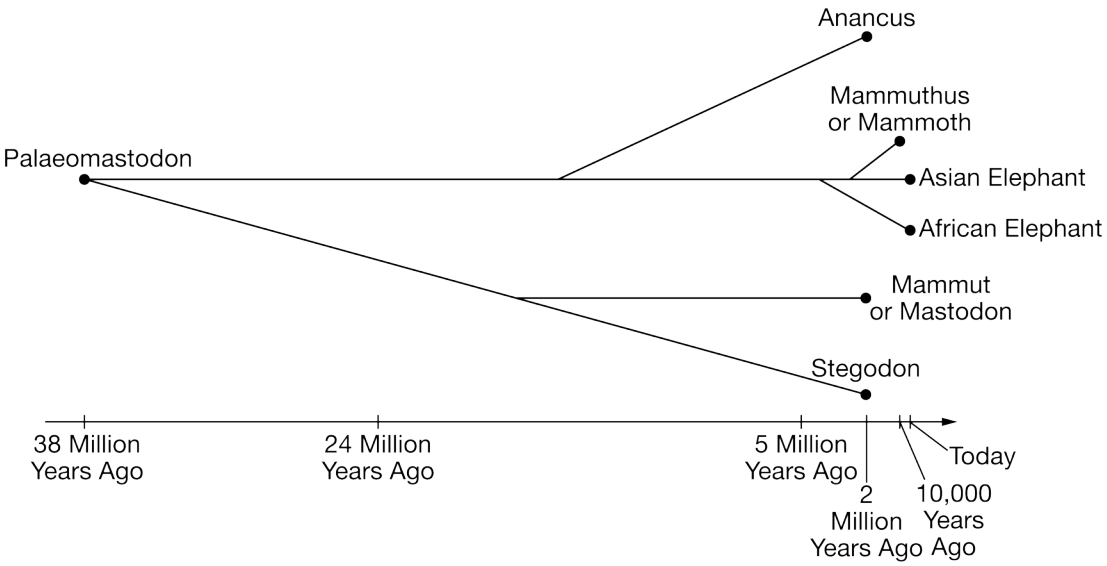
(C) 175,000 ✓

(D) 285,000

Practice Exam 3 MCQ

Answer C

Correct. The data indicate that the population in 2000 was almost 175,000.



The figure shows a phylogenetic tree of various members of the order Proboscidea, which includes modern elephants.

27. Which of the following claims is best supported by the information in the figure?

(A) The Asian and African elephants are the most closely related species shown on the tree.

(B) The mammoth diverged from its most recent common ancestor with African elephants before the mastodon diverged from its most recent common ancestor with Stegodons.

(C) The mastodon and the Stegodon diverged from their common ancestor 2 million years ago.

(D) The common ancestor of the African elephant and the mastodon is the Palaeomastodon. ✓

Answer D

Correct. If the lineages of the African elephant and the mastodon are traced back until they intersect, the species at the point of intersection is the Palaeomastodon.

---

28. Which of the following describes the most direct effect of a mutation in the DNA that encodes a cell's rRNA?

Practice Exam 3 MCQ

(A) The cell's ability to transport the amino acids needed for translation will be reduced.

(B) The cell's ability to transcribe RNA transcripts that will be translated will be reduced.

(C) The cell's ability to properly assemble ribosomes and initiate translation will be reduced. ✓

(D) The cell's ability to modify proteins after they have been assembled will be reduced.

Answer C

Correct. rRNA molecules, along with proteins, make up the structure of the ribosome. Translation is initiated when the rRNA interacts with the start codon in the mRNA.

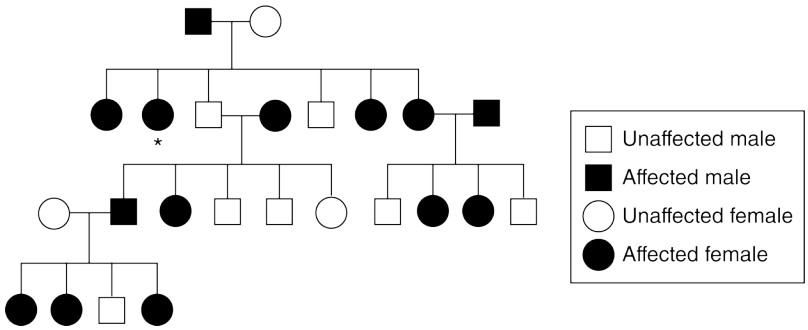


Figure 1. Incidence of protoporphyria in a particular family

Protoporphyria is a genetic disorder characterized by an extreme sensitivity to sunlight. One form of protoporphyria is caused by a mutation in the *ALAS2* gene that results in the accumulation of protoporphyrin, an organic compound, in the blood, liver, and skin. The pedigree in Figure 1 shows the incidence of protoporphyria in a particular family.

29. Which of the following best describes the genotype of the individual identified with an asterisk in the pedigree in Figure 1 ?

(A) Two dominant *ALAS2* alleles

(B) Two recessive *ALAS2* alleles

(C) One dominant *ALAS2* allele and one recessive *ALAS2* allele ✓

(D) One recessive *ALAS2* allele and no second allele for the *ALAS2* gene

Answer C

Correct. The individual must have one copy of the dominant allele because the individual is affected and

one recessive allele because the individual's mother is unaffected.

30. Which of the following best describes the inheritance pattern illustrated in Figure 1 ?

(A) Protoporphyria has an autosomal recessive inheritance pattern.

(B) Protoporphyria has an X-linked dominant inheritance pattern. ✓

(C) Protoporphyria has an X-linked recessive inheritance pattern.

(D) Protoporphyria has a mitochondrial inheritance pattern.

Answer B

Correct. There are unaffected offspring of two affected parents, indicating both had a recessive allele to pass on to those offspring. Additionally, affected males pass only to female offspring, whereas affected females pass to either male or female offspring.

31. One mutation in $ALAS2$ that is associated with protoporphyria is a four-nucleotide deletion. The protein expressed from the mutant allele is 20 amino acids shorter than the wild-type protein.

Which of the following best explains why a shortened protein is produced?

(A) The mutation disrupts the start codon, preventing the ribosome from beginning translation.

(B) The mutation introduces a premature stop codon, causing translation to end early. ✓

(C) The mutation changes the gene's regulatory region, causing unregulated gene expression.

(D) The mutation affects posttranscriptional modifications by preventing the removal of introns.

Answer B

Correct. The introduction of a stop codon will result in the production of a shortened protein.

32. A researcher claims that an individual has protoporphyria, based on a physical exam. Which of the following techniques would most likely confirm the researcher's claim?

(A) Transforming bacteria with the mutant variation of $ALAS2$ to measure gene expression

(B) Culturing cells from the individual in the lab and measuring the cells' growth rate

(C) Using light microscopy to examine the individual's chromosomes during metaphase

(D) Determining the nucleotide sequence of the individual's $ALAS2$ alleles ✓

Practice Exam 3 MCQ

Answer D

Correct. Sequencing the individual's DNA will allow a researcher to confirm that the individual carries a mutant *ALAS2* allele.

---

33. Which of the following is evidence that eukaryotes and prokaryotes share a common ancestor?

(A) All eukaryotes and prokaryotes contain linear DNA.

(B) All eukaryotes and prokaryotes contain ribosomes. ✓

(C) All eukaryotes and prokaryotes use organic molecules as an energy source.

(D) All eukaryotes and prokaryotes are capable of mitosis.

Answer B

Correct. Although the ribosomes of eukaryotes and prokaryotes have a slightly different structure, both contain ribosomes that perform the same function, suggesting that the trait came from a common ancestor.

Wolves, once native to Yellowstone National Park, were hunted to the point of complete extinction in the park. As a result, the elk population in the park flourished, putting extra demands on the carrying capacity of the park. Many other species, such as certain trees and beavers, were negatively affected by the increase in elk.

Years later, wolves were reintroduced into the park. While the elk population decreased after the wolves returned, the beaver and songbird populations began to increase as did the populations of various plant species.

34. Which of the following best explains how wolves are a keystone species in this ecosystem?

(A) Wolves help balance the population sizes of other species, allowing more species to thrive in the wolves' presence than in their absence. ✓

(B) Wolves prey on certain species, increasing competition among other species, which results in a decrease in biodiversity in the ecosystem.

(C) Wolves are not affected by factors that typically limit the population size; therefore their population can grow exponentially, increasing the demand on the parks' resources.

(D) Wolves prey on other species without having any natural predators themselves; therefore their presence significantly decreases many populations within the community.

Practice Exam 3 MCQ

Answer A

Correct. By decreasing the elk population to a size that is sustainable in the ecosystem, the wolves increased the amount of resources available to other species.

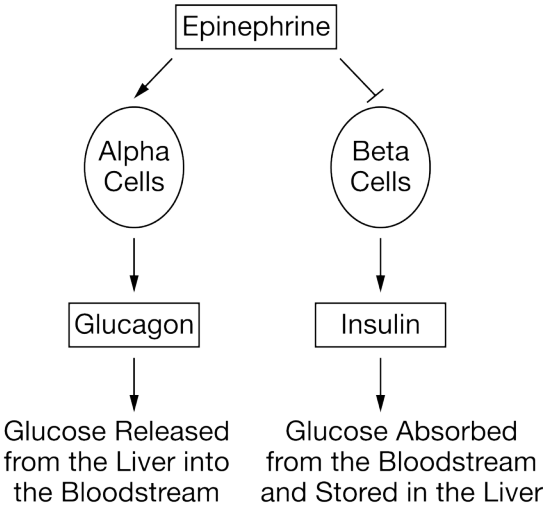


Figure 1. A model of epinephrine signaling

Two types of cells, alpha and beta cells, produce signaling molecules that affect blood sugar levels in opposite ways (Figure 1). Epinephrine is a chemical, often released during periods of exercise, that ultimately causes an increase in blood sugar levels in the body.

35. Based on Figure 1, which of the following best explains how exercise causes blood glucose levels to rise?

(A) Epinephrine inhibits alpha cells, causing the release of glucagon, and activates beta cells, blocking the release of insulin.

(B) Epinephrine activates alpha cells, blocking the release of glucagon, and inhibits beta cells, causing the release of insulin.

(C) Epinephrine activates alpha cells, causing the release of glucagon, and inhibits beta cells, blocking the release of insulin. ✓

(D) Epinephrine inhibits alpha cells, blocking the release of glucagon, and activates beta cells, causing the release of insulin.

Practice Exam 3 MCQ

Answer C

Correct. Epinephrine activates alpha cells, thus activating glucagon release, and inhibits beta cells, which inhibits insulin release.

Pesticides are chemicals that are often sprayed on crops to kill plant-eating insects, preventing damage to the crops. While pesticides are effective initially, many researchers claim that any single pesticide will see reduced effectiveness in as little as ten to fifteen years.

36. Which of the following best supports the claim by scientists that the pesticides will eventually lose their effectiveness?

(A) Insects will evolve to avoid any plant sprayed with pesticides to increase their chances of survival.

(B) Insects that are naturally resistant to the pesticide will survive and reproduce more than the insects that are sensitive to the pesticide. ✓

(C) The insects will build up a tolerance to the pesticides, and eventually the pesticide will not affect them.

(D) The pesticides will increase the mutation rate in the insects, resulting in higher genetic diversity and higher survival rate.

Answer B

Correct. An already-existing variation that makes an insect resistant would increase the insect's fitness, resulting in an increase in the frequency of resistance genes in future generations.

Newborn babies and hibernating animals contain a large amount of brown adipose (fat) tissue (BAT). Certain proteins in the BAT cells increase the permeability of the inner mitochondrial membrane to protons, disrupting the proton gradient.

37. Which of the following best predicts the effect of disrupting the proton gradient in BAT?

(A) The pH of the matrix will increase, allowing the production of more ATP per gram of substrate.

(B) The pH of the intermembrane space will decrease, allowing a steeper proton gradient to form.

(C) Electron transport and oxidative phosphorylation will be decoupled, generating more heat but less ATP. ✓

(D) The number of protons available to pass through ATP synthase will increase, resulting in more ATP.

Practice Exam 3 MCQ

Answer C

Correct. The protons pumped into the matrix because of electron transport will diffuse back into the intermembrane space and will not be available for oxidative phosphorylation. As a result, more energy will be converted to heat and less will be available for ATP production.

In pea plants, flower color and the length of the flower's pollen grains are genetically determined. Researchers studying pea plants crossed homozygous dominant pea plants with homozygous recessive pea plants. The F_1 plants were then crossed, and the number of offspring with each phenotype was recorded. The researchers' observed data, however, differed from the expected data. The researchers did a chi-square analysis and calculated the chi-square value to be 5.5.

38. Based on their calculation, the researchers would most likely conclude which of the following?

(A) The genes that determine these two traits are likely on the same chromosome.

(B) The differences between observed data and expected data are due to chance. ✓

(C) The genes mutated since the researchers began the experiment and now have a different inheritance pattern.

(D) The allele frequencies of the offspring have changed, suggesting the population is evolving.

Answer B

Correct. The calculated chi-square value is less than the critical value for three degrees of freedom (four possible phenotype combinations -1), indicating the researchers will fail to reject the null hypothesis.

A massive increase in the growth of a specific species of algae resulted in record-breaking levels of a potentially dangerous toxin being released into the water. A researcher hypothesizes that the unusual growth of this algal species was caused by an increase in water temperature. The researcher designs an experiment to test the hypothesis.

39. Which of the following is the dependent variable in the researcher's experiment?

(A) The growth of the algae ✓

(B) The temperature of the water

(C) The concentration of toxin in the water

(D) The different species of algae growing in the water

Practice Exam 3 MCQ

Answer A

Correct. The researcher hypothesizes that an increase in water temperature will affect the growth of the algae, making algae growth the dependent variable.

---

40. Which of the following is the most likely effect of a mutation in the gene coding for a DNA repair enzyme?

(A) The cell containing the mutation will divide more frequently because the cell cycle checkpoints will not function properly.

(B) Mutations will accumulate more quickly because the cell will not be able to fix errors in replication. ✓

(C) The mutated gene will not be transcribed because RNA polymerase cannot transcribe mutated DNA.

(D) The cell will immediately undergo apoptosis so that mutated DNA is not replicated in future rounds of cell division.

Answer B

Correct. DNA repair enzymes detect and correct errors that occur during replication. If the cell cannot do so, mutations will be passed on during future rounds of division.

Practice Exam 3 MCQ

Certain chemicals, including sodium fluoride (NaF), are capable of inhibiting specific steps of glycolysis. Figure 1 shows the steps of the glycolysis pathway, indicating where various macromolecules enter the pathway as well as the specific reaction inhibited by NaF.

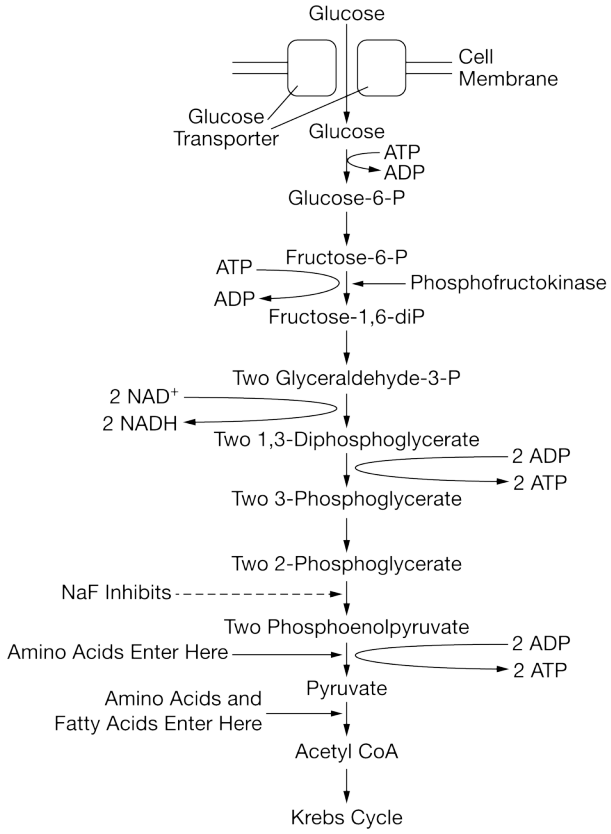


Figure 1. Key steps in the metabolic pathway of glucose

41. If NaF is added to cells undergoing cellular respiration, which of the following will most likely accumulate in the cells?

(A) Glucose Transporter

(B) Pyruvate

(C) Phosphoenolpyruvate (PEP)

(D) 2-phosphoglycerate ✓

Answer D

Correct. 2-Phosphoglycerate will accumulate because it is the last chemical in the pathway prior to the NaF inhibition.

Practice Exam 3 MCQ

42. Based on Figure 1, the net number of ATP molecules produced during glycolysis from the metabolism of a single glucose molecule is closest to which of the following?

(A) 0

(B) 2 ✓

(C) 4

(D) 8

Answer B

Correct. During glycolysis 4 ATP are generated, but 2 are converted to ADP in the process.

43. An increase in the concentration of protons in the cytosol will most likely have which of the following effects on glycolysis?

(A) Glycolytic enzymes will denature as a result of the increased H^+ concentration. ✓

(B) Reaction rate will increase as a result of the increased H^+ concentration.

(C) H^+ will replace phosphorous and inhibit ATP formation from ADP.

(D) The water potential will increase, resulting in a decrease in the rate of glycolysis.

Answer A

Correct. A decrease in the pH could cause enzymes in the glycolysis pathway to become denatured.

44. Which of the following describes why a glucose transporter is needed to move glucose into the cell?

(A) Glucose is nonpolar and requires ATP to move across the membrane.

(B) Glucose molecules are polar and need to move from low concentration to high concentration.

(C) Glucose molecules are charged, and charged molecules are only ever actively transported.

(D) Glucose is large and polar and cannot pass through the phospholipid bilayer. ✓

Answer D

Correct. Glucose is a relatively large and polar molecule and requires a transport protein to cross the membrane.

Practice Exam 3 MCQ

45. Tarui disease is an inherited disorder that is caused by mutations in $PFKM$, the gene that encodes a subunit of phosphofructokinase, an enzyme in the glycolysis pathway. Individuals with Tarui disease produce little or no functional phosphofructokinase in skeletal muscle cells. Based on Figure 1, which of the following best explains why a low carbohydrate diet is recommended for those with Tarui disease?

(A) Carbohydrates are capable of undergoing lactic acid fermentation, and amino acids and fatty acids are not.

(B) Carbohydrate metabolism requires all the reactions of glycolysis, and amino acids and fatty acids do not. ✓

(C) Carbohydrates cannot be used to synthesize important metabolic enzymes like amino acids and fatty acids can be.

(D) Carbohydrates cannot be stored, while amino acids and fatty acids can be.

Answer B

Correct. Noncarbohydrates can provide energy to a cell through other metabolic pathways.

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46. A mutation in the upland cotton plant causes the development of chloroplasts with a single outer membrane and no internal membranes. Which of the following would most likely be observed in chloroplasts of cotton plants with this mutation?

(A) They would be unable to remove waste products, because internal transport proteins would not be present.

(B) They would be unable to generate the ATP and NADPH needed to make sugars, because these processes occur on membranes within the chloroplast. ✓

(C) They would be unable to take up carbon dioxide, because $\mathrm{CO_2}$ is transported into the chloroplast in membrane-bound vesicles.

(D) They would be unable to move within the cell, because only organelles with double membranes are mobile.

Answer B

Correct. Many of the molecules needed to perform the light reactions of photosynthesis are embedded in the membranes of the thylakoids.

47. A researcher is crossing two organisms that are heterozygous for three Mendelian, unlinked traits (XxYyZz). Which of the following is the fraction of offspring that are predicted to have the genotype xxyyzz?

Practice Exam 3 MCQ

(A) $1/64$ ✓

(B) $1/32$

(C) $1/16$

(D) $1/8$

Answer A

Correct. One out of four offspring are expected to be homozygous recessive for each of the three genes. To determine the fraction expected to be homozygous recessive for all three genes one would calculate: $\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{64}$

Mosquitoes are frequently a target of insect control strategies because of their ability to spread disease. One strategy is to introduce guppies, a type of freshwater fish, into areas where mosquitoes are a problem. Guppies can consume large numbers of mosquito larvae, which cuts down on the number of mosquitoes living to adulthood.

Researchers assessing the effectiveness of the guppy solution are concerned that the introduction of guppies might cause more problems than it is fixing. Guppies are hardy, tolerant of a wide range of environmental conditions, and fast-reproducing.

48. Which of the following predicts the most likely ecological problem that would occur if guppies are introduced into new areas to control the mosquito population?

(A) The guppies might have no natural predators, which will result in a dramatic increase in the guppy population and an increase in competition for other native species. ✓

(B) The guppy population will likely evolve to consume another food source instead of the mosquito larvae.

(C) The larvae will evolve a mechanism that will enable them to avoid predation from the guppies.

(D) Predators in the area will consume the guppies, requiring the addition of more guppies on a regular basis.

Answer A

Correct. Guppies are strong and fertile. If they have ample resources and few predators, their population could grow unchecked and create problems for native species.

Practice Exam 3 MCQ

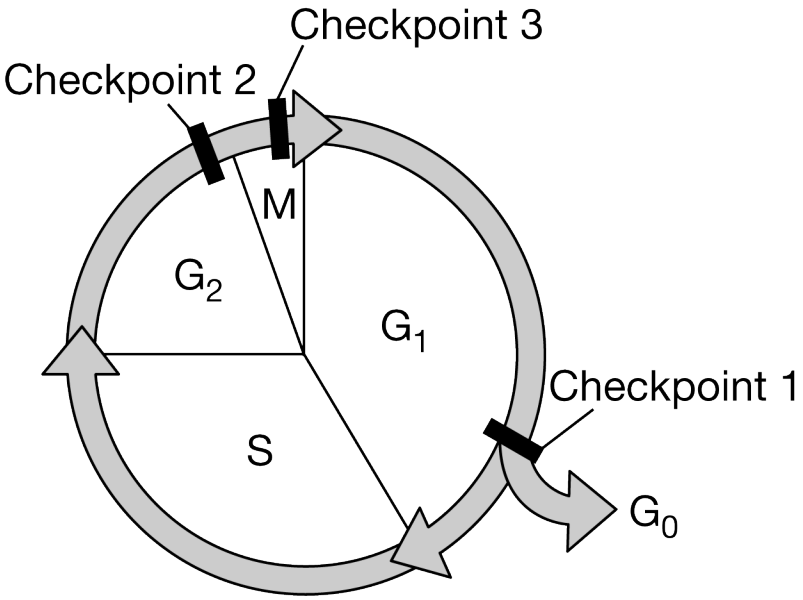


Figure 1. Diagram of the cell cycle with key checkpoints

49. Which of the following describes a mutation that would lead to an increase in the frequency of nondisjunction?

(A) A mutation affecting checkpoint 1 proteins that forces cells to enter G_0

(B) A mutation affecting checkpoint 2 proteins that allows cells to divide with DNA damage

(C) A mutation affecting checkpoint 3 proteins that prevents attachment of spindle fibers ✓

(D) A mutation affecting checkpoint 2 proteins that prevents duplication of the chromosomes

Answer C

Correct. If the cells move into anaphase without the spindle fibers properly attached to the chromosomes, the chances of nondisjunction occurring will increase.

Scientists studying a wild population of mantled howler monkeys found the average birth rate to be 0.22 and the average death rate to be 0.12. At the start of the study, the population consisted of 13 monkeys.

50. Assuming no immigration or emigration, which of the following best describes the change in population size that will occur over the next year?

Practice Exam 3 MCQ

(A) The population size will decrease because a population of 13 is too small to sustain itself.

(B) The population will decrease because more monkeys are dying each year than are being born.

(C) The population will increase exponentially because it is so small that there are likely few competitors and abundant resources.

(D) The population will increase because more monkeys are being born each year than are dying. ✓

Answer D

Correct. The birth rate exceeds the death rate. Therefore, the population is expected to grow.

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51. Which of the following best explains how the prokaryotic expression of a metabolic protein can be regulated when the protein is already present at a high concentration?

(A) Repressor proteins can be activated and bind to regulatory sequences to block transcription. ✓

(B) Transcription factors can bind to regulatory sequences to increase RNA polymerase binding.

(C) Regulatory proteins can be inactivated to increase gene expression.

(D) Histone modification can prevent transcription of the gene.

Answer A

Correct. High levels of protein product can result in a negative feedback mechanism in which a repressor protein is activated, binds to the regulatory sequence, and blocks transcription.

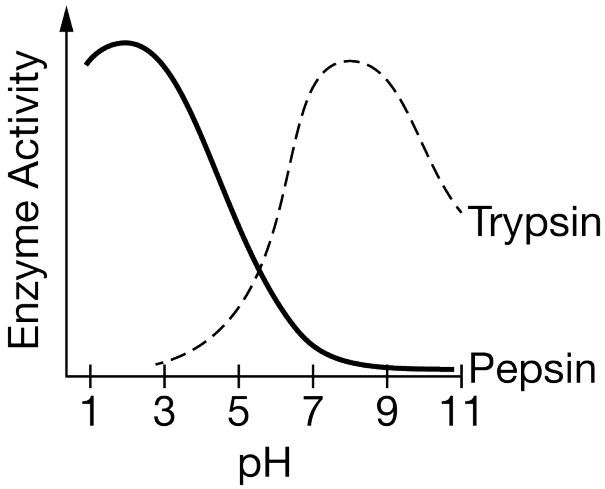


Figure 1. Activity levels of two digestive enzymes over a range of pH

Trypsin and pepsin are enzymes that function in different areas of the digestive tract. One functions in the stomach, where the pH is between 1.5 and 3.5, while the other functions in the small intestines, where the pH is between 6 and 8.

52. Based on Figure 1, which of the following best describes where each enzyme functions?

(A) Pepsin works in the intestines because the optimal pH for pepsin is basic.

(B) Trypsin works in the stomach because the optimal pH for trypsin is basic.

(C) Pepsin works in the stomach because the optimal pH for pepsin is acidic. ✓

(D) Trypsin works in the stomach because the optimal pH for trypsin is acidic.

Answer C

Correct. The optimal pH for pepsin is around 2, and the pH of the stomach is between 1.5 and 3.5, making the stomach an ideal environment for pepsin.

Practice Exam 3 MCQ

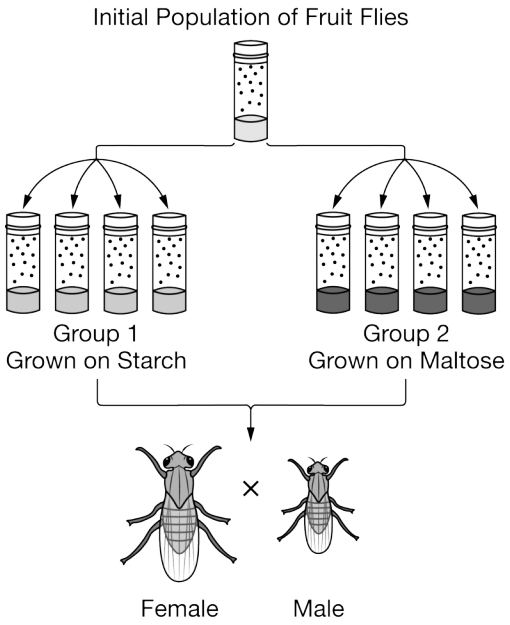


Figure 1. Experimental steps taken to determine whether the nutrient source in a growth chamber affects mating preference

A population of fruit flies (*Drosophila pseudoobscura*) grown on a typical nutrient source was separated into several growth chambers. Each of the new chambers was assigned a different source of carbohydrates for nutrition, either starch or maltose, and maintained for many generations (Figure 1).

After a year, male and female flies were paired up in different combinations. The instances of mating between the males and females grown on the same nutrient source and males and females grown on different nutrient sources were recorded (Table 1).

TABLE 1. MATING INSTANCES BETWEEN MALES AND FEMALES GROWN ON

THE SAME OR DIFFERENT NUTRIENT SOURCES

	Females Grown on Starch	Females Grown on Maltose
Males Grown on Starch	22	9
Males Grown on Maltose	8	20

To test whether the mating preferences were simply the result of being isolated in separate growth chambers, the researchers tested the mating preferences of flies that were both grown on starch, but in either the same or different chambers (Table 2).

Practice Exam 3 MCQ

TABLE 2. MATING INSTANCES BETWEEN MALES AND FEMALES GROWN ON THE SAME NUTRIENT SOURCE IN THE SAME OR DIFFERENT CHAMBERS

	Females Grown on Starch in Chamber 1	Females Grown on Starch in Chamber 2
Males Grown on Starch in Chamber 1	18	15
Males Grown on Starch in Chamber 2	12	15

53. Which of the following best represents the null hypothesis for the experiment detailed in Figure 1 ?

(A) The mating preferences of flies are not dependent on the nutrient source on which they are grown. ✓

(B) Prolonged exposure to a different diet can change behavior in fruit flies.

(C) When crossed, flies grown on the same medium will not produce viable offspring.

(D) Growing flies in a confined chamber will not create a barrier to mating in fruit flies.

Answer A

Correct. The null hypothesis is one that states that the variable tested does not affect the variable measured. In this case the null hypothesis is that growth on different nutrient sources does not affect mate selection.

54. For a population to be in Hardy-Weinberg equilibrium, certain conditions must be met. Which of the following best explains a characteristic of the fly populations that meets a condition of Hardy-Weinberg equilibrium?

(A) The population size of the flies was very small, which increases the likelihood of random mutations having a greater impact.

(B) Since the flies were confined in chambers, they did not experience any migration, so no new alleles were introduced through migration. ✓

(C) Flies have a relatively short generation time and therefore are more likely to pass mutations to the next generation.

(D) The different food sources may have exerted selective pressure on the fly populations, giving an advantage to certain flies.

Practice Exam 3 MCQ

Answer B

Correct. No migration is a condition of Hardy-Weinberg equilibrium, so the fact that the flies were contained met this condition.

55. Which of the following claims is best supported by the data in Tables 1 and 2 ?

(A) The population of flies has undergone sympatric speciation.

(B) The flies have developed a form of behavioral preference that could lead to divergence. ✓

(C) A postzygotic barrier has evolved between the populations of flies that prevents the birth of viable offspring.

(D) The flies have undergone allopatric speciation, since only flies from the same chamber mated.

Answer B

Correct. Divergence might now occur because flies demonstrate differential mate selection based on the food source on which they were grown.

56. Starch is a complex carbohydrate that is digested by the enzyme amylase. A mutation in the gene that encodes amylase prevents the digestion of starch. The inability to digest starch is an autosomal recessive trait. In a population of $1{,}000$ flies, 410 were unable to break down starch.

Which of the following is closest to the number of flies expected to be heterozygous for the amylase mutation assuming all conditions of Hardy-Weinberg equilibrium are met?

(A) 0.13

(B) 0.36

(C) 0.46 ✓

(D) 0.64

Answer C

Correct. The frequency expected to be heterozygous is $2\mathrm{pq}$, where $\mathrm{q} = 0.64$ $\left(\text{calculation:}\sqrt{(410/1000)}\right)$ and $\mathrm{p} = 0.36$ $(\text{calculation:} 1 - \mathrm{q})$.

Practice Exam 3 MCQ

57. Which of the following combinations of flies showed the highest number of mating instances?

(A) Males grown on maltose paired with females grown on starch

(B) Males grown on starch paired with females grown on starch ✓

(C) Males grown on maltose paired with females grown on maltose

(D) Males grown on starch paired with females grown on maltose

Answer B

Correct. There were 22 instances of mating between these types of flies.

The primary function of the kidney is to exchange molecules across a membrane between the blood and the urine. One type of kidney cell has a basic rectangular shape, except for a single surface, which is lined with tiny, finger-like projections that extend into the surrounding extracellular space.

58. Which of the following best explains the advantage these projections provide the cell?

(A) The projections increase the volume of the cell without affecting the surface area, which increases the metabolic needs of the cell.

(B) The projections increase the surface area–to-volume ratio of the cell, which allows for more efficient nutrient exchange with the environment. ✓

(C) The projections increase the speed at which an individual molecule can move, resulting in faster nutrient exchange with the environment.

(D) The projections increase the selectivity of the membrane because the small size of the projections limits the number of transport proteins that can be embedded in the membrane.

Answer B

Correct. These projections add significantly more membrane than they do cellular volume. That results in a greater surface area–to-volume ratio and more efficient nutrient exchange.

Practice Exam 3 MCQ

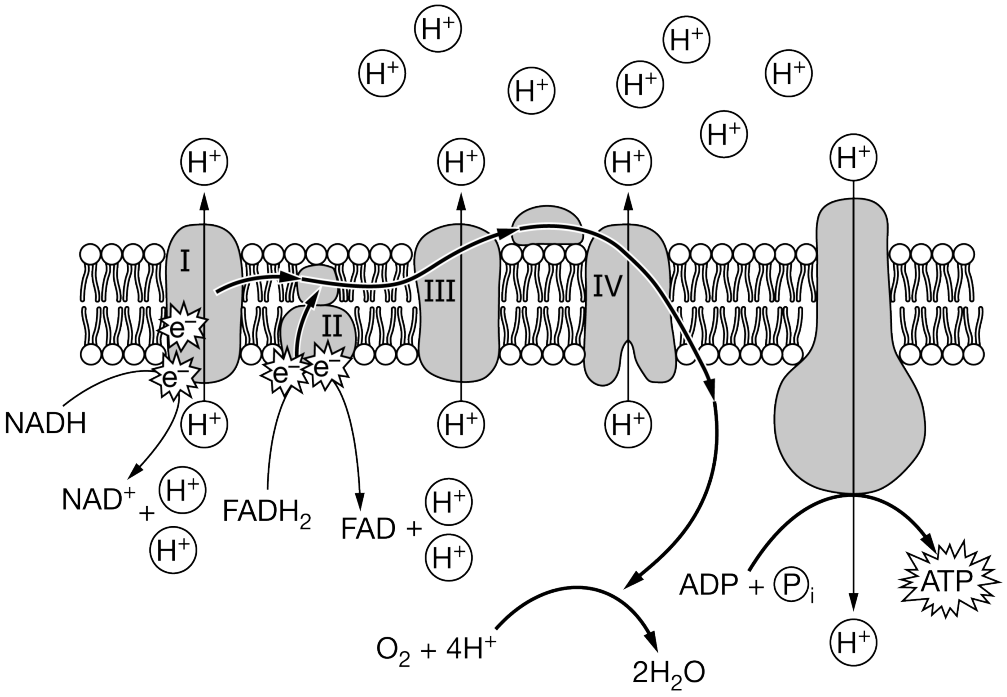


Figure 1. Diagram of the electron transport chain and ATP synthase in the membrane of mitochondria

On average, more ATP can be produced from an NADH molecule than can be produced from a molecule of $\mathrm{FADH_2}$.

59. Based on Figure 1, which of the following best explains the difference in ATP production between these two molecules?

(A) NADH contributes more electrons to the electron transport chain than $\mathrm{FADH_2}$ does and therefore provides more energy to pump protons.

(B) The electrons of $\mathrm{FADH_2}$ are transferred through three complexes of the electron transport chain whereas those of NADH are transferred through all four complexes. ✓

(C) $\mathrm{FADH_2}$ contributes more protons to the mitochondrial matrix, which decreases the proton gradient.

(D) The protons contributed by $\mathrm{FADH_2}$ are combined with $\mathrm{O_2}$ to make water and are not pumped across the membrane.

Answer B

Correct. The electrons from $\mathrm{FADH_2}$ are contributed to complex II rather than complex I and therefore are transferred through fewer complexes and have a lesser effect on the proton gradient than the electrons from NADH do.

---

60. Which of the following best explains why ligase is required for DNA replication?

Practice Exam 3 MCQ

(A) The lagging strand cannot be replicated continuously, and ligase is needed to join the fragments. ✓

(B) Ligase forms the hydrogen bonds between complementary bases in the two strands of DNA.

(C) Ligase facilitates the binding of RNA polymerase to the promoter region.

(D) Ligase enables the newly synthesized DNA to twist into a double helix.

Answer A

Correct. Since DNA is synthesized in the 5′ to 3′ direction, one strand of DNA is synthesized discontinuously, forming fragments that are joined by ligase.

Practice Exam 3 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph hide_paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Cryptosporidium parvum (*C. parvum*) is a single-celled, eukaryotic parasite that infects human cells in the digestive system and causes illness.

Although it is a eukaryote, *C. parvum* does not have functional mitochondria and generates ATP only through glycolysis. *C. parvum* uses the enzyme lactate dehydrogenase to perform fermentation after glycolysis.

Two chemicals, gossypol and FX11, are noncompetitive inhibitors of lactate dehydrogenase. Researchers investigated the effectiveness of gossypol and FX11 as drugs to kill *C. parvum*. In the experiment, human cells were treated with different concentrations of either gossypol or FX11 after infection with *C. parvum*, and the relative growth of *C. parvum* compared with that of control cells was measured (Figures 1 and 2).

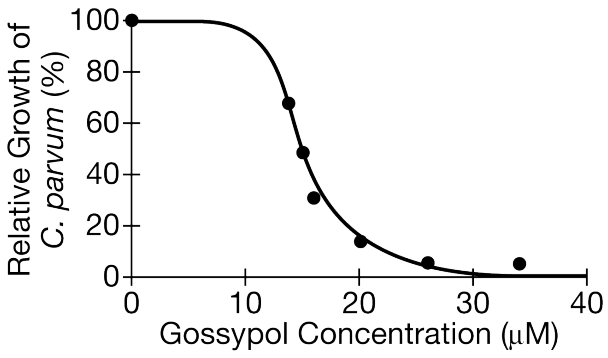


Figure 1. Effect of different concentrations of gossypol on *C. parvum* growth

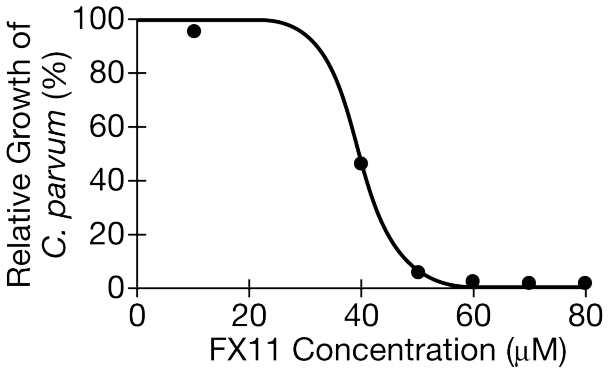


Figure 2. Effect of different concentrations of FX11 on *C. parvum* growth

1. (a) **Describe** how *C. parvum* obtains the glucose it needs for glycolysis after it has infected another cell. **Explain** the role of lactate dehydrogenase in enabling *C. parvum* to continue producing ATP by glycolysis.

(b) **Identify** the independent variable used in the experiment. **Identify** the difference between the control cells and the experimental cells used in the experiment. **Justify** the researchers using a different range of concentrations for FX11 than was used for gossypol.

(c) Based on the data in Figure 1, **identify** the concentration of gossypol that reduced *C. parvum* growth to 50% of that in control cells.

(d) Researchers discovered a strain of *C. parvum* that expresses a functional variation of the lactate dehydrogenase gene. A DNA sequence comparison showed that the variant differs from the normal sequence in the region that codes for the enzyme's allosteric site. **Predict** the effect of FX11 treatment on *C. parvum* cells that express this variant of lactase dehydrogenase. **Provide reasoning** to support your prediction. **Explain** how gossypol and FX11 might be used as drugs to treat *C. parvum* infections in humans without negatively affecting human cells.

Part A

Practice Exam 3 FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following.

· A description that *C. parvum* absorbs glucose from its environment, which, in this case, takes glucose away from its host

· An explanation that lactate dehydrogenase allows fermentation to convert NADH to NAD^{+}, which is needed for glycolysis to continue making ATP

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The response includes all of the following.

· An identification of the concentration of the chemical as an independent variable

· An identification of the characteristics of the control cells as the same type of cells, infected with the parasite but not treated with any chemicals

· A justification that $\mathrm{FX11}$ had minimal impact on the growth of *C. parvum* until it reached a concentration of greater than $30\ \mu\mathrm{M}$, which was the maximum concentration of gossypol that was tested

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response identifies the concentration of gossypol that reduced *C. parvum* growth to 50% of that in control cells as $15\ \mu\mathrm{M}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Practice Exam 3 FRQ

0	1	2	3 ✓

The response includes all of the following.

· The prediction that FX11 will have little to no effect.

· The reasoning that, because the allosteric site is the region of the enzyme where a non-competitive inhibitor like FX11 would bind, it is likely that changing the sequence of this region would decrease inhibitor binding

· The explanation that, while the drugs will prevent *C. parvum* from generating the ATP it needs to survive, human cells have mitochondria and do not rely on fermentation to produce ATP, so they are not likely to be harmed

Practice Exam 3 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Many types of cancer are treated with a combination of therapies. In lung cancer, some tumors respond well to the drug paclitaxel followed by radiation treatment. Paclitaxel is a chemical that disrupts mitosis. Instead of spindle fibers originating from the two sides (poles) of the cell, paclitaxel-treated cells develop three poles and then divide into three cells (tripolar division). Radiation therapy is more effective on tumor cells that have undergone tripolar division than on cells that have undergone normal mitosis.

Researchers treated cancer cells in the lab with different concentrations of paclitaxel for 15 hours. The researchers then determined the average percent of mitotic cells that were tripolar. The results are shown in Table 1.

TABLE 1. EFFECT OF PACLITAXEL CONCENTRATION ON PERCENT OF MITOTIC CELLS THAT WERE TRIPOLAR

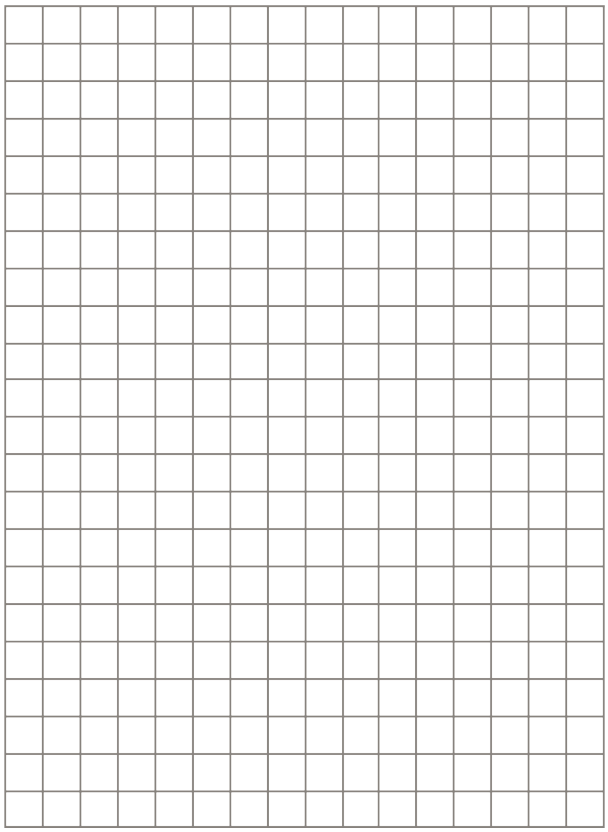
Concentration of Paclitaxel (nM)	Average Percent of Mitotic Cells that were Tripolar ($\pm 2\mathrm{SE}_{\bar{x}}$)
0	0.0 ± 0.0
2	17.0 ± 3.0
4	48.0 ± 3.5
6	65.0 ± 5.0
8	70.0 ± 4.0
10	50.0 ± 2.0

The $AURKA$ gene encodes an enzyme that helps assemble the spindle fibers, which signals the cells to continue through mitosis. When researchers analyzed the levels of AURKA protein in different types of cancer cells, they found that cancer cells expressing high levels of AURKA protein had more tripolar divisions when treated with paclitaxel, than did cancer cells expressing low levels of AURKA protein.

Practice Exam 3 FRQ

2. (a) **Describe** the situations in which a normal human cell would enter the cell cycle and undergo mitotic cell division. **Explain** how spindle fibers help ensure the products of mitosis are two identical cells with a full set of chromosomes.

(b) Using the template in the space provided for your response, **construct** an appropriately labeled graph that represents the data shown in Table 1. Based on the data, **determine** the concentration(s) of paclitaxel that is (are) most effective in causing tripolar cell division.



(c) Based on the data, **identify** the lowest level of paclitaxel that will allow for at least 50% of the cells to be tripolar. From the start codon through the stop codon, the length of the fully processed $\mathrm{AURKA\ mRNA}$ is $1{,}212$ nucleotides. **Calculate** the number of amino acids in the polypeptide chain coded for by the mRNA.

(d) **Predict** the effect of a mutation that prevents the expression of AURKA on a normal (noncancerous) cell.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following.

Practice Exam 3 FRQ

· A description that cells divide by mitosis when the organism is growing or repairing tissues

· An explanation that spindle fibers attach to the center of each duplicated chromosome and assist in pulling one chromatid to each pole of the cells so that, when the cell divides, each daughter cells contains a copy of each chromosome

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The response includes all of the following.

· Correct axes and labeling

· Correctly plotted points and scaling

· Correctly plotted error bars

· The determination that the concentration of paclitaxel that is most effective in causing tripolar cell division is between 6-8 nM

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response includes both of the following.

· An identification of 4 nM as the lowest level of paclitaxel that will allow for at least 50% of the cells to be tripolar

· A calculation of 403 as the number of amino acids in the polypeptide chain coded for by the AURKA mRNA

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the prediction that the cell will be unable to undergo mitosis.

Practice Exam 3 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In mice, fur color is a genetically determined trait. To observe the effects of natural selection on fur color in mice, scientists set up six enclosures with either light- or dark-colored sand on the ground. The enclosures were isolated from all ground predators and wild mice but accessible to predatory birds. The scientists placed equal numbers of light- and dark-colored mice into each enclosure. A total of 500 mice were used in the experiment. After several generations, the scientists sampled the mice and found that populations in the light sand enclosures were, on average, lighter in color than the original population, while those in the dark sand enclosures were, on average, darker in color than the mice in the original population.

3. (a) **Describe** the way the scientists will determine the evolutionary fitness of the mice in the experiment.

(b) **Identify** the independent variable in the scientists' experiment.

(c) **State** the null hypothesis of this experiment.

(d) The scientists claim that the changes in the frequency of fur color were the result of natural selection. **Justify** the researchers' claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response describes that the scientists will determine if the mice survive to successfully reproduce.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response identifies the color of the ground in each of the enclosures as the independent variable.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Practice Exam 3 FRQ

0	1 ✓

The response states that the null hypothesis is that the color of the enclosure ground will not affect the frequency of light and dark mice in the population.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response justifies the claim by indicating that mice with a coat color that helped them blend into the environment were able to avoid predation and reproduce, increasing the frequency of their genes in the population.

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The pinewood nematode is a eukaryote that infects certain species of pine trees, feeds on the cells surrounding the trees' transport system, and ultimately kills the trees. Trees are infected when nematode-carrying beetles feed off the trees and inject the nematode into the trees when they bite through the bark. Once infected, pine trees increase the production of chemicals that serve as a defense mechanism for the trees by negatively affecting the nematodes.

Researchers have found that pinewood nematodes contain symbiotic bacteria that can degrade the pine trees' defensive chemicals. To investigate the role these bacteria play in nematode survival in the presence of these defensive chemicals, researchers pretreated nematodes with antibiotics and then exposed them to α–pinene, one of the defensive chemicals produced by the pine trees.

4. (a) **Describe** the relationship between a parasite and its host.

(b) **Explain** how producing the enzymes that digest α-pinene is beneficial to the bacterial species living within the nematodes.

(c) **Predict** the effect of the antibiotic treatments on the mortality rate of the nematodes when exposed to α-pinene.

(d) Provide reasoning to **justify** your prediction in part (c).

Part A

Practice Exam 3 FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response describes that the parasite will benefit while the host will be harmed.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response explains that, by producing these enzymes, the bacteria increase the survival rate of the nematodes ensuring that the nematodes continue provide a protected habitat and resources for the bacteria.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response predicts that the mortality rate of the nematodes will increase after the nematodes are treated with antibiotics.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes the justification that the antibiotics will reduce the bacterial population and prevent the degradation of α-pinene, resulting in there being more α-pinene to negatively affect the nematodes.

Practice Exam 3 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

A cross section of a chloroplast showing membranes and the spaces between membranes is shown in Figure 1.

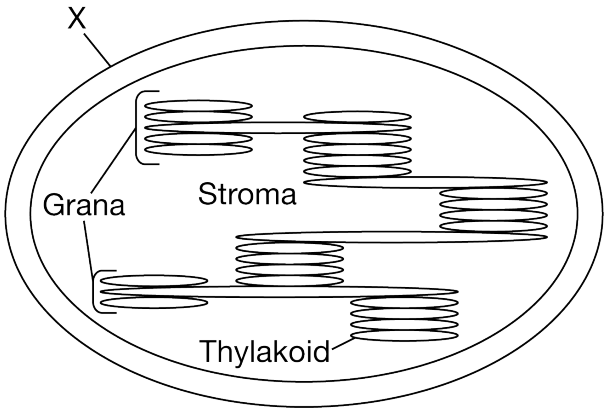
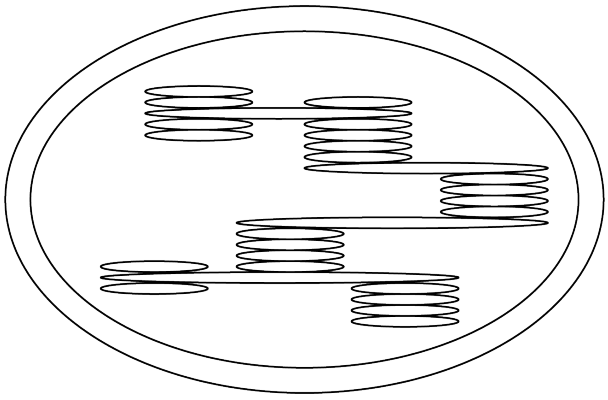


Figure 1. A diagram of the cross section of a chloroplast

5. (a) **Describe** the major process that takes place in this eukaryotic organelle.

(b) **Explain** the function of the structure labeled with an X in Figure 1.

(c) On the template in the space provided for your response, **represent** the location where carbon fixation takes place by writing "CF" and the location of the electron transport proteins by writing "ETP".



(d) **Explain** how the shape and stacking of the thylakoids contributes to the rate of carbon fixation by the chloroplast.

Part A

Practice Exam 3 FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response describes the major process that takes place in the organelle as light providing the energy to fix carbon into organic molecules.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response explains that the presence of organelles allows the compartmentalization of specific structures to more efficiently carry out specific metabolic functions.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response includes both of the following.

· The stroma is marked with "CF ".

· A thylakoid membrane is marked with "ETP ".

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response explains that the thin and stacked nature of the thylakoids increases the surface area of the thylakoid membrane, allowing for a greater number of chlorophyll molecules and electron transport proteins.

Practice Exam 3 FRQ

Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Drosophila melanogaster (*D. melanogaster*) is a species of fruit fly frequently used by researchers in genetic studies. Members of this species have two of each of four different chromosomes: the sex chromosome (flies have X and Y) and three autosomes (chromosomes 2, 3, and 4). Researchers studying *D. melanogaster* conducted genetic crosses to investigate a particular X-linked recessive trait encoded by a single gene (Table 1). Affected flies have the trait.

TABLE 1. CROSSES PERFORMED AND THE PHENOTYPES OF THE RESULTING OFFSPRING

Cross Number	Phenotype of Females Used in the Cross	Phenotype of Males Used in the Cross	Percent of Total Population That Are Affected	Percent of Male Offspring That Are Affected	Percent of Female Offspring That Are Affected
1	Unaffected	Affected	0	0	0
2	Affected	Unaffected	50	100	0
3	Unaffected	Unaffected	25	50	0
4	Affected	Affected	100	100	100

6. (a) **Identify** the genotypes of the male and female flies used in cross 2.

(b) **Identify** the cross in which the female parent was most likely heterozygous.

(c) The researchers hypothesize that crossing any unaffected female and an affected male will result in a 0% chance of producing an affected male offspring. **Evaluate** the validity of the hypothesis.

(d) **Explain** how the results exclude the possibility that the trait is encoded by a mitochondrial gene.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

Practice Exam 3 FRQ

The response identifies X^rX^r and X^RY as the genotypes of the male and female flies used in cross 2.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response identifies cross 3 as the cross in which the female parent was most likely heterozygous.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response determines that the hypothesis is not supported because an unaffected female could be heterozygous and pass on the recessive allele.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response explains that traits encoded by mitochondrial genes are inherited from the female parent only, resulting in offspring that all have the same genotype as their mother.